

Table of Contents

What Came Before Knowledge? (Ferris 2004).....	4
Introduction to Geometry.....	6
How Geometry Got Its Name.....	6
Story Epilogue.....	8
Review of Primary Experiences with Geometry.....	9
Principles in Geometry Work at the Elementary Level.....	11
Congruency, Similarity and Equivalence & Metal Insets.....	13
Congruency, Similarity and Equivalence & Constructive Triangles.....	21
Triangle Box.....	21
Small Hexagonal Box.....	22
Large Hexagonal Box.....	24
Combined Boxes: Small Hexagonal and Triangular Box.....	25
Small Rectangular Box.....	27
Introduction to Polygons.....	28
Introduction to Angles.....	31
Types of Triangles by Angles.....	34
Types of Triangles by Sides.....	35
Angles.....	37
Parts of an Angle.....	37
Measuring an Angle.....	38
Addition and Subtraction with Angles.....	41
Regular Protractor.....	46
Bisecting an Angle.....	49
Lines.....	50
Lines Nomenclature.....	50
Position of Two Lines.....	53
Intersecting Two Lines.....	55
Relationships Using Angles and Lines.....	56
Relationship Between Angles.....	56
Adjacent and Linear Pairs.....	57
Complementary and Supplementary.....	58
Vertically Opposite.....	59
Relationship of Angles Formed by Parallel Lines and a Transversal.....	60
Interior and Exterior Space.....	60
Alternate Interior and Alternate Exterior.....	61
Corresponding Angles.....	63
Size Relationships of Angles Formed by a Transversal.....	65
Pythagoras and Euclid.....	68
Pythagoras' Plate I.....	68
Show Congruence of Corresponding Parts.....	70
Pythagoras' Plate II.....	72
Count Length of Sides.....	74
Write Theorem.....	74
Pythagoras with the Constructive Triangles.....	75

Another Way to Calculate Pythagoras' Rule.....	78
Equivalence with Euclid's Plate.....	79
Red Square and Two Parallelograms.....	80
Yellow Square Yellow Parallelogram.....	80
Yellow Parallelogram and Yellow Square.....	82
Yellow Parallelogram and Large Red Rectangle.....	83
Blue Square and Blue Parallelogram.....	85
Small Red Rectangle and Blue Parallelogram.....	86
Polygons.....	88
Nomenclature of Polygons.....	88
Triangles.....	89
Diagonals.....	92
Trapezoid.....	94
Nomenclature of regular polygons.....	95
Center.....	96
Bisect.....	96
Radius.....	97
Apothem.....	97
The Sum of Angles in Plane Figures.....	98
Sum of the Angles in a Triangle.....	98
Equivalence with the Iron Material.....	101
Triangle-Rectangle - Equivalence.....	101
Rhombus-Rectangle - Equivalence.....	102
Triangle-Rectangle - Reasoning.....	104
Reasoning level For Triangles, Rhombi and Rectangles.....	106
Trapezoid - Equivalence Level.....	108
Trapezoid - Reasoning Level.....	110
Decagon – Equivalence.....	112
Decagon – Reasoning.....	115
Area of Plane Figures.....	119
Concept of Area Using Yellow Material.....	119
Rectangle.....	119
Parallelogram.....	124
Triangles.....	126
Acute.....	126
Doubling Area.....	126
Dividing Height.....	127
Dividing Base.....	127
Right-Angle.....	129
Doubling Area.....	129
Dividing Height.....	130
Dividing Base.....	130
Obtuse.....	132
Doubling Area.....	132
Dividing the Height.....	133
Dividing Base.....	134

Deriving Formulas for Area – Yellow Material.....	136
Rectangle.....	136
Parallelogram.....	138
Acute Triangle (Doubling Area, Split Height and Base).....	139
Right Triangle.....	143
Obtuse Triangle.....	145
Summary.....	148
Deriving Formulas for Area - Iron Material.....	149
Triangles and Rectangles.....	149
Rhombi and Rectangles.....	150
Different Rhombi Orientations.....	151
Trapezoid.....	155
Decagons And Tall Rectangle.....	157
Decagon and Long Rectangle.....	159
Special Polygons, Circumference and Area.....	162
Nomenclature of the Circle.....	162
Relationships Between Line Segments & Circumferences, Two Circumferences....	166
Area of a Circle.....	174
The Circle as a Special Polygon.....	174
Measuring the Circumference.....	176
Formula for Area of a Circle.....	179
Relationship Between Apothem and Sides of a Polygon.....	185
Area of a Regular Hexagon.....	186
Solid Geometry.....	187
The Concept of Volume.....	187
Equivalence as Related to Solid Figures.....	190
Three Important Dimensions.....	192
Equivalence Between Prisms with Various Bases.....	198
Derivation of the Formula.....	202
Solids of Rotation.....	205
Volume of a Pyramid.....	207
Volume of Cylinders and Cones.....	210
Polyhedrons.....	212
Nomenclature.....	212
Surface Area.....	215
Lateral Area.....	216
Total Area.....	216
Formulas.....	217
Geometry Charts.....	223
Geometry Nomenclature Material.....	225
Geometry Commands.....	226
Geometry Summary.....	227
Appendix A.....	228

What Came Before Knowledge? (Ferris 2004)

There was a time when learning was more valuable than gold. There were no schools. If one was sick and did not know about medicine that was available in another land, one died. Even gold could not save him. And learning was so rare that humankind had many questions but few answers. Even today, while we have so many wonderful inventions, humankind still does not always know how to live together peacefully. So, there are important things to be discovered. Maybe you will be the one to discover them.

In all times when we have questions we can not answer; there are always people who will say: “We must accept that there is no answer. It is not for us to understand.” And so they can never find the answer. But there are also those who continue to ask the question. And to question is to observe. One cannot ask a question without looking toward its answer. To ask the questions about what we do not know – is the beginning of understanding what we do not know.

We have heard of Pythagoras – who's name is used to describe an important natural law about squares. If you have not heard his name, you will before too long. People sometimes think he discovered the fact we call the Pythagorean theorem. But in fact, he was the first to prove it – not the first to observe the fact.

All knowledge is like this. You and I are always building onto the knowledge that came before us. And this knowledge, though it is taught to you as true – is sometimes not true. Perhaps you are the one who will show that something is not true that others accept as true. Because we believe it is true, we teach it to you and every generation that follows. We say it is true *to our best knowledge*.

But there is a different kind of knowing we use when we ask a question, when we bring our whole attention to solving a question sincerely. Pythagoras had a teacher named Thales (624-545 B.C). Thales lived over 2500 years ago. In those days, when learning was more valuable than gold, there were lots of unanswered questions. And people like Thales traveled all over the world for the sake of learning. He was a great athlete too and applied himself vigorously to everything he did.

One day while in Egypt, Thales asked himself how he could know the height of the pyramids he saw. He had no previous knowledge of how this could be known. And he asked the question, without any knowledge to draw upon. But he had noticed that at a certain time of day, his shadow was the same length as his true height. He wondered if the same was true for the pyramids. He had no theorem or proof; his intelligence led him to wonder about this – with no prior knowledge to use. But in asking the question and bringing the answer he found back home, he introduced geometry to Greece.

Today, he is considered one of the Seven Wise Men of Ancient Greece – not because of all he learned, but because of the way he looked carefully at a problem and asked questions to understand the nature of solving that problem. This kind of intelligence - that does not rely only on information and reason - we sometimes call intuition; and every human being has this other kind of knowing.

Even old Pythagoras used this kind of intelligence. His father was a

merchant around the Mediterranean Sea and how often Pythagoras must have been on the sea shore waiting for his father to return on the great sea ships of that day! Do you know that Pythagoras was the first to teach that the world was round? Can you imagine what a ship in the great distance looks like as it approaches the sea shore? Why do we see the mast of the ship before seeing the bottom of the ship? What blocks our view of the bottom of the ship and why? In those days, people learned the world was flat. Maybe you can see, using your intuition, why Pythagoras may have suspected the world was round.

So, Pythagoras, though everyone around him said otherwise, led the world in understanding that the world was round – not flat. Everyone must have looked at him strangely! Especially, since his old teacher, Thales, had taught that the world was flat and floated on water. Thales thought that an earthquake was caused by crashing waves below the ground. Though he could not prove it, it was an understanding which was based on ideas of logic. So we can see that intuition and logic are both useful if we understand that error is always present, and we therefore seek to prove and test what we call truth.

How do we know about Thales? He left no writings for us. We know of him through later writers only. But in his day, he was quite famous. He studied the skies constantly and wrote down the movement of the stars and planets. In 585 B.C., he announced the sun would disappear for a time on May 28th. When on that day, the sunlight was blocked by another planet passing between the sun and the earth, Thales became famous overnight.

How did he know what was going to happen? How do you use the knowledge passed on to you, your observation, intuition and logic - in what you know? How do you test what you think you know? Can you see why people become upset to learn something they thought was true - is not true?

Thales is known as the founder of Greek Philosophy – a word meaning love of truth. If we learned the world was a triangle, not flat or round, how could we use other knowledge given to us, our observation, our intuition and our logic to find out for ourselves? What important questions will you and I wonder about, even after we have been given an answer – because maybe there is more to know? Maybe we can predict the next solar eclipse of our day. Perhaps we can find answers worth even more than gold and clever inventions, that will be passed on to generations after us, by using our independent discovery of the whole truth about the universe we live in.

Thales showed us that simply asking a question, using other's knowledge but always testing it for oneself, honestly dedicating ourselves to a problem and having a faith in our own minds, are the first steps to all understanding. Whether it is measuring the height of pyramids or seeing ships sail over the distant horizon, **each of us has the ability to question and understand directly for ourselves.**

November 4, 2004

Introduction to Geometry

How Geometry Got Its Name

We know from our study about the Egyptians, that life in Egypt centered around a very important river – the Nile river. Egypt has what has been called the “gift of the Nile.” Egypt has also been called the mother of geometry.

Century after century, year after year, the Nile overflowed its banks and washed down the black mud from the surrounding hills. That very fertile soil is what gave the Egyptians the rich land they needed to be such successful farmers and to sustain their lives.

Those floods each year washed out the markers that the farmers put up to mark off the corners of their fields. So every year, the land had to be measured again, so that each farmer would know exactly where his land began and ended.

This **measuring that went on every year taught the Egyptians a great deal about geometry**. An Egyptian surveyor was called a harpedonapta and he walked along, followed by 3 slaves. One of the slaves **carried a rope that had two knots** tied in it. When stretched and held by the 3 slaves, it **formed a right triangle**.

(call three children, giving them a rope which has two knots. They hold the rope taut connecting its ends, and they realize that they formed a triangle)

These rope stretchers would go around and measure off the corners of the fields in this way and the corners of the pyramids in this way. Just this one idea had a lot of different uses.

Geometry means **land measuring** and it is from Egypt that it got its name.

Hundreds of years later, a Greek **merchant visited Egypt** and he became very interested in the wealth of knowledge that the Egyptians had. His name was **Thales**

and he was a very successful businessman. In fact, he became so successful and so rich that after a few years, he was able to retire – and he did. And then he devoted his time entirely to studying earth measurements, mathematics and astronomy. We'll probably talk about Thales again when we talk about the Babylonians, because **he learned about astronomy from the Babylonians**.

It's not for his work in astronomy that we best remember Thales. But rather it is for his contributions to mathematics and especially to geometry. Thales was very curious about the reasons behind information. And this **led him to a system of logical reasoning which gave birth to the new science of geometry**.

Pythagoras is another well-known historical figure in the area of geometry. He **studied under Thales** and he probably visited Egypt on his master's advice, so that he could study the earth-measuring that was going on there. When Pythagoras was about 50 years old, he went to live in Crotona, a town in Southern Italy. He quickly gathered together a **group of followers and they became known as Pythagoreans**. Today, many of the beliefs of these Pythagoreans might strike us as a little far-fetched. They're an interesting group – but they did study geometry. They made no practical use out of geometry; they were **only interested in the abstract side of mathematics**. They greatly expanded upon the work that was started by Thales. They used **deductive reasoning and they found logical proofs** about certain geometric facts.

For example, the Pythagoreans are credited with proving, using a right triangle in the proportions of 3 to 4 to 5, that the area of the square drawn on the longest side is always equal to the sum of the area of the squares drawn on the other two sides. This is the **Pythagorean Theorem**. It **proved why it worked for the Egyptians**; they didn't know why it worked but had discovered it and used it only.

Moving on to the year **400 BC**, about a mile outside of Athens, there was a walled **garden called Akademia, after the man who owned it – Akademos**. This is where the great philosopher **Plato** taught his philosophy. He felt that there **should be games to help learning and that problems should be about things that were interesting** to the students. Plato had some very strict rules that he set down for his grown-up followers. The inscription over the doorway into his school was this: "Let No Man Destitute of Geometry Enter My Doors."

Infinity was a puzzling problem to Plato. So, in order to avoid constructions that might involve infinity, he insisted on the hard and fast rule that only a ruler and a compass could be used for making geometrical constructions. Around **300 BC**, a man named **Euclid** wrote the book that still remains the best-selling book ever written on geometry. It's still in print today. His work consisted of 13 books, each written on a separate piece of parchment. The entire work is called "The Elements." This book

helped to train the mind of one of our American presidents. At the age of 40, **Abraham Lincoln studied and mastered the first 6 books of The Elements** – solely as an exercise in training his mind. If we go to a book store we can still find The Elements today. That's my story for today. On another day, we'll have another story about some of the people involved in the story of geometry.

Story Epilogue

Through stories, you want to build up for the children a general progression of the people involved in the history of geometry. Some of those are some of the famous people we already know. Some of them are less known. This is for the children, not a research paper.

For albums, we will write an original story on one of these people. It could be one of the people mentioned above and developed into a story just about that person, or it could be about someone new. If we want to be organized about this, you can share them afterwards – and have 25 stories available to all.

We start off with this historical perspective for two reasons. The children love stories and this is the type they find interesting. The second reason is to enable you to reflect on the relationship between the child's development and historical developments. What is useful in everyday life is generally what comes first and the rules come later – and are based on practical experience and knowledge. Children follow this same path to understanding.

Geometry deals with shapes, surfaces, lines, angles and the relationships between these things. At first glance, people sometimes think that these ideas are separate from everyday life. But if we stop and think about it, everything around us has shape or lines or angles or surface. We are surrounded by these things. And the child has been surrounded by them from the beginning of his life. We shouldn't overlook these impressions when we're working with children, regardless of what level we are working at. The child has been building up a sub-conscious storehouse of geometrical knowledge just by living in the environment; so this has been going on a long time.

Review of Primary Experiences with Geometry

Now let's think about the kinds of geometry experiences that the child can

ideally have in a Montessori primary environment. The primary teacher needs to recognize that the children coming into their classes, come with geometry experiences. The sensorial materials then help the children classify and make clear the forms found in the environment, through the geometry cabinet, a piece of material that gives the child experience with plane geometrical forms.

First, the child has a **muscular experience with tracing**. There's a mechanical control of error in that the forms will only fit into the insets in a certain pattern. Language comes later. The language goes from the general, general names like triangle, circle and so on – to more details – specific kinds of triangles for example.

Ideally, the child **should know the names of all or nearly all of the shapes** in that cabinet by the time she leaves the primary class. As the child begins working with the card material, he becomes less dependent on the muscular memory to know the shapes and can now recognize them using the visual sense. There's experience with similar shapes in the geometry cabinet – the circles. We don't speak of it specifically though similarity is introduced. The child gets a solid understanding of solid geometry with the geometric solids, as well as with the cylinders, the pink tower, the brown stair and the red rods.

With the **geometric solids, the child can see the relationship between the plane form and the solid form**, by superimposing the bases of solids onto the plane figures. This enables to the child to have a sub-conscious impression about their relationship.

With the **constructive triangles, the child has the opportunity to explore the triangle as the constructor** of all straight sided plane geometrical forms. The different boxes emphasize the different shapes – the rectangle, the triangle, the hexagon. Superimposed geometrical figures, where you can build up concentric figures or you can build up concentric and inscribed figures. So, the children should have these experiences in primary. That activity is related to the designing work that the children should be doing with metal insets. And they can begin to decide for themselves what they find harmonious and what they find inharmonious.

The large metal insets include samples of the basic shapes. Through working with them, the child is able to gain further knowledge of what it is that makes up various shapes. In the geometry cabinet, the shapes are already formed for the child. With the metal insets, the child is constructing the shape for himself. Every time he goes around a corner, he is feeling that angle. As he superimposes and does designing work with them, he notices more and more details about the different shapes. Additional work that the children might be introduced to in the primary include the small metal insets, classified card material and the opportunity to do handwork related to geometry – such as using a

ruler.

In terms of the small **metal insets, they include 3 sets: circle, square and triangle**. They can be used in the primary class for designing. The child builds up symmetrical designs; just doing this means we figure out which plates match to make sure we put things away in the right frame. The children can do this on top of a plain piece of paper, carefully tracing around the pieces. He traces an exposed edge and then removes the piece, moving on to the next shape, coloring the entire figure in when he is done. The next step is that he traces piece by piece, all or part of a piece is added as he goes, so that his work becomes quite elaborate. **On a basic level, this aids handwriting. It develops their fine motor control. This work continues in the elementary classroom. In introducing it in the elementary, take the time to do it carefully, setting the model that work is done to one's best ability**, thorough and carefully carried out. Getting the children doing designing work in the beginning days of school is one of the ways in which, informally, we can begin to find out what the children know about geometry.

For classified card material, these are essentially nomenclature cards – showing the parts of a circle, for example. It would be worked with just like the other nomenclature cards. First is just the picture and they learn the name. Then there are just the labels that they can read and match with the pictures and finally, use the control cards (pictures with labels) for control of error.

For handwork, they can be shown how to draw a straight line with a ruler, how to measure with a ruler or how to construct angles. They might be shown how to use a compass to construct a circle, for example.

The child who comes from the 3-6 class into the elementary class should already have a good concrete foundation in geometry. He **may already know the nomenclature material** and be ready to go on. Unfortunately, many will arrive without a complete experience with geometry. What should they come with? What is the minimum?

If they don't have a minimum level of experience, we will need to work with primary teachers to know what is a minimum foundation and why. The children should have had basic work with the geometry cabinet and the cards that come with it. They **should know the basic language: circle, square, triangle, some of the names of the regular polygons, the names of the figures in the mixed drawer such as Rhombus, Oval, parallelogram, curvilinear triangle and quatrefoil. They should have had some exposure to the trapezoid and the common quadrilateral or trapezoid. The children should have worked with the geometric solids and they should know their names. They should have had the opportunity to match them up with their bases, so**

they have at least an intuitive understanding of the relationship between solids and plane figures. They should have had some work with the constructive triangles and superimposed geometrical figures and some design work with the large metal insets.

From most primary class, you should find most of the children coming with this amount of experience. No matter how little or much they have, it's important to see our work as a continuation of their experience level, wherever he is, and to help him move forward. We are surrounded by geometry, so even a child who arrives with little primary experience has some knowledge to build on.

When a child enters the elementary class, one of the first things to **find out is what the child already knows.** In geometry, one way of doing this is through the design work. Another way is to simply look around a classroom or look around the outdoor environment, and see what geometric shapes the children can find. You can do this informally in a game type format – so as not to put anyone on the spot. Another thing one can do is to have a **collection of postcards. With a small group, pass out a postcard to each - “who has a circle? Yes, look at that!”** The child who never raises his hand is the one to assist first. Informally, we can discuss shapes, angles, lines and so on in this manner – and find out what they know in this way too.

Whatever we do, do it informally without making anyone feel badly. As soon as you become aware of children who don't know all the names in the geometry cabinet or of the geometric solids, start to teach them to the children. We use them for different purposes but we have them in the elementary environment as well. Get a small group together and do a 3-period lesson.

Principles in Geometry Work at the Elementary Level

When we think about the psychological characteristics of the 6 to 9 years old child, we remember that they're **interested in analyzing and in finding out relationships** – related to the factual information they learned in the first plane of development. They want to know the reasons why.

Hopefully, the child's primary experience has given him a base from which he can now carry out a reasoned geometrical exploration. Our work will then be more natural and go more easily.

Six to twelve year olds take great **joy in intellectual activity.** The study of geometry is therefore very well suited to this plane of development. The geometry

materials are creative by their very nature, because they give ample opportunity for the children to create their own abstractions. Although the laws that govern geometry will be given principally at the high school level, if they take up study of geometry there, the geometry materials that we'll use will enable these young children to create demonstrations of many of the major theorems. We won't look at theorems like one does in high school; rather **we'll look at concrete sensorial representations of many of the theorems so that the children will have that as their foundation** for later work.

Just as in the other subject areas, we're not going to be presenting geometry just for its own sake, but rather principally **for the purpose of providing the children with a structure for intellectual development**. Geometry is a wonderful format for providing the children opportunities for logical thinking, deductive reasoning and in forming abstractions. It's for the development of the mind. You'll see that manipulative activity will continue as an aid to the child's intellectual development. Geometry also provides a wonderful scope of variety in follow-up work.

The work we do in geometry is going to center around 6 topics. The **albums will not organize into these topics**. They are:

congruency, similarity and equivalence

polygons

angles

lines

area of plane figures and

solid geometry

You will see once again that here in geometry that many of these topics will be worked with in a parallel fashion. A child who begins working with angles does not have to finish all the angle work before beginning the work with lines. They can go back and forth. Also, as with the other subject areas, we'll **follow the child's interest, the prerequisite knowledge required and making sure we present what is required for the public school curriculum**.

November 4, 2004

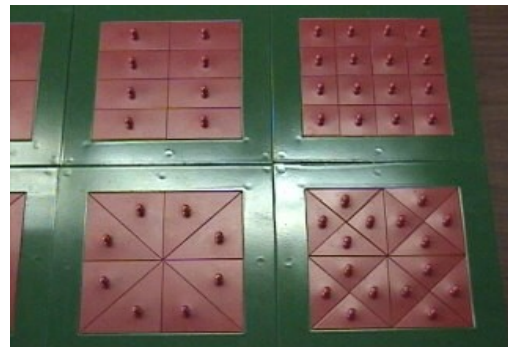
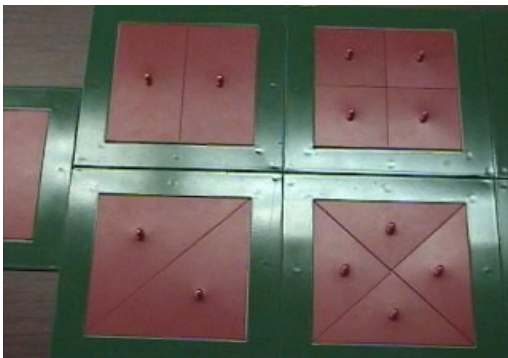
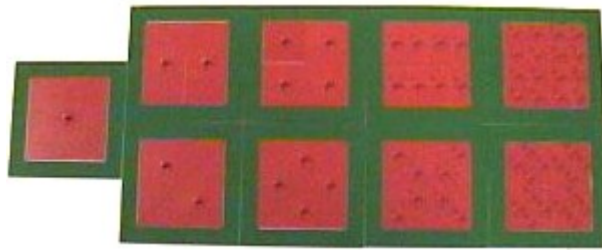
Congruency, Similarity and Equivalence & Metal Insets

Age

This can be introduced very early in the school year – this is for 6 year olds. Concepts might be reviewed by the 9-12 age group but this presentation is for lower elementary.

Material

The square set of the small metal insets arranged as follows:



Detail from left: whole square, halves, quarters split horizontally (top) and diagonally (bottom) followed by eighths and sixteenths horizontally (top) and diagonally (bottom).

Prologue

This is generally where we start with all of the children. From this topic, there are some different directions in which we could go for variety. We use circles for fractions. This material is used for designing.

The material is set out in the formation displayed. The solid unit (square) is set out on the far left, spanning both rows – not aligned with either top or bottom row. This is the original square from which others are abstracted. The material has green

frames and a white base.

Going across the top row from left, we have the same square divided into 2 equal rectangles. The next one is divided into 4 equal squares. The next is divided into 8 equal rectangles, $1/8^{\text{th}}$ the original. The last in the top row to the far instructor-right is divided into 16 equal squares – so each is $1/16^{\text{th}}$ of the original square.

The second row, from left, is: the first is split into 2 equal right isosceles triangles – each $1/2$ the original. The next is divided into 4 right isosceles triangles, each being $1/4^{\text{th}}$ of the original square. The next is divided into 8 equal right isosceles triangles – each $1/8^{\text{th}}$ the original. Lastly, we have the original divided into 16 right isosceles triangles – each is $1/16^{\text{th}}$ of the original.

There is also a triangle set of iron material used here. It includes the following: the first is an equilateral triangle; the second is divided along the height into two equal right-angle scalene triangles; the third is divided by angled bisectors (divide the angle in half) into 3 equal obtuse-angle isosceles triangles; the fourth is divided by lines that connect the midpoints of the sides (one in the center) into 4 equilateral triangles.

This is a small group (2-3) presentation. Sit down to present.

Begin

Congruency

(put two $1/2$ rectangles back to back, feel the edges of the rectangles and then separate them again as the children watch)



These two rectangles are exactly the same. One fits exactly over the other. We say that they are **congruent**. They are exactly the same.

(now choose and compare $1/2$ triangular pieces)



These two exactly fit together also. These 2 pieces are congruent. They are exactly the same. There is no part that is sticking out when we place them back to back, therefore we say they are congruent

(give each child in the group a chance to find 2 pieces that are congruent)

How do we know if they are congruent? We put them back to back.

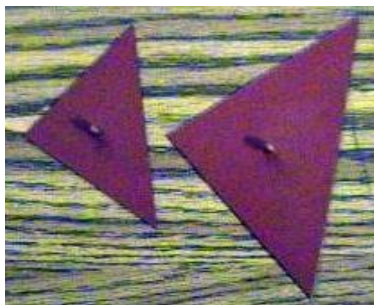
(we could allow the children to trace the pieces and manipulate the pieces, color, cut-out, mount on paper – and break off the lesson at this point. We could also continue...)



Similarity

Let's take these two.

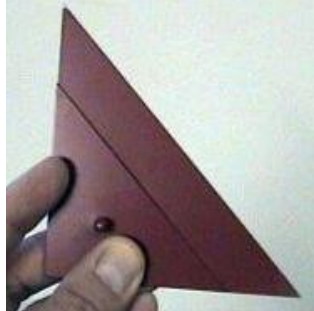
(pick 2 different but similar triangles – $\frac{1}{4}$ and $\frac{1}{2}$)



I wonder if these two are congruent. I'll place them back to back.

(line up a vertex of each)

We can see there is all this part of the triangle sticking out. So these two pieces are not congruent.



They have the **same shape but are different sizes**. We say they are **similar**.

(pick up two differently sized rectangles)

Let's try these two. Are they congruent? No. when we put them back to back, one of the squares sticks out from the other's edge.

But we can say that these two are similar. These two squares are similar. They are the same shape but different sizes.

(invite each child to find 2 pieces that are similar. Now we can mix up the 2)

Choose one and set it down. Ask a child "Can you find one that is similar"? Why is it similar"?

Now can you find one that is congruent? Why?

(again, you might suggest they mount pairs that are similar, make a booklet or chart, tracing the figures and so on)



Equivalence

(select $\frac{1}{2}$ rectangle and $\frac{1}{2}$ triangle pieces - from top and bottom rows, first one at left, one piece taken from each frame)

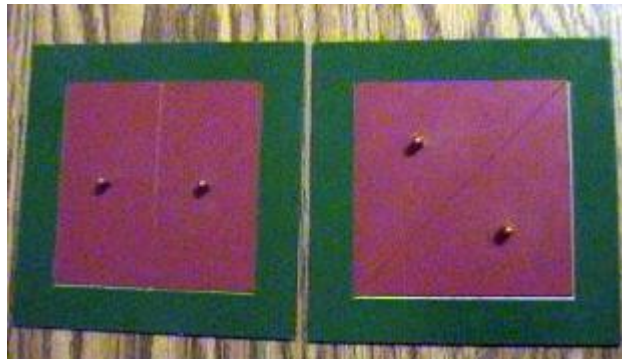


Are these two similar? Why not? Not the same shape, are they?
 Are they congruent? No – there are these pieces sticking out.



(swap the triangular and quadrilateral pieces into each other's frames.)





We can say these two pieces are **equivalent**. They have the same value. **They are each $\frac{1}{2}$ of the same whole.**

(repeat swapping sequence in next two frames. Each time setting one piece of each fraction set (triangular and rectangular) on the table; at end, take out the original two pieces and lay them on the table, e.g., $\frac{1}{4}$ th quadrilateral and $\frac{1}{4}$ th triangle)

We can say these two pieces are equivalent. **They have the same value. They are each $\frac{1}{4}$ th of the same whole.**

(invite child to repeat procedure with the next two frames to the right)

Can you find two pieces that are equivalent? Where do we have two that are equivalent?

(child chooses)

How do we know these two are equivalent? You've been counting the pieces. This is $\frac{1}{8}$ th of the original and this is $\frac{1}{8}$ th of the original. Right?

(we could also say that both squares have been divided into 8 equal pieces, therefore each will have the same value)



Can you find two pieces that are congruent?

(two pieces from the same frame that, placed back to back, fit exactly)

Can you find two that are similar?

(two pieces that are the same shape but of different sizes)

Can you find anything here that is equivalent to this one?

(choose one and place it on the table – it is the same number of pieces within the two frames – must choose $\frac{1}{2}$ and $\frac{2}{4}$ to find equivalence with the triangle set.)

End

Epilogue

Make sure you stay within the confines of the material. Don't look around for examples in the environment yet; stick to the material at this point. Through the 2nd period repetition, for similarity there is usually more than one possibility. Whereas for congruency, they will only find it in the same frame the original piece has come from.

The work we do with area is based on equivalence. The more experience you can give the child with equivalence, the more follow-up work in this, the better since it is used down the road.

If the child has done fractions in the primary class or if he's been introduced to fractions in the elementary class prior to this (often this is the case), then we can make use of fractions here. Otherwise, modify your language to avoid fractions.

Somewhere along the way, bring out the triangles and show them to the children. All three concepts can be looked at with this set of triangle plates as well, though there is not as much you can do with them as with the square plates.

Dr. Montessori wanted to show that the unit can be different shapes; it doesn't have to be a square. It can be a circle or triangle. The iron triangle set reminds one of the constructive triangles. These are called iron plates versus constructive triangles.

Now that we've presented all three concepts provide as much 2nd period practice as they need. Can you find similar and how do you know. Who can find two that are congruent and how do you know. Same language is used with equivalence.

The children will often do a 1-1 correspondence but will quickly discover they don't have to stop at just one piece. If I take a rectangle out and ask for something that is equivalent to it, we can array 3 pieces forming $\frac{1}{2}$ frame as our counterpart. If they've had fractions, we can verbalize that $\frac{1}{2}$ is equivalent to $\frac{1}{4}^{\text{th}}$ plus $\frac{1}{4}^{\text{th}}$. Maybe they'd like to mount them, showing they are equivalent to one another. Including fractional values is needed if they've had fractions already. It provides practice for them in both domains.

We also don't have to stick with just geometric shapes. We can build non-

geometric shapes that are equivalent. For example, we can combine triangles to form a flag (or an odd shape) but still find a combination of counterpart inset pieces that will form an equivalence. (one might form a flag and the second forms stairs). This generally will intrigue the children if they've not yet shown enthusiasm for this work.

Equivalence leads to much other work like Pythagoras, Euclid, area, volume – so the more experience the children get with equivalence, the better; they'll have a solid foundation for that later work. So encourage as much follow-up manipulation as you can.

November 4, 2004

Congruency, Similarity and Equivalence & Constructive Triangles

Triangle Box

Prologue

We can look at the same 3 concepts with the constructive triangles. We'll start with the triangle box, taking all the pieces out and spreading them on the table at random.

Begin

Do we have any triangles here that are congruent? Shelly, which ones? The two green triangles – yes, they are congruent because they are exactly the same.

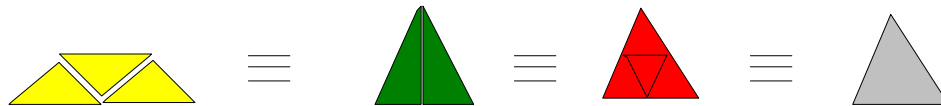
Do we have any pieces here that are similar? Carrie, can you see some that are similar? Yes, the large gray and smaller red triangle are similar. They have the same shape but they have different sizes.

Now let's consider equivalence. Let's see what we can do here with equivalence.

(superimpose large gray onto two large green, the gray over the 3 yellow, the gray over the 4 red)

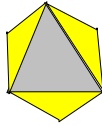
They are the same, right?

Therefore, can we say this **trapezoid (form with 3 yellow) is equivalent to this triangle (green)?** Yes, because they are **both equal to the gray triangle** – so they have the same value.



(form figure - large gray in center with 3 yellow surrounding to form a hexagon. Fold 3

yellow triangles upward at obtuse vertices – toward center – to form 3-dimensional figure)



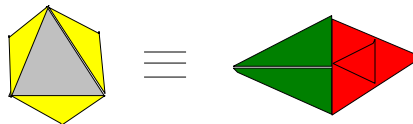
(remove gray from center)

I can also say this yellow triangle outline is the same as this gray.

(superimpose the 4 red onto the gray – see first illustration)

We can say that these four have the same value as the gray.

(form 1 gray and 3 yellow as one hexagon and the 2 green and 4 red as one rhombus)



Therefore I can say this hexagon is equivalent to this rhombus.

End

Epilogue

Just show the children enough to get them started. We'll go through and explore each box, one at a time. After that we could combine the boxes. And all of the boxes could also be combined and explored together.

Return in order from bottom:

1. gray equilateral
2. two green right
3. 3 yellow obtuse isosceles
4. 4 red equilateral – at top

Small Hexagonal Box

Prologue

If constituent pieces are equivalent and you have equal number of parts in each constructed figure, you have equivalent figures – no matter the shapes created. Again take all pieces out and set them at random on the table.

Begin

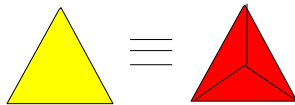
Who can find two triangles that are congruent? (chooses 2 gray equilateral)

Yes, we have several pairs that are congruent, don't we.

Who can find two that are similar? How do we know they are similar? Yes, the shape is exactly the same but the size is different.

What can we do with equivalence?

(superimpose large yellow equilateral onto 3 red isosceles obtuse)



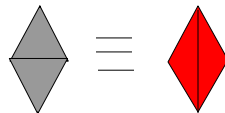
Well, 3 of these exactly fit on this one.

(make **something new and odd** with the red pieces)

So we can say that this (point to red) is equivalent to this (point to yellow)

Anyone else have an idea for finding an equivalence?

(two gray equilateral and two red isosceles forming two rhombi)



Since we formed two rhombi with two pieces each, we can also say each half triangles have the same value.

This means that **anything we build out of an equal number of equivalent parts – is also equivalent.**

End

Epilogue

Return in order from bottom:

1. red obtuse and yellow equilateral
2. hexagon of gray equilateral
3. 2 red equilateral and 3 green equilateral
4. 3 red obtuse isosceles - at top

Large Hexagonal Box

Prologue

Begin the same way as previous, layout pieces on table. This collection includes 1 large yellow equilateral and 10 obtuse isosceles of the same size but different colors.

Begin

Do we have any that are congruent? Sure these two (yellow or gray pairs).

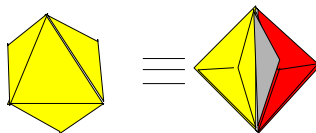
We have lots that are congruent here.

Who can find two that are similar? No! We have many congruent but none similar.

Who can find equivalent pieces?

(superimpose yellow equilateral onto a set of obtuse isosceles)

(constructs non-geometric shapes after showing constituent pieces are equivalent)



Now, maybe you'd like to make more equivalent shapes of your own.

End

Epilogue

Return in order from bottom:

1. 1 large yellow equilateral
2. 6 yellow obtuse isosceles
3. 2 red obtuse isosceles
4. 2 gray obtuse isosceles

Combined Boxes: Small Hexagonal and Triangular Box

Prologue

Begin the same way as previous, layout all pieces on table.

Begin

Who can find two pieces that are congruent? (she chooses yellow and gray equilateral)

Nope, the sizes are different aren't they? But they are similar.

Want to try again (chooses red and gray equilateral).

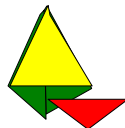
Yes, these are congruent and we have lots of them here.

What is the difference between the first two she chose? How can we find out?



We know this gray is equivalent to the two green right triangles. Watch.

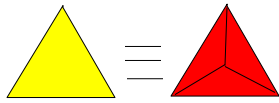
(turn one of the green and invert it too – **put hypotenuses together to form a kite**)



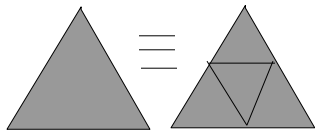
The yellow is smaller than the two green. But this red is enough to cover what remains.

Let's see what we can find out using what we know about equivalence. First of all we can

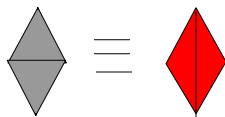
see that our 3 red isosceles triangles are equivalent to our yellow equilateral triangle.



Next we can see that we can use our smaller gray equilateral triangles to superimpose against our large gray equilateral triangle. So they are also equivalent.



Lastly, we can see that if we form rhombi from the red isosceles triangles and gray equilateral triangles, we get two figures that are equivalent also. Therefore, the red isosceles and gray equilateral triangles must also be equivalent.



(This rhombi equivalence is the basis for knowing the $\frac{3}{4}$ ratio)

How many of these small gray triangles make up the large gray triangle? **Four.**

How many of the small red triangles are used to make up the larger yellow triangle?
Three

Anything built from an equal number of equivalent parts is also equivalent.

We know the small gray and red triangles are equivalent, so we can say the **gray triangle is $\frac{4}{3}$ of the yellow triangle**; the **yellow triangle is $\frac{3}{4}$ of the gray triangle**.

End

Epilogue

Remember that constructive triangles are coded on the back. Gray dots go in the triangle box, green dots go in the hexagonal box. See coding when you put them back. Try not to mix up the two sets either – since there are two complete sets in this classroom.

Small Rectangular Box

Prologue

Begin the same way as previous, layout pieces on table.

Begin

In this box we have 12 right scalene triangles.

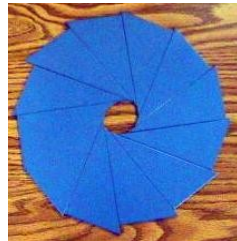
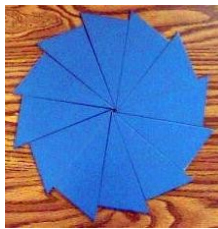
By how these are stacked in the box, we can see that any two figures we build out of an equal number of pieces in each is going to be equivalent.

Suppose we start with 4 and build something like that. (build virtually anything)

With another 4 I build something like this.

What can we say about these two shapes? They are equivalent. They have equal value.

Suppose I use all twelve. (any even number can be used to make these shapes)



End

November 5, 2004

Introduction to Polygons

Material

The box of geometry sticks. The sticks are ordered by length color-coded as follows: 1 – brown, 2 – purple, 3 – orange, 4 – red, 5 – black, 6 – tan, 7 – green, 8 – pink, 9 – blue, 10 – yellow; plus non-painted sticks with various colors. Measuring angle, plumb and other various materials.

Prologue

Once we introduced congruency, similarity and equivalence you could introduce polygons as one choice. There could be other directions as well such as angles. It is easy to take up polygons since we've just worked with constructive triangles.

Remember that the children should have had many of the names in primary. If they've had them, make use of that knowledge here. If there are names they don't know, you can bring them in over time.

This is a small (3-4) group presentation. This is the basic presentation for polygons. We've presented the concept of triangles, quadrilaterals and so on. There is one more area to clarify (see the follow-on at end).

Begin

(form a polygon with the sticks)

This is a polygon

(undo one end)

This is not a polygon.

(reattach the end)

This is a polygon.

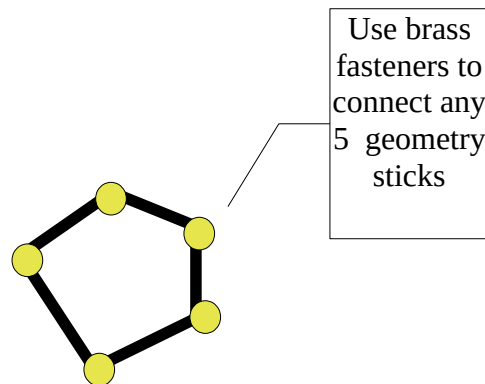
(detach again)

Now it is not a polygon.

(add 2 more segments – 7 total, attach ends)

This is a polygon.

(let the children form 3-sided, 4-sided and more polygons. Cover the table)



What is the minimum number of sides we needed to make a polygon (3)

What do we call a 3 sided polygon? (triangle)

Some of our polygons have 4 sides. We call that a **quadrilateral**. For now, we'll call ones with more than 4 sides a **polygon**.

Look at this! (hold up a triangle)

No matter how I hold it, it keeps its shape. Let's try this with other polygons.
(she tries a triangle, it still holds its shape)

How about trying this one?
(she chooses a non-triangle shape and it collapses as she holds it up.)

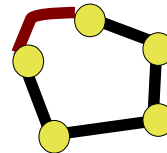
What do we see in this? **Only the triangles keep their shape**. They are the stable figure.
Do you suppose this is why the triangle is the constructor?



There's just a bit more I'd like to show you about polygons.

(build a figure with a curved side)
This is enclosed space.

(undo one fastener and separate)
This is not enclosed space.



(reattach)
This is **enclosed space**. But it is not a polygon. What's different about a polygon?

Ah! Yes, **polygons do not have curved sides**.

(construct a polygon)
This is also enclosed space – this time the enclosed space makes a polygon.

(detach one connection)
This is not a polygon.

(reattach)
This is a polygon.

End

Epilogue

In the box of geometry sticks, we have sticks that go from 2 cm. To 20 cm. The definite length of the sticks is from end-hole to end-hole – not from the tips of the sticks but from the holes. The sticks increase in length by 2 cm. Each starting at 2 cm. Some sticks have holes only at the ends while others have holes every 2 cm. In addition to the 10 compartments of colored sticks, at the bottom are some uncolored sticks; these are used to form the hypotenuse of isosceles right-angled triangles.

We have thumb-tacks, one compartment of thin nails, some carpet tacks (similar to thumb tacks but with longer stem), some brass fasteners, a triangular shaped “measuring-angle”, four curved lines and a plumb line. Small figures of children. These are the contents of the box of geometry sticks.

This material enables them to construct geometric figures long before they are able to, with a pencil, compass, ruler, protractor, etc. So, in many ways, it's like the moveable alphabet in primary which allows the children to write before their hand is ready to do so on paper. Using this material, the children are constructing their own figures. They become more aware of the characteristics of those plane figures.

There's an element of trial and error involved with this material; not every stick will work for the particular thing we might be trying to make. The child is prompted to find out what will work. In this way, this material is perfect for the reasoning mind.

After the presentation, the children can explore polygons with the box of sticks, especially making those with many many sides. With the 9-12 children, then we can ask them to find its name. They can draw polygons and label them and explore the geometry cabinet. They can classify the polygons. They could find all of the triangles, all quadrilaterals and so on. They can attach names to all the polygons. They can explore the environment, inside and out, for polygons.

In the follow-on, we are just refining our idea a bit showing enclosed space and the fact that polygons do not have curved sides.

Extensions

- With Straws, Blunt needle, yarn. You can string together your own polygon using straws. You can size the sides you want, cut the sides and take them home.
- Rack D is the nomenclature material for polygons. It's constructed just like nomenclature materials in other domains. Let the children write their own definitions for various figures: polygon, triangle, etc. Then ask them to are their definition to the one in the booklet.

November 5, 2004

Introduction to Angles

Material

Box of geometric sticks, a planar board (like a cork board without sides), tacks to hold paper onto the board, large pieces of paper (of newsprint) and pencil.

Prologue

This is the basic presentation on angles. Tack a sheet of paper down into the plane and we choose two sticks.

Begin

Let's choose two sticks; one longer and one shorter. The shorter is set on top of the longer one.

Now I need to tack down one end of both of them together and then tack down the end of the longer stick (**both sticks are laid in the same direction horizontally**).



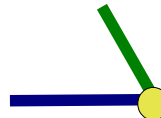
(separate sticks to make an angle)

This is an angle (point to the narrow gap between the sticks in the angled corner)



(move the short stick further away from the longer stick with a pencil)

This is an angle (point)



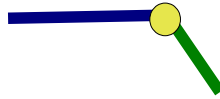
(move short stick again to 180 degree mark)

This is a **straight angle**.



(move stick further around to make an angle of about 200-250 degrees)

This is an angle (point to the narrower gap between the sticks in the angled corner)



(move short stick all the way around by this point to the original location atop the longer)

This is a **complete angle**. It is also called a **whole angle**.

(now repeat with variations)



(move to 90 degrees and take the measuring-angle from the box and lay it into the angle)

This is a **right angle**.



(move to 145 degrees)

Is this a right angle? No, it's bigger than a right angle. This is called an **obtuse angle**.



(move to 45 degrees and take the measuring-angle from the box and lay it into the angle)

Is this a right angle. We better check (set triangle in corner)



No, this is an **acute angle**. It's smaller than a right angle.

(use some 2nd period repetition here)

End

Epilogue

It does not matter where we stop or how many times we stop; the only exception is the positions for complete angle and straight angle.

After introducing the various angle types, let them form the different kinds, making sure they use the measuring-angle from the geometric box to prove whether an angle is obtuse, acute or right angle.

After they've had plenty of 2nd period repetition, do the 3rd period repetition. The children can search the environment for angles. Furniture, doorways, playground equipment and so on can be material for classifying angles.

The geometry cabinet can again be used for exploration, sorting out figures according to what kind of angles they have. The children can draw angles and label them.

If they confuse a 270 degree angle with a 45 degree angle, you can introduce the term **reflex angle**. An obtuse, by definition, is greater than a right angle and less than a straight angle. So this all fits together and we can clarify if needs be. Reflex angle is greater than a straight angle and less than a complete angle.

The children now should explore the triangles in the geometry cabinet to classify each of the triangles. This is an application of the angle work here. We can also make triangles with our box of geometry sticks, according to the kinds of angles and according to the kinds of sides of the triangles.

November 5, 2004

Types of Triangles by Angles

Material

Geometric box of sticks.

Begin

(construct a triangle with the geometric box – using red, green and blue sticks)

Let's look at the angles in this triangle. What kind is this? (acute)

How about here (also acute)

And what about here? (also acute)

This is an **acute triangle**.

(2 black and 1 uncolored stick – only one fits with each color – create new triangle)

I made another triangle. Let's look at the angles. What kind is this one? (acute)

What kind of angle do we have here (point) (an acute angle – gradually open the angle)

What kind is this angle? (right angle)

We better check. (use measuring-angle from box) Yes, this is a right angle.

We call this a **right-angle triangle**.

(choose 3 more sticks – yellow green and red – create a new triangle)

Here, I made another triangle. Let's see what angles we have here.

How about here? (acute)

How about this one? (obtuse) We better check. (use the measuring-angle from the box)

This is an **obtuse-angle triangle**.

Maybe now you'd like to make your own triangles with different types of angles.

End

Epilogue

We identify each angle of each triangle. This is the first time they've had this material, so don't rush through.

November 5, 2004

Types of Triangles by Sides

Material

Geometric box of sticks.

Begin

(construct a triangle with the geometric box – using pink, green and blue sticks)

Today we're going to look at the sides of our triangles.

What do you notice about the sides of this triangle (they're all different)

Yes, each is a different color and a different length.

This is called a **scalene triangle**. Scalene comes from the **Greek word scalenus**, which means **uneven**. In this case, the sides of my triangle are uneven.

Let's make another triangle. This time I'll use two red sticks and an orange stick.

Look at the sides of this triangle. What do you notice?

Yes, two sides are equal – they're the same aren't they.

This is called an **isosceles triangle**. **Isosceles** comes from the **Greek word isosceles** – meaning **same legs**.

Let's make one more triangle. This time I'll use three green sticks.

Look at the sides of this triangle. What do you notice?

Yes, all three sides are the same.

This is called an **equilateral triangle**. Equilateral comes from two **Latin words**, **aequums and latus** – together they mean **same sides**.

Now you can make your own triangles – scalene, isosceles and equilateral.



Now we're ready to make triangles according to sides and angles.

Judy, I would like you to make an obtuse-angle isosceles triangle.

Barb, I would like you to make an equilateral triangle.

Judy what did you make – yes, an obtuse isosceles triangle. How do you know it's an obtuse isosceles triangle. (because two sides are the same length and one angle is greater than a right angle). Maybe you'd better check your angle with the measuring-angle.

(repeat for each child – what did you make, how do you know, check it to be sure)

End

Epilogue

Same exploration apply here. Booklets and so forth can be compiled. They can explore the geometry cabinet or constructive triangles, the environment inside and outside the classroom.

Follow-on with creating triangles introduce trial and error with the geometric sticks. This is more repetition and refinement of concept. How many possibilities are there with angles. How about with sides. How many different kinds of triangles can you make? These are questions to get them thinking and exploring.

Natural colored sticks are for the isosceles right triangle. All the types are isosceles right, isosceles acute, isosceles obtuse, scalene right, scalene acute, scalene obtuse and equilateral triangles.

November 15, 2004

Angles

Parts of an Angle

Material

Geometric box of sticks. Work board.

Prologue

This lesson prepares the child for some later work.

Begin

Today, we're going to look a bit more into angles.

On my board, I've made an angle. Each part has its own name.

These lines formed by my sticks are called the **sides**. They are the sides of the angle.

This spot here where the two sides come together is called the **vertex**.

And this space inside the vertex is called the **amplitude**. **It's the size** of the angle. I can make the amplitude greater like this:

(spread sides to make angle larger)

Or I can do this to make the amplitude less (move sticks sides closer to make the angle smaller)

(do 2nd period repetition and then 3rd period – on parts of an angle)

November 15, 2004

Measuring an Angle

Material

A Montessori protractor. Circle fraction iron inset material. Any of the other iron material from the cabinet of iron material. Paper and pencil.

Prologue

Be sure to point out that the vertex of the inset must line up with the end of the white line on the Montessori protractor. This lesson starts with a story.

Begin

The people of old Babylon were living about 4000 years ago in a place called Mesopotamia – which is a place between two rivers.

Today, it's a country called Iraq. But its the same place.

These Babylonians were great star-gazers or astronomers. Just think of this very early time, 4000 years ago, when so much of what we know today wasn't known.

These Babylonians were clever people and they made some important discoveries. They noticed that every 360 days, the sky looked the same at night.

Remember they didn't have streetlights like some people do. There was no smog to cover the sky from us. They had no tall buildings that block the sky either. So they could clearly see the stars in the sky at night.

They noticed that every 360 days, the same constellations were in the same places in the sky. So they thought the stars went around the earth in a great circle and that it took them 360 days to make a complete trip. Today we know it takes a little longer.

Today, we know it takes closer to 365 days – but those Babylonians were very close – to only be off by 5 days. They used that number 360 because it was easily divisible by 60, which was the base of their number system.

So they took that number, 360, and divided a circle into 360 equal pieces as you can see here on our Montessori protractor

(hold up Montessori protractor)



Montessori protractor

Each one of these equal pieces is called a degree. If you look here at the top, we start with zero and we can count around, we pass 340, which is the last number printed on our circle, and when we arrive back to the top at 360 – which is the same position as we started – at zero.

Using this Montessori protractor and some of our iron material, we can measure some angles.

Suppose I take this piece from my circle frame (choose 1/3 circle piece and hold it up to see). I want to know how many degrees are in this angle.

Looking here, coming down from the zero is a white line. We use it to line the piece along the line. We put the piece from the frame onto the protractor, lining up the piece along the white line.

We look to the other side of our angle, along our circle, and we see the other side aligns with the mark on the circle at 120 degrees. The vertex of the angle meets the end of the little white line.

(Give each child a chance to measure an angle and have understood how to use the

Montessori protractor.)

Now I'd like to show you how to write down how we measure an angle. (hold up the 1/4th iron angle circle)

Let's put it into our Montessori protractor, line it up along the white line and now we can see our answer over here. How many degrees does this angle measures? Yes, 90 degrees.

So on our paper, we can write $\frac{1}{4} = 90^\circ$. This little circle is how we write what we call degrees. Now you can measure any angle you'd like to measure and you can write down any of the number of degrees you find.

Epilogue

We can use any of the iron material. These pieces do not fit the same way as the circle iron figures will but the other shapes can be laid on the white line as before – even though the pieces don't fit the same way. Show the children how to lay the piece onto the Montessori protractor. Show the child how to extend the side of the triangle to meet the edge of the protractor, using a straight edge (ruler), if the figure being measured is too small to reach the edge.

Once you get into measuring other shapes, you follow-up tracing the triangle, measuring the degrees of the angles and writing that measurement at the point of the amplitude on paper. This is a source of handwork for them.

We want the children to have enough experience with the Montessori protractor and our materials before going on to using a regular protractor. Extend their experience as much as possible before going on.

November 15, 2004

Addition and Subtraction with Angles

Material

A Montessori protractor. Circle fraction iron inset material. Any of the other iron material from the cabinet of iron material can be used for the extended work. Paper and pencil.



Begin

So far we've worked with the Montessori protractor, one piece at a time. Today, we'll start working with more than one piece at a time.

I'm going to start with this piece ($\frac{1}{6}^{\text{th}}$). First, I want to know how many degrees are in this angle. So, let's measure it. Let's align our side with the white line and put the vertex at the end of the line. We can see this angle equals 60 degrees.



Now I want to know how many degrees are in this angle ($\frac{1}{4}^{\text{th}}$) We see this angle is equal to 90 degrees.



Now I want to know how much these two angles equal together. So let's put them both in the protractor together. Let's put the first piece in.

(place first one in, index finger of left hand anchors it and holds it steady.)



Now we've added the second piece. To see how much they equal together, we look at where the side meets the edge and we see that together they equal 150 degrees.

(verbalize how much each equals and then in total).

(give each child a chance to do this on their own. After 3-4, show how to write it.)

So we write this: $60^\circ + 90^\circ = 150^\circ$ Now as you do more problems, you can write them down like this. Now I'll show you how to do subtraction and then you can work independently on your own.

Subtraction

We're going to start with this angle ($1/3^{\text{rd}}$) and I'm going to take this angle ($1/4^{\text{th}}$) away from it. First we need to start with our large angle and measure it so we know how many degrees it has. How many degrees does it equal? Yes, 120 degrees. Let's remember that.



Now we measure the second angle, the one we're taking away. How many degrees does this one equal? Yes, 90 degrees.



Now, let's place the side at the 120 degree mark, the measured angle of our first angle. Then we look to the top side of the angle to find our answer. It aligns with 30 degrees.



Now, we write this: $120^\circ - 90^\circ = 30^\circ$

Epilogue

Be sure to show the children to begin with the larger angle first in subtraction, to avoid negatively measured angles.

November 15, 2004

Regular Protractor

Material

A Montessori protractor. Circle protractor and the half-circle protractor.
Ruler, compass and paper and pencil.

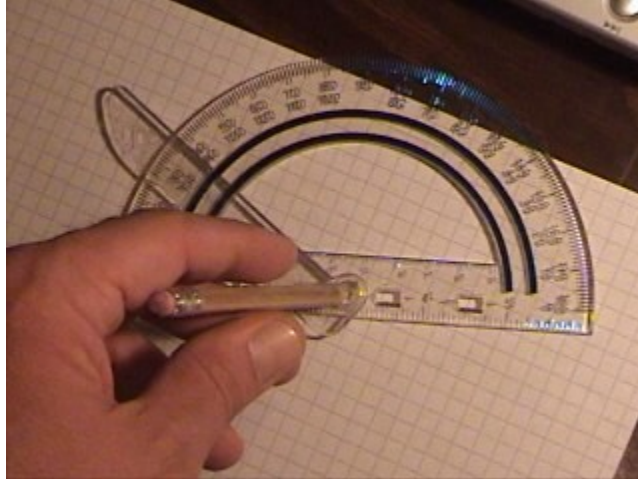
Begin

This Montessori protractor is very special. Dr. Montessori developed it just for us to use in Montessori classrooms. Other people don't have a Montessori protractor. I'll show what they use.

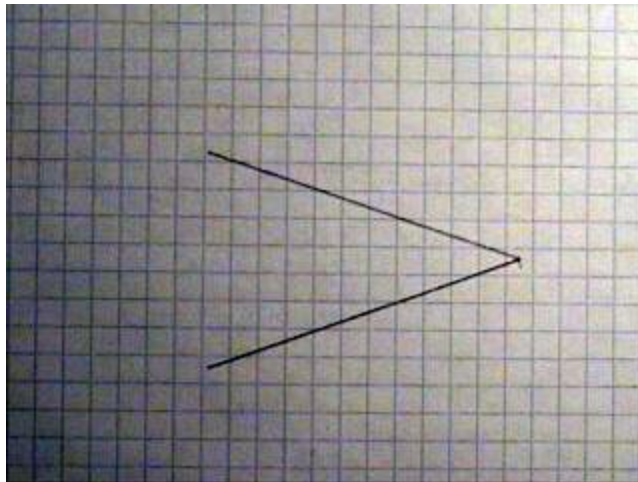
(start with **full circle protractor** if you have one. show numbering on the outside of the protractor; the zero on both sides, two number sets – one for the outer circle and one for the inner circle. They go in opposite directions to allow one to read the degrees in any orientation. Also note, the black line that comes down to the straight edge – is where we line up the vertex of the angle – at the base of the black line. One side of the angle will be along the straight edge that connects across the middle of the circle.)

(**half-circle protractor** again has inner and outer numbers also – to 180 degrees. Show children this again for this protractor.

The straight edge on the bottom, there is still a black line but also a hole. The vertex of the angle goes at this point on the hole. One side of the angle goes along the black line.)



Let's draw an angle. (take paper, ruler, pencil) We're drawing any angle now. I wonder how many degrees are in my angle. First, let's use our circle protractor to measure it.



Angle made with a ruler, pencil and paper.

Vertex lines up at the base of the black line. One side of the angle lines up here through the zero point. (look at which way your numbers are going). So, we can see this angle is 33 degrees. This is how we use the circle protractor to measure this angle.



(measure same angle with the half-circle protractor to demonstrate its use)

Now you can draw any angle you want and measure it with any protractor you want to use. We can also draw an angle with a fixed amplitude. Pick a number, like 220. Let's draw an angle of 220 degrees.

(start by drawing your first side.)

I'll need my circle protractor in order to do this. The end of my line is at the vertex of my angle. I must align the straight edge, through the zero. I start at zero and I look along the outside of the circle, making a small mark with my pencil when I reach 220 degrees. Now using my straight edge, I'll connect that point with the vertex to complete the side. I'm going to label this angle in the amplitude. 220 degrees.

November 15, 2004

Bisecting an Angle

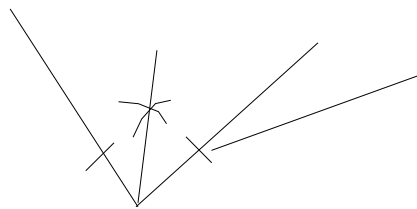
Material

A compass, ruler, paper and pencil.

Begin

Today, we'll use a compass. The point of the compass goes first onto the vertex of our angle (mark each side).

Then we'll place the point of the compass (same compass position) at each point on the sides we just created – creating a mark in the center of the angle.



Point of compass begins on the vertex to make marks on sides. Then point of compass is placed on the each of the marks on the sides to find the intersection of the angle.

The two side-based marks intersect in the middle of the angle. Use a straight edge to connect the vertex point with the point created by the intersection of the two side-based compass measurements.

Epilogue

Drawing angles of fixed amplitude and bisecting angles are the applications the children will continue to experience.

November 15, 2004

Lines

Lines Nomenclature

Material

A string, 2 pair of scissors, jar of colored water.

Prologue

We started with shapes (polygons and angles) since there is more follow-up possible. Here we come back to lines. It is for the most part nomenclature material. This lesson is with a small group of children. We can begin to incorporate the term **plane** into our presentations at this point.

Begin

Close your eyes for a minute. I want you to feel the top of your table. Now feel the edge of the table. Do they feel the same? That edge is a **line**.

Does a line end at a certain point? This is a line (hold up a string pulled taut.) This is a straight line (pull ends – hands - further apart).

This is a line (move ends further apart and rotate hands to provide impression that a line can go any direction. repeat several times alternating the orientation of the hands and the distance between them).

We can see a straight line goes on and on forever.

(ask for a helper and hold up the taut string)

Could you put a mark on our string. Now take our scissors and cut it at the mark. (done)

Now, my line goes from only one end. We call this a **ray**.

(ask for a second helper)

Now, each of you are going to make a spot on the line. Now take our two pairs of scissors. When I say go, cut at the spot you have marked. Go. (done)

Now my line cannot go further at either end. Now we have what is called a **line segment**. My line stops at both ends. These (hold up ends where the two cuts were made) are called the **endpoints**.

How many endpoints does my line segment have? Yes, two. How many endpoints does my ray have? Yes, there is one endpoint for a ray.

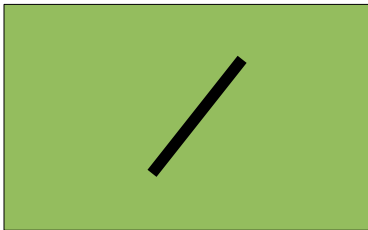
(hold up a plumb line - string with a weight)

This is a **vertical line**.

Have you ever been to the shores of Lake Michigan or the ocean, and looked at that point where the water meets the sky? Have you noticed it's like a line where they meet. Here I have a **jar of colored water**. You have to get down at eye level with the water to see that same sort of line here but you can see it. This is called a **horizontal line**.

(put a stick on the work board at an angle to the board's sides.

Is this line horizontal? No. Is this line vertical? No. This is called an **oblique line**.



(ask for helper)

I'm going to hold the board and you can help me make a **vertical line**. I'll hold the **plumb line and you place the stick aligned with it**. I can set my board down again and we can still say this line is vertically placed.

(ask for helper)

I'm going to make a horizontal line now. I'll hold my board up. We'll use our jar of water in front of the board to help us. Now, **line your stick up with that horizontal line you see at the top of the water**. Run your long stick behind the jar so you can line it up with the line at the top of the water.

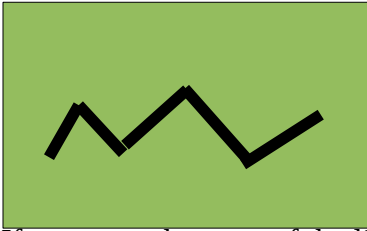
Now we have a horizontal line on our plane. If I set this down on my table, we can still say that this line is horizontal.

Epilogue

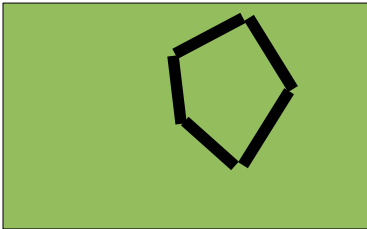
Each child needs repetition making lines. Use labels to label vertical and other kinds of lines.



There's one more thing I want to show you. Suppose we have something like this. This is called a **broken line**.



If we move the parts of the line to connect the ends, now we have a polygon.



(hold up taut string)

This is a **straight line**. This is a **curved line** (allow the string to droop in the middle). Straight line (hold taut). Curved line (drooping). Maybe you'd like to explore the geometry cabinet and group all of the shapes bounded by straight lines (polygons) and group together all those that are bounded by curved lines. (named individually, ellipse, etc.)

November 15, 2004

Position of Two Lines

Material

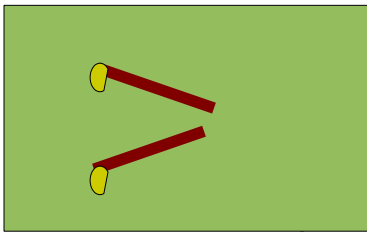
Three matching pairs of sticks, work board and tacks for the sticks.

Begin

(attach one stick to the board. **Use a shorter stick as a gap-measure at both ends of the two sticks.** Lay second stick parallel to the first, using the measurer stick to enforce parallel relation between them)

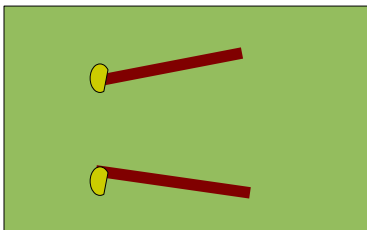
These are called **parallel lines**. (see illustration below but *make lines parallel*)

Now let's tack down another stick. (add second separate stick that runs convergent) Now I have two rays. They each have one endpoint (each attached to board). If I extend the lengths of the sticks, they will cross each other. We call these **convergent lines**.



(tack down endpoints of third pair of sticks)

Again we have two rays. If we extended these, they would go further and further away from each other. They are called **divergent lines**.



To help you remember these names, we have little children in our box. The children are

happy when they meet; we use these for convergent lines.



These two divergent lines go further and further away from each other – so we have sad faces on the children for these types of lines.



And here we have the parallel lines – always the same distance from each other. So, these children have never met and never will meet, so they have neutral faces.



November 15, 2004

Intersecting Two Lines

Material

Two matching **pairs** of sticks, work board and tacks for the sticks.

Prologue

If for any reason, you went to lines before angles, you need to do some work with angles before this presentation.

Begin

Today, we'll investigate two convergent lines. (tack one stick down)

If we have convergent lines and we extended them, what would happen? They would cross over each other and keep going. Maybe they would cross over like this.

(tack a second line over the first)

These are called **intersecting lines**. Let's use our measuring-angle and see if we have any right angles. (check each of the 4 angles made at the intersection).

No, there are no right angles – these are just intersecting lines.

Now, let's take two more lines. Let's tack the first one down. We'll add the other one so we have two convergent lines. If we extended them they would meet and cross over as before. But suppose they came together like this? (**use measuring-angle to form a right angle at the intersection**). Let's check to see what angles we made.

Here I have a right angle. And we find a second right angle here. And the last is also a right angle. We have four right angles in all.

When we have right angles at the intersection, we call these **perpendicular lines**. Maybe now you'd like to make some intersecting lines, drawing some perpendicular lines and some other intersecting lines – making a booklet of your collection perhaps. You can label the number of degrees in the angles made by your lines.

November 22, 2004

Relationships Using Angles and Lines

Relationship Between Angles

Material

Box of geometry sticks and the large work plane (board). Measuring angle.

Prerequisites

Presentations on names of main kinds of angles, how to measure angles and how to bisect angles. They must know the parts of an angle and how to add and subtract angles. After the children have the basic presentations on **angles and lines**, they can work on aspects of them parallel to each other. This presentation **needs both of those concepts – angles and lines**.

Prologue.

It's only when we were adding and subtracting angles that we worked with two angles at a time. Now we're going to consider more than one angle at a time.

We have more nomenclature that we'll introduce here. We introduce **adjacent** angles (share a side and a vertex), **linear pairs** (two that form a straight line with 2 sides, one from each triangle), **complementary** (two angles forming a 90 degree angle), **supplementary** (two angles equaling 180 degrees) and **vertically opposite**.

Difference between complementary and supplementary – any two angles of 90 degrees are complementary. Any 2 angles which equal 180 degrees are **supplementary – complementary and supplementary angles don't have to be adjacent angles. A linear pair must be adjacent.**



Begin

Adjacent and Linear Pairs

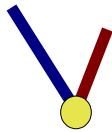
(make one acute angle with 2 sticks – pinned down)

I'm going to start by making this angle.



What have I made? Yes, an acute angle.

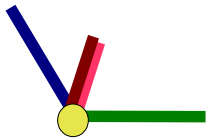
I'm going to make another angle. (add 2 more sticks – two in middle are the same color)



What have I made here? Yes, another acute angle. Now, I'll put my two angles together.

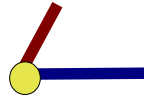
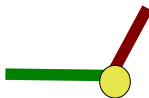
(take care to maintain the amplitude of the second angle as you move it)

What do they now share or have in common. (common side and vertex) Yes, they share a common side and a common vertex. (sticks are pinned, not tacked)



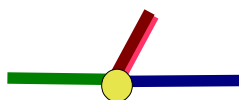
We call these **adjacent angles**.

What about these two angles? (create two more angles – one obtuse and one acute. colors of the two sticks in the middle don't matter - but they need to be the same color)



What have I made here? (acute and obtuse) Yes, an acute angle and an obtuse angle.

Now what do my angles have in common. They have a vertex and side in common. But these two other sides form a straight line - so from an obtuse and acute angle I have formed what we call a **straight angle**. Together, the two angles form a **linear pair**.



(invite children to form more adjacent angles and linear pairs as a **second period**.)

Complementary and Supplementary

(make an acute angle)

What have I made here? Yes, an acute angle.

(make another pair.) How about these? What are they? Yes, another acute angle.

I can put them together to form adjacent angles.

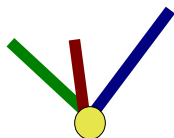
What do our **adjacent angles have in common** – yes, **a side and a vertex**.

I'm going to do this. (pull off one of the two sticks that created the one shared side)

Using our **measuring-angle**, let's check to see **what these angles equal all together**.

What do they make? They make a right angle. (measuring with angle)

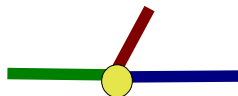
These are called **complementary angles** – two angles together which together measure 90 degrees.



Now, I'll form another angle.

(form acute angle) What kind is this? An acute angle. I'll add a second obtuse angle now.

(Add two sticks creating an obtuse angle) When we put them together like this, they're still adjacent. They have one side and one vertex in common. So we don't need this anymore (take off extra side). These two angles, together, make a straight angle.



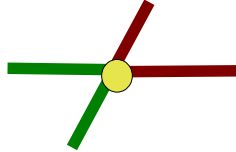
We call these angles **supplementary angles** Two angles which together make 180 degrees.

Vertically Opposite

There's one more thing to show you. (form another acute)

(add another acute angle pointing in the opposite direction – pin them down)

What do these two angles share? Yes, a vertex. Is there anything else you notice about them? Their sides form two straight continuous lines. These angles are called **vertically opposite** angles.



Epilogue

They might want to label the angles, draw them on paper. If they've had the lesson on using a regular protractor they can measure the sizes of the angles.

November 22, 2004

Relationship of Angles Formed by Parallel Lines and a Transversal

Materials

Geometry sticks, green plane board, large sheet of paper and some crayons.

Purpose

Not to memorize lots of new names – but to provide the child with sensorial exploration that will help in later studies in geometry.

Prologue

We'll look at angles formed when you have parallel lines and you cut them by a line going across them – which is called a transversal. This is a three-period lesson.

Alternate (interior or exterior) angles are two angles formed along a common transversal. Vertically opposite angles meet at their vertex.

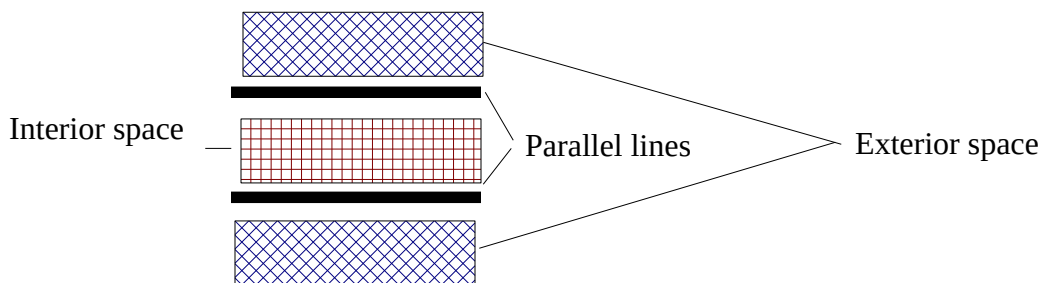
Begin

What do we know about parallel lines? They never meet. So the first thing I'll do is form two parallel lines. (use one short stick as a measurer at both ends of each parallel line)

Interior and Exterior Space

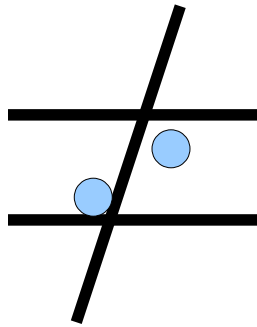
Now I have parallel lines on my paper. When we have parallel lines, we have what is called the **interior space** – we'll color that one color.

The space here and here (above and below two horizontal parallel lines) is called the **exterior space**. I'll color that with a different color.



(lay line across diagonally)

This is called a **transversal**. By putting on a transversal, we've made a lot of angles! We still have our interior space, can you point to it? We still have our exterior space; can you please point to that?



I'm going to use these *colored thumb tacks* to mark *pairs* of angles. Let's look at these angles.

Are they on the same side or opposite sides? Opposites.

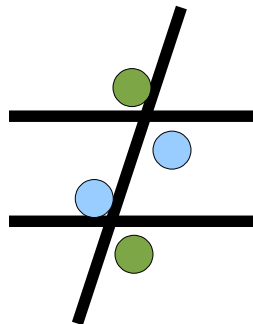
Are they in the interior or exterior space? Both in the interior space.

Are they on the same parallel lines or different parallel lines? Different parallel lines.

Alternate Interior and Alternate Exterior

These are called **alternate interior angles**. ()

Now I'll make two more angles. Are they on the same or opposite sides of the transversal. Opposite. Are they in the interior or exterior space? Exterior. Are they attached to the same parallel lines or different ones? Different ones. These are called **alternate exterior angles** ()



(form third and fourth angle pairs – opposite the first two)

Ask series of questions as above - opposite, interior and different lines These are alternate interior angles.

Ask series of questions as above - different, exterior, opposite sides of the transversal– another pair of alternate exterior angles

(we have now made two pairs of alternate interior and alternate exterior)

(do

Second Period

now, children form the various angles)

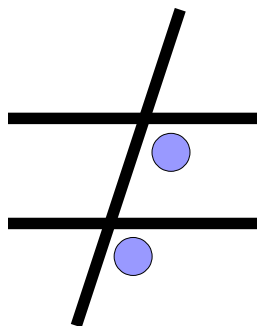


I'm going to mark a pair of angles. Let's look at these.

Are they on the same side or opposite of transversal? - same

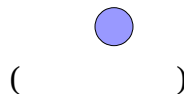
Are they in interior or exterior space? – one is interior and one is in the exterior space

Are they attached to the same parallel line or opposite? – opposite parallel lines.



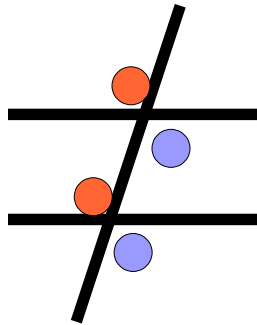
Corresponding Angles

These are called **corresponding angles**.



Let's make another pair. Same side of transversal, both interior and exterior space, attached to same or different parallel lines – different.

We have another pair of corresponding angles. ()



(add third set of colored thumb tacks)

are they on the same side or opposite of transversal - same

are they in interior or exterior space – one is in interior and one is in the exterior space

are they attached to the same parallel line or opposite – opposite parallel lines.

So we have another pair of corresponding angles.

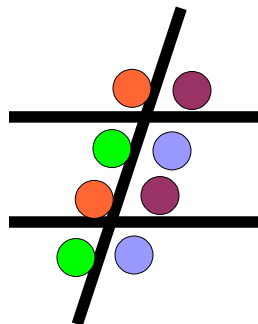
(add fourth set of colored thumb tacks)

are they on the same side or opposite of transversal - same

are they in interior or exterior space – one is in interior and one is in the exterior space

are they attached to the same parallel line or opposite – opposite parallel lines.

So we have a fourth pair of corresponding angles.



Second Period

Becky, can you mark a pair of alternate interior angles. Melissa can you form a pair of corresponding angles?

Epilogue

We can have more than one pin in one angle during the second period repetition, when we form multiple angle pairs. Keep only one pair of thumb tacks, make different angles and ask what are they for a **third period**.

November 22, 2004

Size Relationships of Angles Formed by a Transversal

Material

Piece of white paper, ruler, lead pencil, crayons and a scissors.

Prologue

We find, through sensorial exploration, that vertically opposite, corresponding, alternate interior and alternate exterior angles all will have the same.

Begin

On our piece of paper, we'll draw two parallel lines.

(draw one line, then measure equidistance from several points, then connect to form the 2nd parallel line)

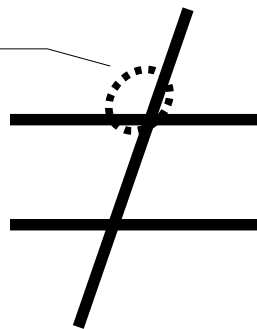
Now I have interior and exterior space. I'll color them different colors so we don't lose track of them.

Now we're going to add a transversal

I'll highlight the amplitude of the upper left hand angle.

Now I'm going to cut out that upper left hand angle. I'll cut two sides and rip the angle (third dotted side) out.

Cut sides along transversal and parallel line. Rip the dotted side along the amplitude



I wonder which other angles here are the same size as this one.

(work down left side and up right side of the transversal)

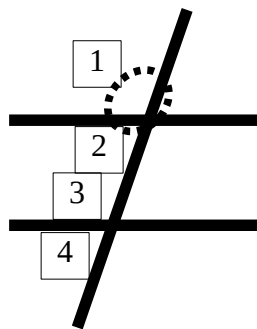
(angle is ripped out; move it around to each other angle to see which angles are of the same size as the one we've ripped out)

(rip out next angle – **work down left side of transversal**)

Is this angle the same size as my first ripped out angle? No. (continue to compare)

(rip out third angle; compare to the ripped out angles separated from the paper first, then proceed to compare with other angles on the page)

(rip out fourth angle, etc.)



investigate each angle, in the order above, comparing all to angle 1

So how many different sizes of angles do we have here? Only two! All these angles but only two different sizes.

Let's match up the one's we've cut out with the angles still on the paper on the right side of the transversal. (**match up cut-out angles with those angles on the right side**)

Epilogue

Through this sensorial exploration, they realize vertically opposite and corresponding angles are equal, as are alternate interior and alternate exterior. All of these angle pairs are equal.

Follow-up for this lessons repeats the lesson. They will want to try a

transversal that is very slanted and one that is perpendicular. They might want to use a regular protractor and measure the angles in degrees, writing them into the amplitude of the angles.

If doing this with young children, to simplify, draw vertically opposite angles and do the same kind of exploration. You'd have only 4 angles – half as many.

November 22, 2004

Pythagoras and Euclid

Pythagoras' Plate I

Prologue

Show triangles right and left are equal. Do same for all squares. Then switch parts between squares on legs and squares on the hypotenuse. Should understand squaring.

Material

Plate I with two sides each with similar right isosceles triangles, with squares built on each of the sides. Squares on right are in pieces and the squares at left are whole.

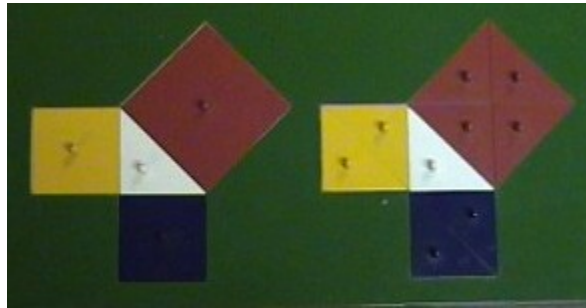


Plate I

Plate II has a right scalene triangle, proportion of the triangle is 3, 4 and 5. The squares are divided into 1 cm. Squares.

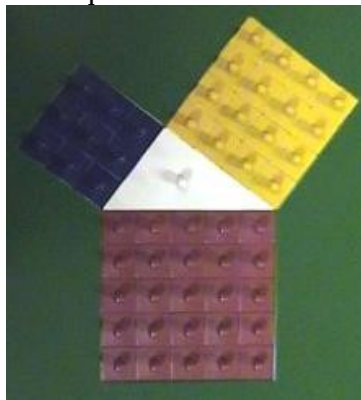


Plate II

One also needs a measuring-angle, and box of sticks. Be sure to return solids

to left side and pieces to the right side – as we found it.

Prologue

Pythagoras proved that the sum of the squares, built on the legs of the right triangle is equal to the square built on its hypotenuse. **This first plate will allow us to show the sensorial relation of the equivalence. The second plate allows us to show sensorially but also numerically, the same thing.** Prior to presentation of this plate, make sure the children know something about Pythagoras. These presentations should be done with a small (2-3) group so they can manipulate the materials.

Prerequisites

Make sure the children know the basic nomenclature – such as hypotenuse (appears opposite right angle) and legs (the other two sides). They **should have done work with equivalence and squaring.**

Age

About 8.

Begin

Pythagoras was a Greek born about 570 BC in Samos, Greece. When he was about 50, he went to live in Southern Italy. Most of the work for which Pythagoras is known was done when he lived in Naples.

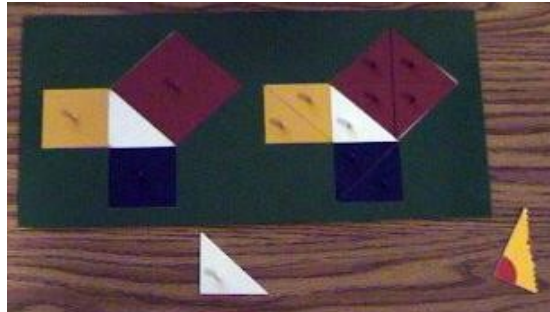
He formed a religious order in which he contemplated the divine order and believed that everything could be explained mathematically. He studied the harmony expressed in music. He and his followers were very interested in why the Egyptian rope worked.

So they did a lot of work with right triangles. Remember the Egyptians used the right triangle in the proportion of 3 to 4 to 5. This became known as the **Pythagorean triplet** because of all the work that Pythagoras did with these triangles.

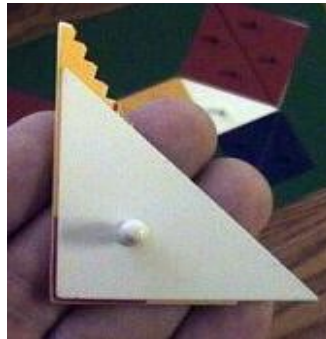
We could **explore the box of sticks and see what triangles we can make that are in that proportion – 3 to 4 to 5.** Certain ones will work but not all of them. (orange, red and black are the 3, 4 and 5 sticks)

Show Congruence of Corresponding Parts

Let's look at these pieces of iron material. We have a special triangle here (choose white right) The two legs are equal so we call it isosceles.



Let's measure to make sure it's a right triangle. Yes – it's a right angle.



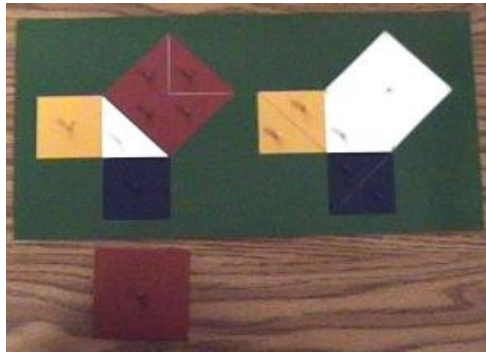
My two triangles look alike. Let's see if they aren't equivalent.

(put one piece into frame of the other)

(show congruency of right triangles)



Let's look at the square that is built on the hypotenuse. (Place one square in the other to see if they also fit as we did for the triangle – do so for all – 3 squares)



(finally, show congruence of the blue and yellow squares between the two plates also)
(we have now shown congruence of corresponding parts between the two like plates)

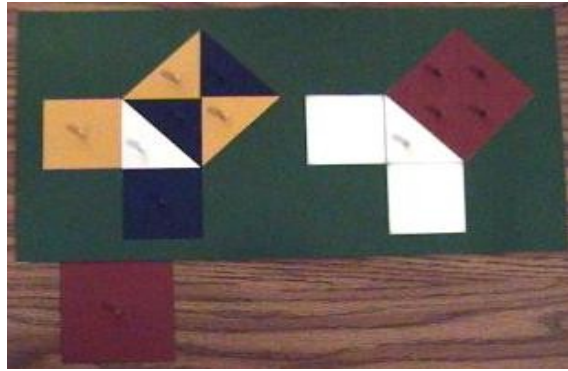
What Pythagoras found is that the sum of the squares built on the legs is equal to the square built on the hypotenuse.

I have a hard time showing that with these whole pieces, but we know these whole pieces are equivalent to these cut out pieces. Let's see how that helps me.

I can show the red square built on the hypotenuse is equivalent to the squares built on the legs.



And I can show the squares built on the legs together equal the square on the hypotenuse.



Because these pieces are equivalent, I can say the square built on the hypotenuse is equal to the sum of the squares on the legs.

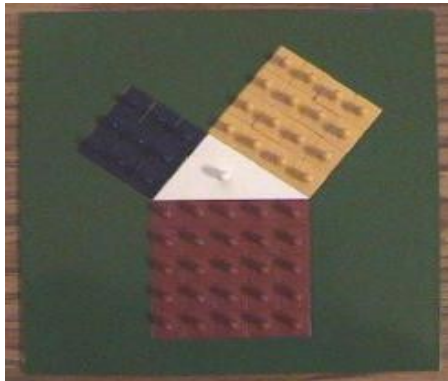
(we've swapped the cutout pieces for the whole piece of the hypotenuse square.)



Pythagoras' Plate II

Begin

This looks a bit like the plate we used the other day.



Let's look at this triangle. This appears to be a scalene right triangle.

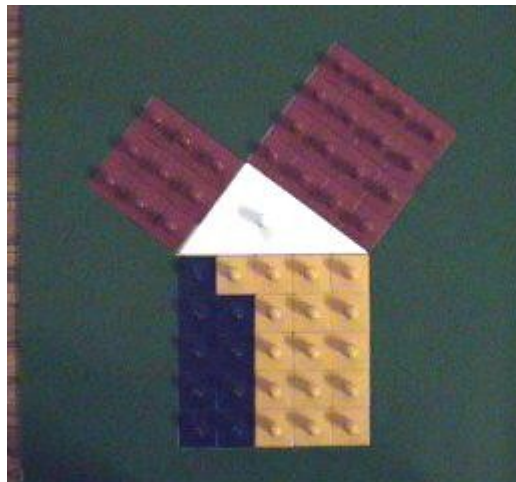
Let's check with our angle measurer. Is that a right angle? (test 90 degree angle)



Yes, it is a right angle.

What did Pythagoras prove? (area of square on the hypotenuse equals sum of areas of the squares built on the legs)

We could look at this and count the squares but let's do this. (swap, piece by piece, the parts from the sides, with the hypotenuse side)



(we've swapped all pieces now to show / prove the Pythagorean theorem)

So we can see its true. The sum of the squares built on the two legs is equal to the square build on the hypotenuse.

Count Length of Sides

We can express that with numbers here. Let's look at this square (smallest) This side is 1, 2, 3 in length. This side is also 3 (count adjacent side). **So this square size is 3 squared.**

Let's look at this square (medium) This side is 1, 2, 3, 4 in length. This side is also 4 (count adjacent side). **So this square size is 4 squared.**

Let's look at this square on our hypotenuse. This side is 1, 2, 3, 4, 5 in length. This side is also 5 (count adjacent side). **So this square size is 5 squared.**

Write Theorem

So 3 squared plus 4 squared, 9 and 16, equals our 5 squared which is 25. (write it)

$$3^2 + 4^2 = 5^2$$

$$9 + 16 = 25$$

$$25 = 25$$

Epilogue

This is the Pythagorean Theorem. Be sure to return parts to the correct locations again and inform the children about how to do the same.

November 23, 2004

Pythagoras with the Constructive Triangles

Material

Triangular, small Hexagonal and large hexagonal box of constructive triangles. Three teacher-made triangles colored yellow (size of the yellow triangle in the small hexagonal box – stored in hexagonal triangles large envelope).

Prerequisites

Experience with the equivalence work with the constructive triangles and plates I and II of Pythagoras.

Prologue

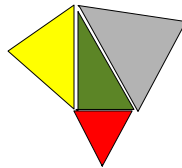
Small group of children to attend.

Begin

When we did the Pythagoras plate, what did we say he proved?

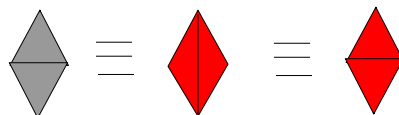
(Sum of the squares of the legs of a right triangle is equal the sum of the square on the hypotenuse.)

I wonder if this will work with triangles. (set out all triangles on table, random)



form triangles around the green right triangle

Do you remember what we showed about these rhombi? We showed they were equivalent.



Since these are equivalent rhombi, each half is also equivalent

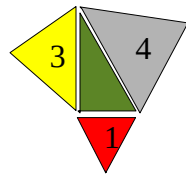
So what can I say about these triangles (gray equilateral and red isosceles). They are

equal.

This means we can also say they have the same value. We could say this triangle has a **value of one.** (pointing to the red equilateral triangle)

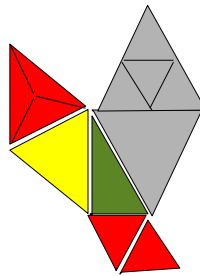
Then we could say this triangle has a **value of 3.** (because the 3 reds make the 1 yellow)

This gray triangle then has a **value of 4** (since 4 small grays make the 1 large gray)
(point to red) 1 + 3 equals 4 (point to yellow for 3 and gray for 4)



So **the sum of these sides (point to the legs) equals the side on the hypotenuse.** So Pythagoras' ideas apply to triangles also!

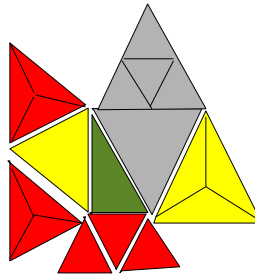
I wonder if this will work with Rhombi. (form 3 rhombi adding extra triangles)



So let's set the value of this triangle as 1 (red again). So this Rhombus (bottom) has a value of 2 and this rhombus (left) has a value of 6 and this rhombus (top right) has a value of 8. 2 + 6 equals 8.

So this rhombus (2) plus this rhombus (6) equals this one (8). So Pythagoras works with rhombi.

So let's see about this trapezoids.



(Add new triangles to form 3 trapezoids)

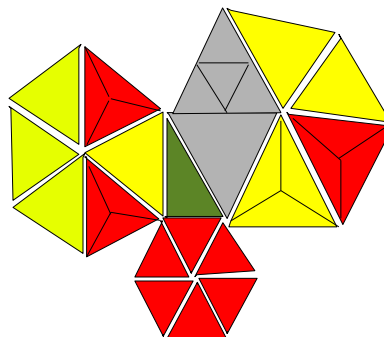
So let's look at the values. The smaller trapezoid (bottom center) here has a value of 3. This trapezoid (bottom left) has a value of 9. This trapezoid has a value of 12 (top right).

$$3 + 9 = 12.$$

(again remember the smallest red triangle started as value of 1. build up the values based on the history above)

Can we keep going? Let's try a hexagon.

(add triangles from large hexagonal box and use extra pieces to build hexagons)



When I built the hexagons by doubling the sizes I doubled the values didn't I. So we have 6 (bottom), 18 (left), and 24 (top-right) for our values. $6 + 18$ equals 24. So this works with hexagons too.

Epilogue

There is one other configuration we could make with the materials we have.



Another Way to Calculate Pythagoras' Rule

Material

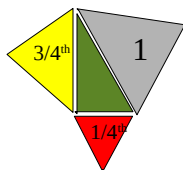
Triangular, small Hexagonal and large hexagonal box of constructive triangles. Three teacher-made triangles colored yellow (size of the yellow triangle in the small hexagonal box – stored in hexagonal triangles large envelope).

Prerequisites

Must be secure with fractions.

Begin

Today, let's say this triangle has the value of 1 (large gray)



What would be the value of the red ($1/4^{\text{th}}$)

How about our yellow triangle? Yes, $3/4^{\text{th}}$ s. $1/4^{\text{th}} + 3/4^{\text{th}} = 1$. So the **sum of the value of the sides built on the legs equals the value of the side built on the hypotenuse.**

(build figures on top of these as part of the first lesson; **calculate values and expressions**)

(refer back to the *rhombi and hexagons* we made when the red was equal 1; **recalculate using our new values for the rhombi and hexagons** – i.e., with red equal $1/4^{\text{th}}$)

(after plates I and II, make up word problems. E.g., the area of a square built on a hypotenuse is 25. The length of one leg is 3. What is the length of the other leg?)

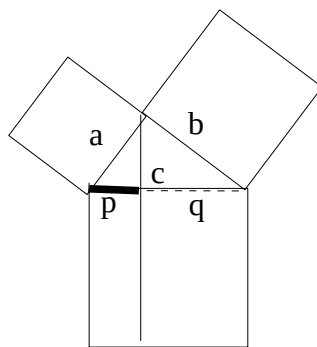
December 15, 2004

Equivalence with Euclid's Plate

Prologue

Euclid took Pythagoras' idea regarding right triangles and extended it. Remember Pythagoras said the two squares built on the legs of a right triangle equal the square on the hypotenuse. Euclid dropped down the height of the triangle – dividing the hypotenuse into 2 pieces. If we call the legs a and b and the hypotenuse c , and label the two pieces of the hypotenuse p and q , Euclid proved that $a^2 = pc$ and $b^2 = qc$.

Our material provides a **sensorial impression of these facts**.



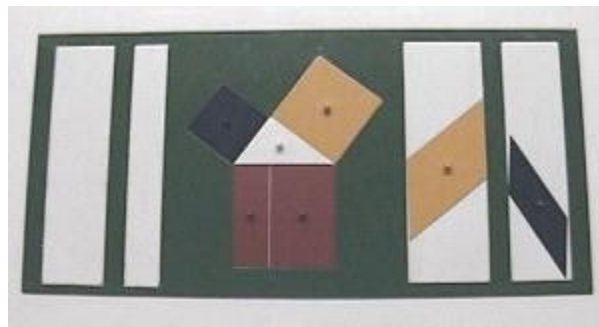
$$a^2 = pc$$

$$b^2 = qc$$

Hypotenuse c is split into p and q

Material

Euclid's Plate. Begin with parallelograms on the right.



Begin

Remember that work we did about Pythagoras. Euclid took that work even further. We'll look at Euclid's work today.

What do we have here? (hold up right scalene triangle – white)

(review parts of the triangle – sides, etc.)

We also have other shapes here. What is built on each side of the triangles? (squares)

But did you notice anything about the red square? It's divided into two rectangles.

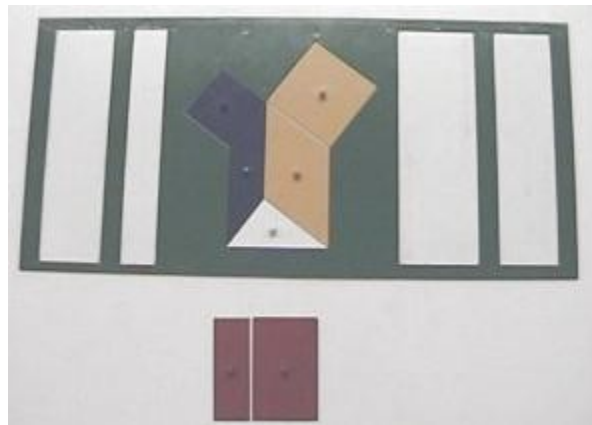
Our square on the hypotenuse is divided into 2 rectangles. Those 2 rectangles together make up the square.

We also have some other shapes. What's this? (2 parallelograms)

Red Square and Two Parallelograms

The first thing we'll look at is the relationship between the red square and the 2 parallelograms. I can show that like this.

(take left square piece and right square piece out and place parallelograms into square area along with the triangle)

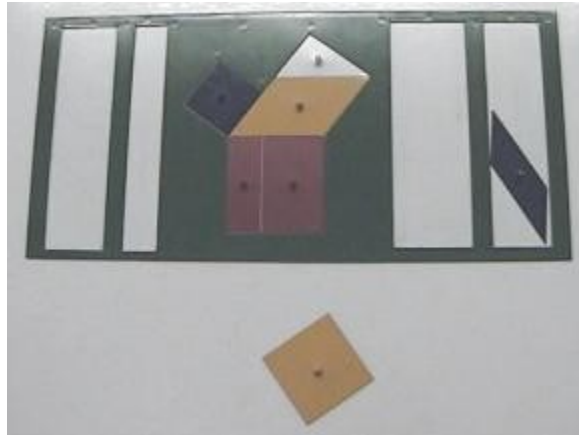


They are equivalent.

Yellow Square Yellow Parallelogram

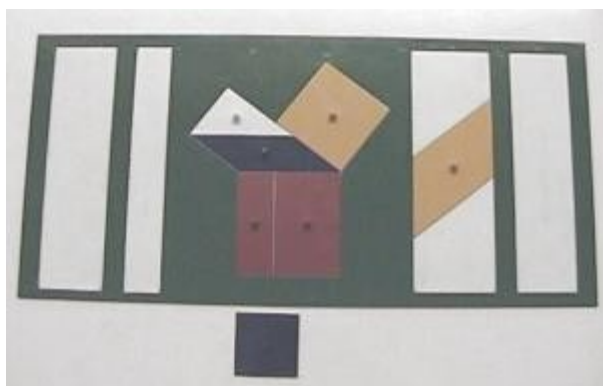
Now I want to look at the relationship between the yellow square and the yellow parallelogram. (put things into starting position again)

Yellow square and yellow parallelograms



They are equivalent!

Blue square and blue parallelograms



They are equivalent!

So we found the blue square is equivalent to the blue parallelogram. The yellow square is equivalent to the yellow parallelogram and the two parallelograms together are equivalent to the red square.



(this was sensorial – now we switch to the reasoning level)

Begin

Today, we'll show the **yellow square is equivalent to the larger red rectangle**. We'll show that by proving that **both are equivalent to the yellow parallelogram**.

Then we'll show the **blue square is equivalent to the smaller red rectangle**. We'll do that by showing that **both are equivalent to the blue parallelogram**.

Remember we worked with equivalence with the iron material? We worked with the rhombus and the rectangle.

We saw the rhombus and rectangle are equivalent if they have base and height in common.

A rhombus is just a special parallelogram – so the same thing is true for parallelograms. **A rectangle is therefore equivalent to a parallelogram if they have the same base and height.**

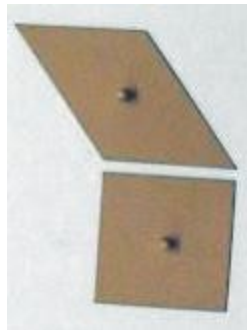
Yellow Parallelogram and Yellow Square

(parallelogram and square on table – both yellow)

(Bases of Yellow Parallelogram and Yellow Square)

We know these are equivalent if they have the same bases and heights.

We want to compare the bases. We can do that like this



(hold up base of one to the base of the other)

So the bases are the same.

(Height of Yellow Parallelogram and Yellow Square)

What about their heights?

To do that we do this.



(turn plate 90° and place square and parallelogram into empty white area)

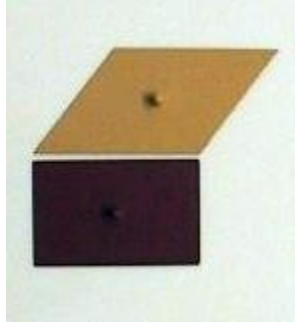
We can say the heights are the same. (area to right is used, turn whole plate 90 degrees)

So we can say they have equal heights and equal bases – so they are equivalent.

Yellow Parallelogram and Large Red Rectangle

(Bases of Yellow Parallelogram and Large Red Rectangle)

Now we need to compare the larger red rectangle to the yellow parallelogram in the same way. We can see the bases are the same

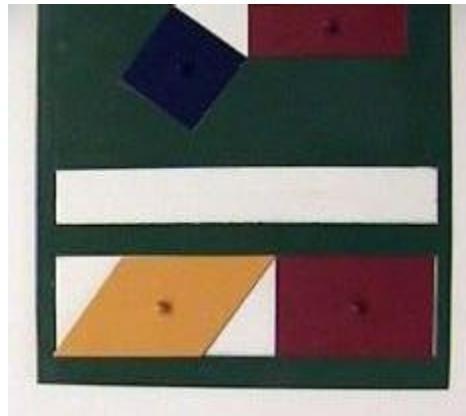


(compare large red square and yellow parallelogram bases as we did above)

(Height of Yellow Parallelogram and Large Red Rectangle)

Now we must compare the height. Again, we turn our frame – this time these two fit here.

(use left side of plate this time – again turning place 90 degrees left before placing figures into the white empty space at far left (bottom after it is turned))



So what did we prove about this red rectangle and this yellow square? They are equivalent!

We showed this large red triangle is equivalent to this yellow parallelogram and we proved this yellow parallelogram is equivalent to this yellow square – therefore what do we know about the large red rectangle and yellow square? They are equivalent!

I don't need my yellow parallelogram now and I can put it away.

Blue Square and Blue Parallelogram

Now we'll do the same thing with the **blue square** – showing it is equivalent to the

smaller red rectangle – by showing that **both are equivalent to the blue parallelogram**.

(Base of Blue Square and Blue Parallelogram)

We start with the blue square and blue parallelogram.

(orient parallelogram with same base as the base of the square)

We know they are equivalent if the bases and heights are the same, right?

Let's compare the bases. (hold up bases next to each other to show same length)



The bases are the same! Now what about the heights?

(Heights of Blue Square and Blue Parallelogram)

(fit both blue square and blue parallelograms into far right empty white space and rotate plate 90°)

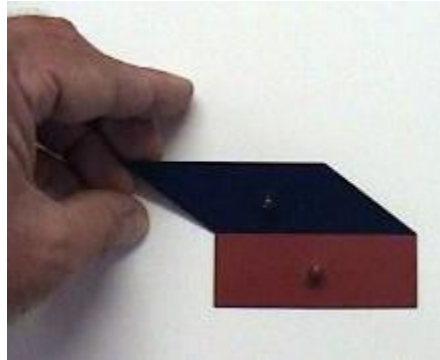


What can we say about the heights? (they are the same)

So we've shown this blue square and blue parallelogram are equivalent.

Small Red Rectangle and Blue Parallelogram

(hold bases together as shown below)



So we can see the bases are the same.

(Height of Small Red Rectangle and Blue Parallelogram)

What about height?

(turn the plate left 90 degrees)

(second from left of start position of plate or second from bottom after turning)



Now my two pieces fit here.

We showed the blue square is equivalent to the blue parallelogram and we showed the blue parallelogram is equivalent to the small red rectangle.

So the blue square and the small red rectangle are also equivalent!

So we don't need the blue parallelogram anymore – we can put it away.

Epilogue

We looked at relationships between these pieces – not at formulas or rules. Prior to this, the children would have worked a lot on equivalence, the concept of a rectangle and parallelogram, all the work with Pythagoras. So this plate comes in the early 9-12 class very often due to all the work that needs to come before it.

Note this lesson outline is:

- $A \text{ (square)} = B \text{ (parallelogram)}$
- $B \text{ (parallelogram)} = C \text{ (red rectangle)}$ therefore ...
- $A \text{ (square)} = C \text{ (red rectangle)}$

... demonstrated for both yellow-large and blue-small combinations.

November 23, 2004

Polygons

Nomenclature of Polygons

Material

Box of geometric sticks. Tall pin fasteners.

Prologue

Some of this, the children should have had in the primary class. We need to find what they know and don't know. Ideally we go into a lesson knowing their level of knowledge. Often, you find they don't know a term and you have to quickly incorporate the term into your lesson.

We'll start with the parts of the polygon. We go from general nomenclature to more specific nomenclature. We've chosen specific colors because we know certain colors are going to work; we begin with 2 green, 2 brown and 1 black to begin.

Begin

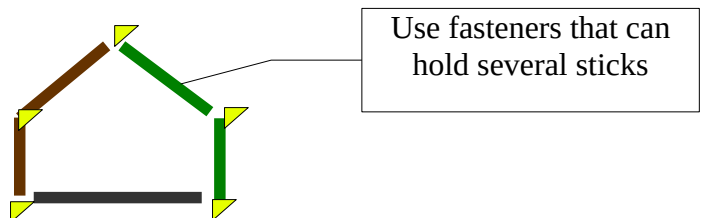
What have I made? Yes, it's a polygon. There are some names to the various parts of a polygon.

Remember that a polygon completely encloses some space. Anyone remember what this inner part is called? We could call it the **area** but we also call it the **surface**.

There's a name for each one of these sticks. Each one would be called a **side**.

What would each of these be called? (point to a vertex) These are called **angles**.

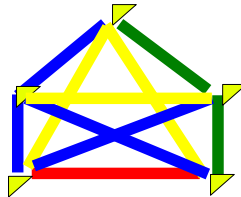
There's a name for all of the sides together. We call that **perimeter**.



Now, we're going to take some more sticks. I'll set one stick like this. This line is called a **diagonal** (add yellow stick such that it connects two vertices)

My polygon can have more diagonals than just this one. There's one here and here, here ... (add 5 **total diagonals** – **only laid across** the tacked together pentagon)

Each of these is a diagonal. How many diagonals could I fit onto the pentagon? Yes, 5.



I wonder how many diagonals a hexagon would have. We better try it. Or how about an octagon? Why don't you try these and see what we get?

(when you do **exploration**, use **shorter sticks for the sides** and use the **longer sticks for diagonals**)

Epilogue

This is the general nomenclature to review and/or teach. Do a **second period** on these terms; invite the child to make his own polygon and identify the parts as he goes. Also make sure to do a **third period** lesson after a time.

Diagonal is any line that connects two non-adjacent vertices.



Triangles

Material

Triangle box of constructive triangles, **plumb line from the box of sticks and a stand used for the triangles**. The stand is sold by Nienhuis but a piece of clay placed near the edge of the table to stand a triangle upward, this works too.

Begin

Here we want to look at the triangle and some of its specific names. Let's take all of our

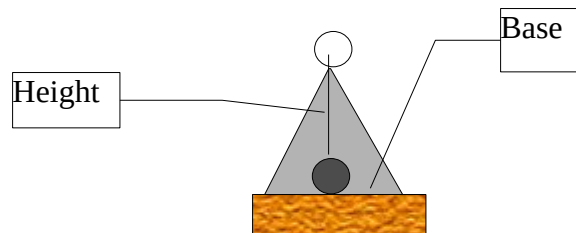
triangles out of the box, placing like-triangles on top of like-triangles (exact fits)

We'll start with this one (**gray equilateral**). Do we know anything about these sides? They are all the same so we have an **equilateral** triangle. We also know something about these angles right. They are **acute and all equal** too.

What do we know about the sides all together. We call them the **perimeter**. How about the inner part of our triangle? Yes, we call that the **surface**.

With a triangle, when we stand a triangle up like this, there's a name for this part here (point to bottom). It's called the **base** – **a triangle has a base**.

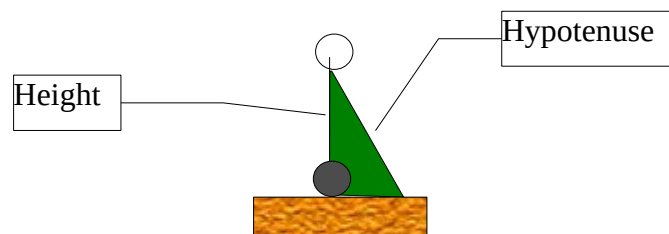
(put triangle standing up in a stand)



Using our plumb line, we're going to extend a line down from the highest point of our triangle straight down. This is called the **height** of the triangle. It's sometimes called the **altitude but we'll call it the height**. It's the line that goes from the highest point of the triangle, straight down to the base.

Let's try one of our other triangles. Let's try this one (green right scalene). What kind is this? Yes, a right angle scalene triangle. (review terms we need and then...)

This side here is special. It's called the **hypotenuse**. We often call the other two sides the **legs**.

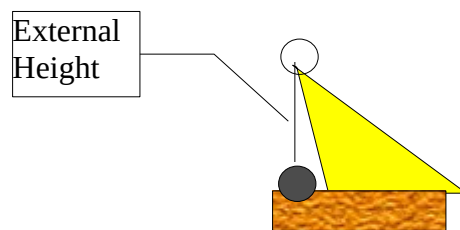


When we put this triangle in our stand we align it like this. With my plumb line, I'll again extend it down from my highest point of my triangle – down to the base. We again call this line the height. Where is it found on this triangle? Right on the side, right?

(show plumb line along the vertical leg)

We also have this yellow triangle. What kind is this? Yes, obtuse isosceles triangle. We could put it in our stand like this to find the height but you can do that on your own.

I'll put it in like this. We'll again hold our plumb line at the highest point to find the height. What do you notice this time?



It's not on the triangle. It's what is called an **external height**.

Epilogue

Any other nomenclature can be reviewed and reinforced. Normally we have the children at our sides.



Prologue

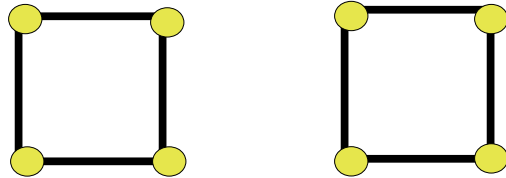
The parts that will be new are **major diagonal** and **minor diagonal**. Begin by choosing 8 sticks that are all the same – black here.

Diagonals

Begin

We go back to our box of sticks and look at a rhombus.

Join them together into two squares. I'll use the measuring-angle to make sure I have right angles.

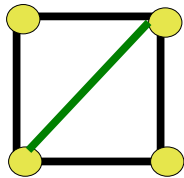


If I arrange my sides like this what shapes have I made? (we've made two)

They are both squares. What can you tell me about squares?

Yes, **4 equal sides**. Yes, there are **4 equal angles** of 90 degrees. Altogether, the sides would be the perimeter.

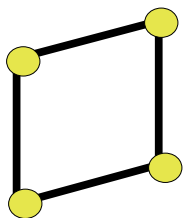
Now, if I put a line like this (green stick) I make a **diagonal**. In fact my square can have 2 diagonals. I'll even **attach one** because – what does the diagonal do for the square?



A **diagonal stabilizes** the square doesn't it? If I pick this one up, it's not a square any more is it?

(pick up square w/o diagonal and watch it collapse)

Now, I'm going to do something to my other square. I'll change it like this. (pull opposite ends to form a rhombus).



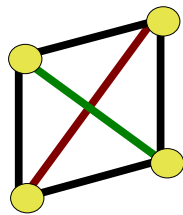
Now what shape do we have here? Yes, a **rhombus**. What do we know about a rhombus. It has **two pair of parallel sides**. What else? Yes, there are **2 obtuse angles and 2 acute angles**.

The sides are still equal.

Suppose we put these on. (put two diagonals on the rhombus). How many diagonals does a rhombus have? Yes, 2.

What's the difference between diagonals on a rhombus and a diagonal on a square?

Yes, there are two different color sticks for different lengths on a rhombus.



The **longer** diagonal is called the **major diagonal**. The **shorter** one is called the **minor diagonal**.

You could attach both of the diagonals or one of them. What would it do for the rhombus? Yes, it would stabilize it, just like for the square.

Now watch what I'm going to do to my rhombus. (**pull the opposite ends even further for a more dramatic shape**)

It's still a rhombus but let's see if my diagonals fit. No, they don't fit now. What's happened? **The major diagonal is now too short and the minor diagonal is now too long.** (note these are the two original diagonals from the first rhombus)

What about the **diagonals in a square? They are always the same**, right?

With our rhombus we can also do this (make a square again). Now let's see if my diagonals fit. Now – my original major diagonal is too long. The minor diagonal is now too short.

Epilogue

Have a little fun changing the major and minor diagonals on a rhombus showing they become different lengths – contrast with a square whose diagonals never change.

There is classified nomenclature material that the children can work with to review.



Trapezoid

Prologue

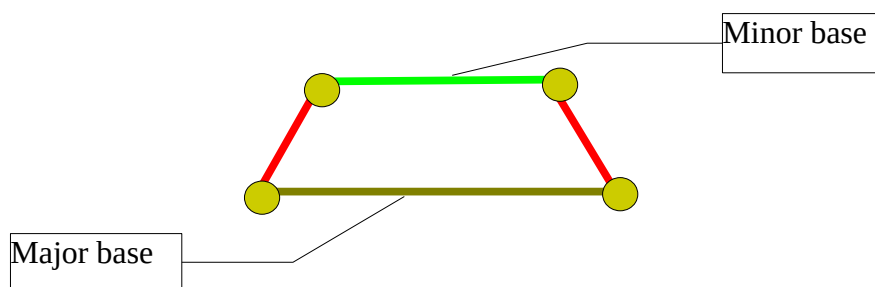
Next we look at a trapezoid.

Begin

Let's make a trapezoid.

What parts do we know of a trapezoid. Where are the **sides**? What do we call all of the sticks together? **Perimeter**. And how about the inner part (surface). And what do we call these? (**angles**). We have two acute and two obtuse angles.

What is special about a trapezoid is that it's considered to have 2 bases. One that is longer which is called the **major base** (point to bottom) and one that is shorter which is called the **minor base**.



(turn over the trapezoid) This is the major base and the minor base (point)

Now you have lots of figures you can draw, label the parts of, write definitions of the parts, and are your definitions to the nomenclature material.



Nomenclature of regular polygons

Material

Solid pentagon iron inset, a piece of white paper, a metric ruler, a compass, lead pencil and some colored pencils.

Prologue

Since our materials are based on metric system, we need to use a metric ruler for exactness.

Begin

What do we call this? A polygon – but also a pentagon. This is called a **regular pentagon** or regular polygon.

A **regular polygon** has **all sides the same length and all angles equal**. So we can verify easily we have a regular polygon by doing this...

(pick up the inset, rotate it one-side worth and replace it into the frame).

All angles are equal and all sides are equal. We can rotate it and replace it into the frame.

What names do we know for this polygon? Sides, perimeter, surface, angles. Good we know these.

Center

Who knows what we would call this center point? We call it the **center**. The knob is on the center, isn't it.

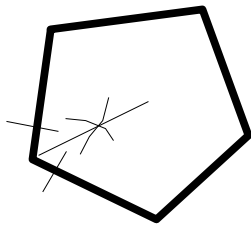
Now, I'll trace my polygon onto my paper. (use lead pencil to trace figure on the paper)
I now have a pentagon on my paper.

Can you tell me where the center is? Not exactly.

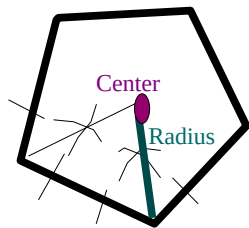
Bisect

But do you remember the lesson we had on bisecting angles? This is why I brought my compass and ruler. I'm going to bisect an angle.

(make 2 points on adjacent sides using the compass to determine length from vertex, use compass to find the intersecting point on the radius and connect the vertex with the new point)



Can you show me where the center is now? No, not yet. Let's bisect another angle – maybe that will help.



Now I have bisected a second angle. Now can you see where the **center** is? Yes!

I'll take one of my colored pencils and label it.

Radius

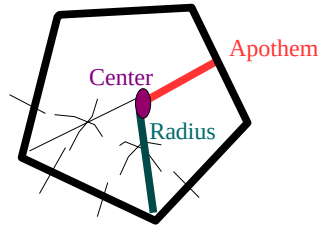
Do you see these **lines that go from the center that bisect the angles**? Those lines have a special name. It is called a **radius**. With another colored pencil, I'll label that too.

(radius is any line that connects the center to a vertex)

Apothem

Now, I'll measure one of the sides and find the mid-point of the side. And we'll mark it. I'll use my ruler for this. (do so)

Now I'm going to **connect the center of my polygon with the mid-point**. This line is called an **apothem**. I'll use another colored pencil to label this line.



Now you can practice drawing your own polygons, find the center, radius and the apothem.

December 13, 2004

The Sum of Angles in Plane Figures

Material

Two envelopes, one labeled triangles and the second labeled quadrilaterals.

In “Triangles”, there is a variety (kinds, sizes, colors) of paper triangles.

Amplitude of the angle is marked an equal distance outward from each vertex of each angle.

In “Quadrilaterals”, there is also a variety by sizes, colors and types. The amplitude of each angle is again marked an equal distance out from each vertex.

Prologue

This is teacher-made material. We need enough for the presentation and then we show the children how to replenish the envelopes for use with the next child.

Sum of the Angles in a Triangle

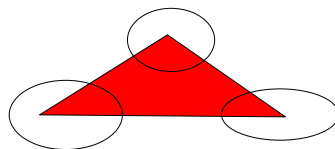
Begin

Today we'll look into how much the 3 angles of a triangle make altogether.

(dump all triangles on the table and spread them about)

We'll choose this triangle (red right isosceles) What kind is this?
answer)

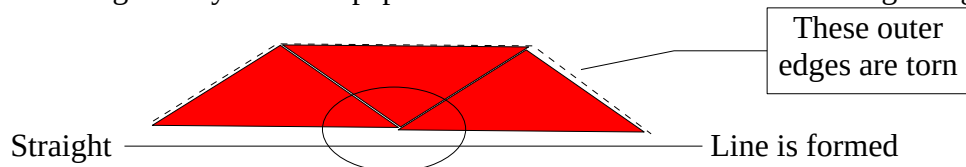
(get



I want to know **how much these 3 angles equal altogether.**

All we have to do is this.

(tear each angle away from the paper – turn each vertex to form one straight angle)



What kind of angle have we made? Yes, a straight angle of 180 degrees.

(repeat with 1 or more additional triangles)

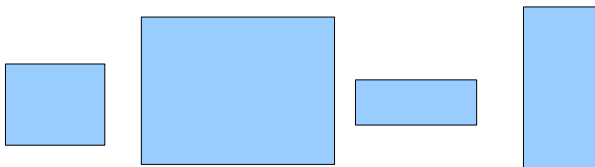


I wonder what would happen if we tried the same thing with quadrilaterals. I have another envelope here for that.

(spread all out on the table)

Let's pick this one. (does not matter which – pick four)

How many angles do we have here (four)

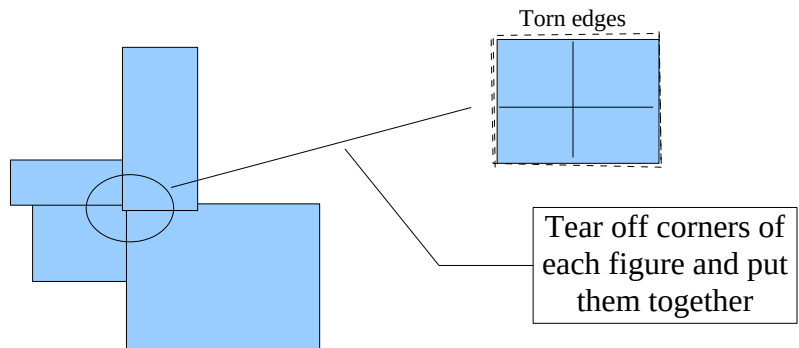


I wonder what these four angles equal altogether?

This is a **complete angle - 360°**.

(repeat)

You can go on and try any of the other quadrilaterals in the same way to see what the sum of the angles make.



What do you think would happen if we did the same thing with a pentagon, or hexagon. You can do this same thing to any polygon you want to.

Epilogue

Between the iron material and the geometry cabinet, we have more shapes the children can use to explore. Hopefully, through the explorations, they will arrive at

the rule for the sum of angles for plane figures.

We measured a triangle and got 180. We measured a quadrilateral and got 360. If we were to do a pentagon, we would get 540 – or 3 straight angles.

The sum of the angles of a plane figure is the (number of sides - 2 times 180 degrees). Now we can use this generalization for any polygon. We hope the children will arrive at this rule on their own. Be ready to cut an angle in order to make a complete angle or straight angle if the children come up with irregular figures.

December 13, 2004

Equivalence with the Iron Material

Material

Trapezoid, Rhombi, Decagon and Rectangle plates

Prologue

This is work that will lead to deriving the formulas for area. **When we introduced equivalence, we mentioned it was used in the work with area; we are laying that foundation still.**

There are 3 levels of work with this material; the first two are strict equivalence work and reasoning work. The reasoning level may come several months after the equivalence work to allow practice and experience with the material.

The third level is deriving the formulas for area; this may come a year after the lesson's first introduction. It is the earlier handling of the material and their understanding of equivalence that will make area work go smoothly later on. We want to encourage lots of manipulation and handwork with the materials.

Here we're using the logic **if $A=B$ and $B=C$, then $A=C$.**

Prerequisites

Nomenclature of polygons.

Begin

(we do here the first two shapes at level 1 equivalence, then stop and do reasoning level for those 2 shapes – third level follows next semester)

Triangle-Rectangle - Equivalence

(start with triangle, first level – equivalence)

What do we have here? (triangle)

What kind of triangle? (equilateral)

What do we have here on the right? (a rectangular space)

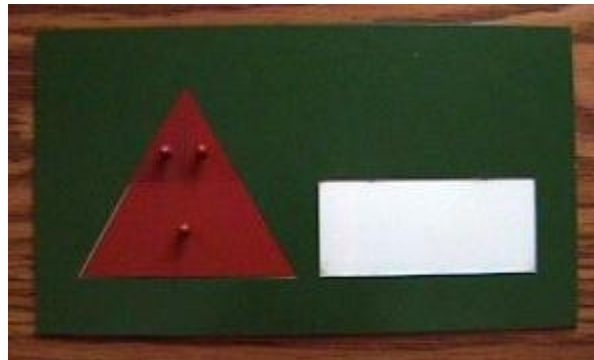
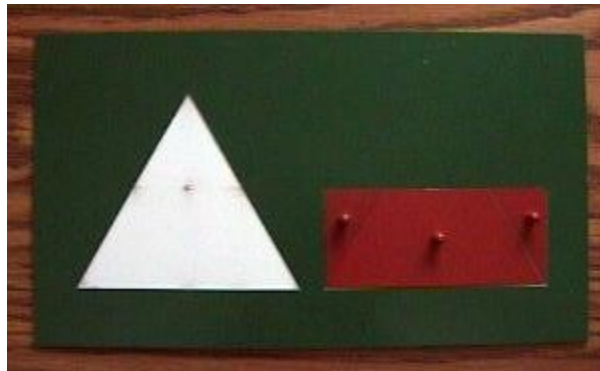


Plate A

I want to know what the relationship is between this triangle and this rectangle.
Can I do this? (move all pieces from the triangle to rectangle)



They all fit! We can say **they are equivalent**.



Rhombus-Rectangle - Equivalence

(If solid rhombus is equal the divided rhombus and the divided rhombus is equal the solid rectangle, then the solid rhombus is equal to the solid rectangle)

What shape is this? (rhombus - see plate B below)

What shape do I have over here? (another rhombus - see plate C below)
And what is this (rectangle – see place C below)

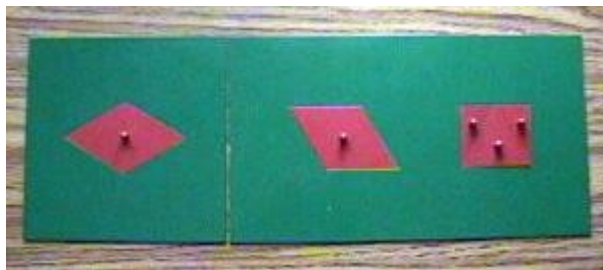
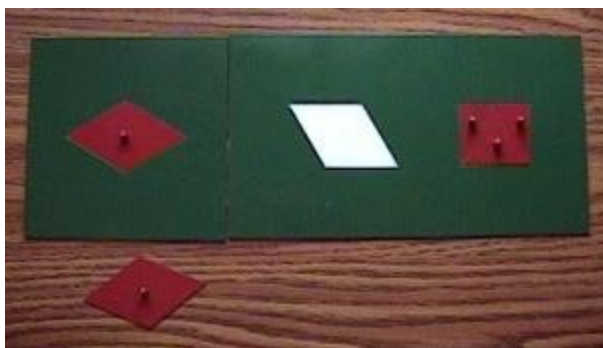


Plate B (left) and C (right)

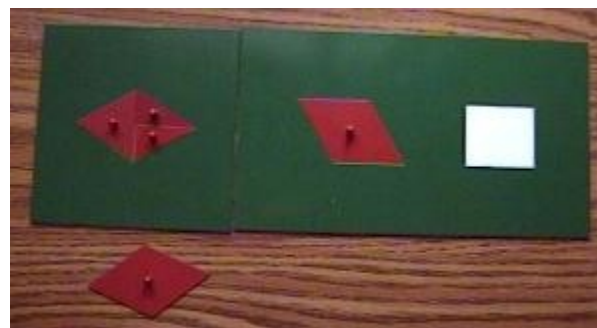
Let's see if we can switch these two (rhombus of plate C switch with rhombus plate B).



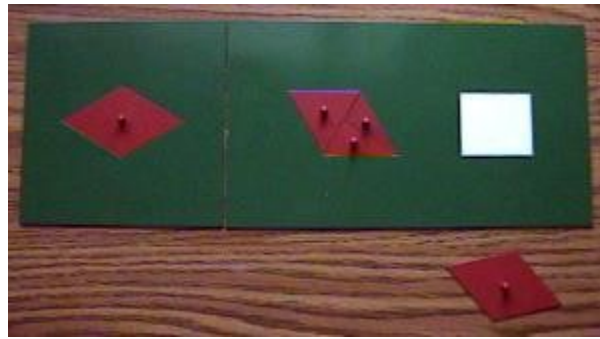
They fit so they are equivalent.

Now let's see if I can switch the rectangle with these rhombi.

The rectangle pieces fit into the rhombus.



But do the rhombus pieces fit into the rectangle? (no)



Because these pieces from the rectangle fit into the rhombus, what can we say about these shapes? They are equivalent; the **rhombus and rectangle are equivalent**.

We can also verify the pieces from one rhombus will fit into the other one. (do so)

You can trace, color or cut-out these shapes on your own. Maybe you want to make a booklet of the shapes that are equivalent to one another.

Triangle-Rectangle - Reasoning

What's the relationship between the rectangle and triangle. We saw they were equivalent. How did we prove that? We took the cut up pieces from the triangle and put them into the rectangle.

We know some parts of these figures.

What do we call this part of the triangle (base)

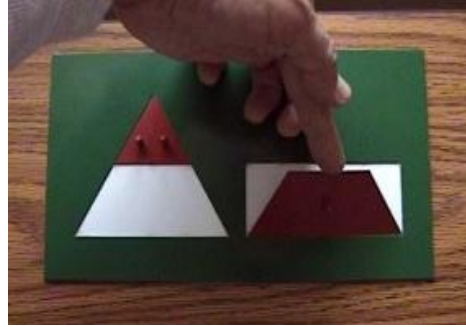
What do we call this part of the rectangle (base)

What would we call this part of the rectangle (side) (altitude or the height)

Where would we find the height of the triangle? (down the middle)

Yes, would you like to see it?

Plate A – comparing bases



What's the relationship between the base of the triangle and the base of the rectangle?

(they are equal – demonstrate by moving the triangle-bottom inset into the rectangle area)

What's the relationship between the heights of these two figures? (the height of the rectangle is half the height of the triangle)

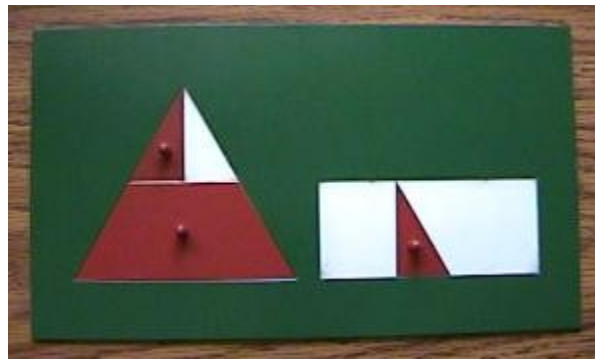
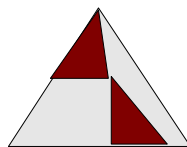


Plate A – comparing height



Shows the height of the rectangle is half the height of the triangle

So we can say that a triangle and a rectangle are equivalent if the bases are equal and if the height of the rectangle is half the height of the triangle.

I wonder if we can find relationships like this for any other figures we've worked with.

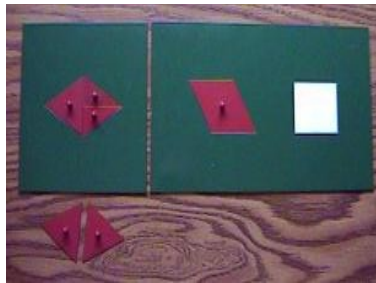
Reasoning level For Triangles, Rhombi and Rectangles

Prologue

In addition to the two plates used above, we have a third plate: from left, there is a rectangle, followed by 2 rhombi. Underneath, there is a rectangle which spans the width of the 3 figures above.

Begin

What is the relation between the rectangle and rhombi in this plate and that in this other plate?



(place pieces from rectangle (plate C) into both rhombi figures plate B)

So we can say these figures are all equivalent.

Now looking at this new plate (see plate D below), I wonder what the **relationship between each of these triangles (the different types)**.

(move pieces about among the 3 top areas to show that they are all equivalent)

How do we know this? (they are each half of equivalent figures, rhombi)

I want to **compare each of these triangles to the others**. First the bases – where do we find them? (base is at the bottom)

If you look here at the inset, we can see we have the base here of this triangle and the

base of this one.

I'm going to take one of them and compare it with the other bases.

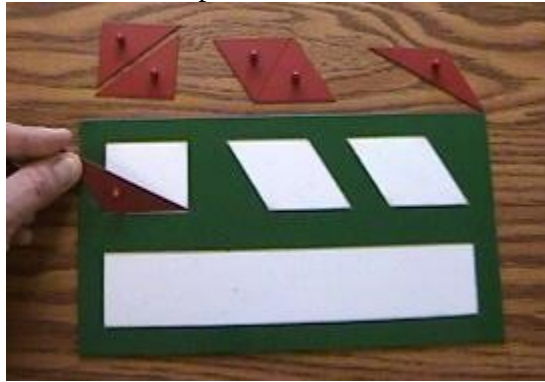
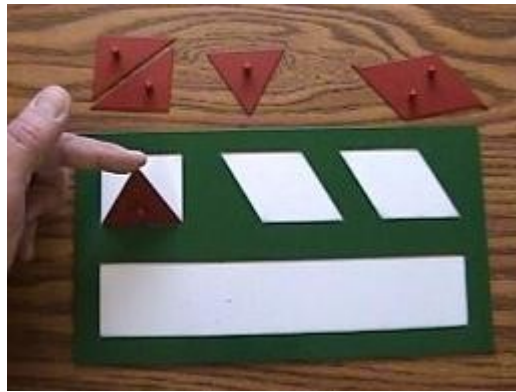


Plate D – compare bases among the triangles.

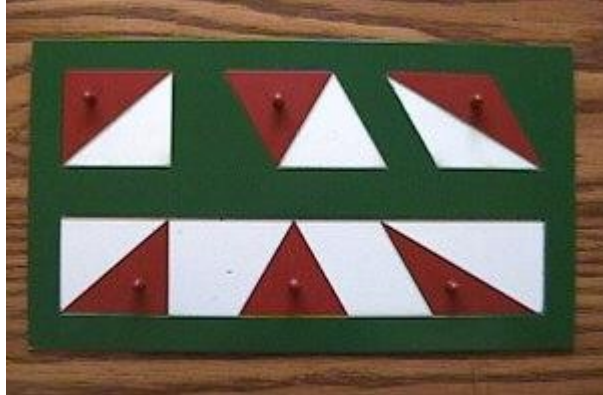


The bases are all the same.
bases)

(hold each base next to each of the other triangle

What about the height? That is what this rectangle here is for. (point empty space below)
(place each triangle in the same rectangular area below showing height is same in all cases)

Plate D – compare heights of each triangle type



What can we say about the heights? They are all the same, right?

So we can say triangles are equivalent if they have equal bases and equal heights.



We can say a rectangle and a rhombus are equivalent if they have equal bases and equal heights – because in our demonstration in our triangles we saw that height and base, if equal, made the triangles equivalent.

Epilogue

When finished, put pieces in original locations. Try to draw the wording out of the children from their own experiencing for why figures are equivalent and other points of reason.



Trapezoid - Equivalence Level

Begin

What do we have here? Two trapezoids, yeah – and how about down here?
(an empty rectangular space)

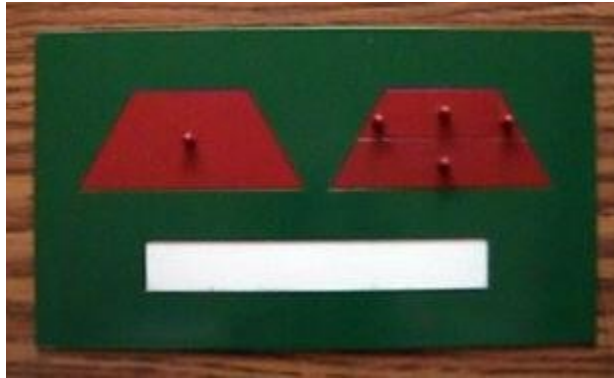


Plate E

I wonder if the two trapezoids are the same.

Let's see. (swap pieces from first to empty-second frame (set aside))

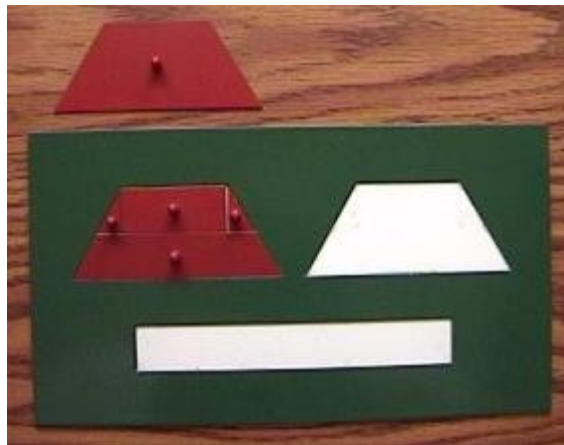


Plate E – compare trapezoids

I wonder what's the relationship between this trapezoid and this rectangle. Let's do this.
(move pieces from trapezoid to fill area of rectangle)

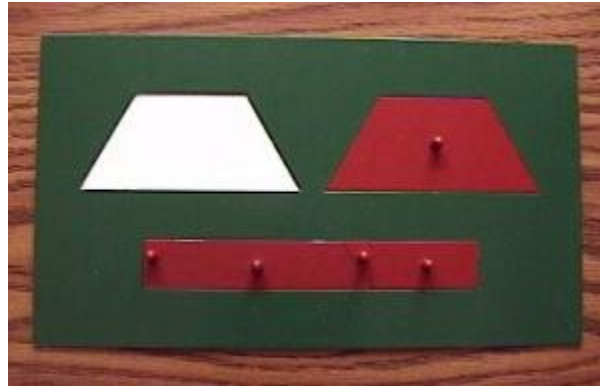


Plate E – compare rectangle to trapezoids

So what can we say about the trapezoid and the rectangle? (they're equivalent)



(months later)

Trapezoid - Reasoning Level

Begin

Remember we worked with this plate. What was the relationship between the trapezoid and the rectangle (equivalent)

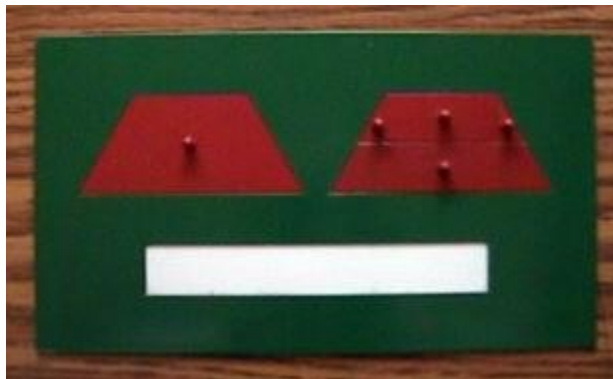


Plate E

Remember we learned nomenclature on the trapezoid; let's review that.

What's this? (surface)

Each of these are the .. (angles)

All the sides together make up the .. (perimeter)

What do we call this special part (major base)

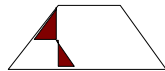
And how about this part up here? (minor base)

Does a trapezoid have a height? (yes)

Where would we find the height?

We drop a perpendicular line from the minor base to the major base to get height.

We can see from these two pieces that the height of the trapezoid is twice the height of the rectangle.



Where would we find the base of the rectangle?

Where would we find the height of the rectangle?

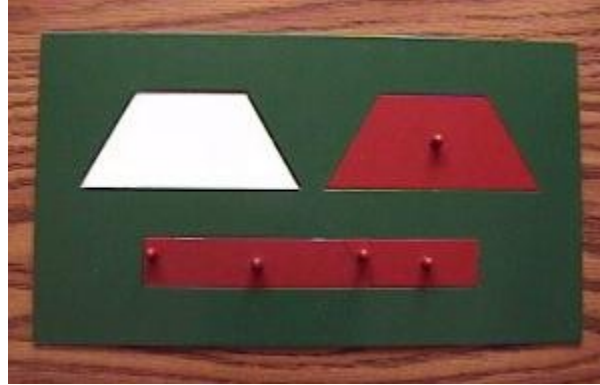
Now we want to look at the relationship between the trapezoid and the rectangle.

What can we say about the relationship between their bases?

What is the relationship of the base of the rectangle and base of the trapezoid. The **base of the rectangle is the sum of the major and minor bases of the trapezoid.**

What's the relationship between the heights of these two shapes? The **height of the rectangle is half the height of the trapezoid.**

Plate E – using base and height determine trapezoid and rectangle are equivalent



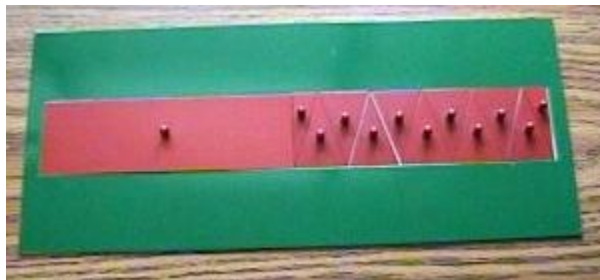
So we can say that a trapezoid and a rectangle are equivalent if the base of the rectangle equals the sum of the major and minor bases of the trapezoid and the height of the rectangle is half the height of the trapezoid.



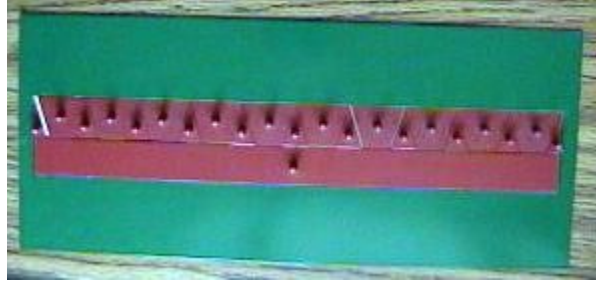
Decagon - Equivalence

Prologue

The base of the short rectangle is equal half the perimeter of the decagon. The height of the short rectangle is equal half of the decagon's apothem (plate H).



Base of long rectangle is equal the perimeter of the decagon. The height of the long rectangle is equal half the apothem (plate I).



Begin

Compare Decagons

We have a solid decagon and we have one divided into 10 equal pieces.

Remember we learned some nomenclature we can apply to the decagon?

What parts do we remember? (review **sides**, **perimeter**, **center**)

Each line from the center to a vertex is called what? (**radius**)

The line from the midpoint of a side to the center is the **apothem**.

The whole face is called the **surface**.

I wonder what the **relationship is between my two decagons?**

(swap pieces – whole to the side, 10 pieces from the other into empty frame)

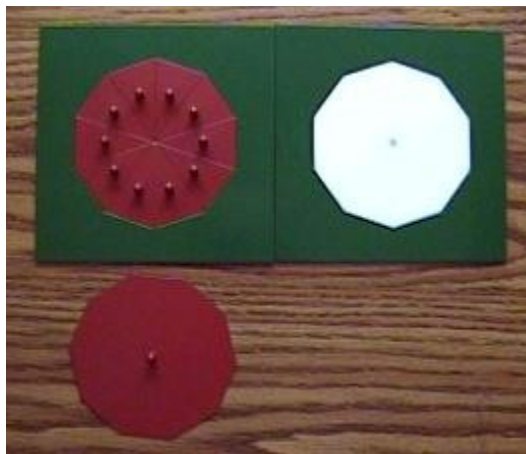


Plate F (left) and G (right) – pieces of G have been place into plate F.

We can say these decagons are equivalent – since they exactly fit.

Compare Rectangle with Decagon

We also have this. Let's look at the relation between this half of the rectangle and a decagon. (remove H1 and place pieces from G into the rectangle)

(remove one whole rectangle at left, and put the decagon's triangles along the bottom of the big empty space standing upward. Use the rectangle pieces to fill in the spaces above.)

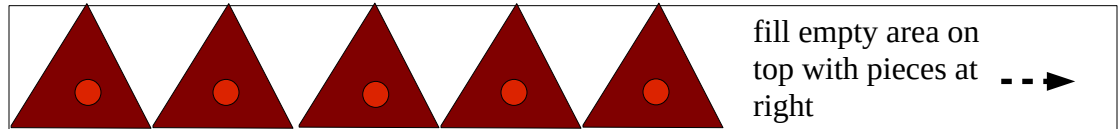


Plate H – Decagon plate G pieces have been moved into the rectangle half at left.

So, what do we have? (2 decagons)

I'm going to take one of them away now.

(put the triangle pieces from the left side back into the decagon and replace the large rectangle whole into its place)

So what is the **relationship between this rectangle and this decagon? They are equivalent.**

What shape do we have here (hold up half rectangle piece plate I)

(rectangle)

I want to know the relationship between this rectangle and the decagon.



Plate I – split half rectangle has been moved into the empty decagon frame

I'll put the pieces from my rectangle into the empty decagon frame to find out their relation.

(now push lower bar in rectangle frame upward and replace pieces onto lower half)

So the relationship between this rectangle (lower half) and this decagon is that they are equivalent.



(this shows the two rectangle portions are equivalent)

Decagon - Reasoning

Begin

Plate H

What did we find out about the relationship between this decagon and this large rectangle piece on the left? (equivalent)



Plate G decagon in pieces



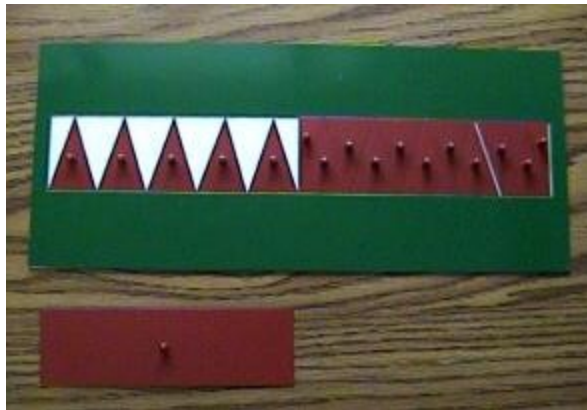
Plate H rectangle half and in pieces

What can we compare and find in common between these figures?

Where do we find the base of our rectangle? (across the bottom)

What's the relationship between the base of the rectangle and the decagon?

(the base of the rectangle is equal to half the perimeter of the decagon)

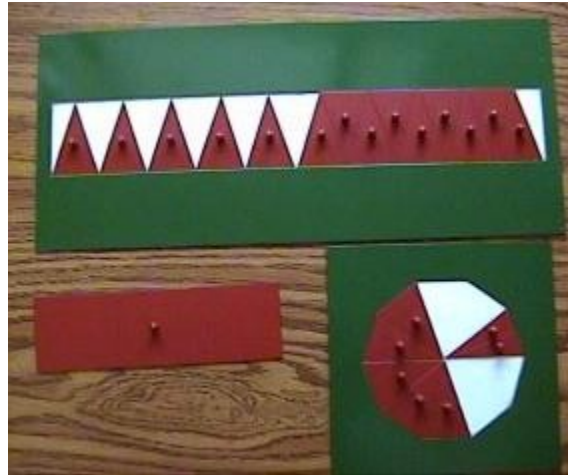


(this is determined by putting the pieces from decagon into rectangle – showing the base is equal the 5 pieces (laid next to each other on their bases) of the decagon)

Where do we find the height of the rectangle? (along the side)

What's the **relationship** between the height of the rectangle and the decagon?

*(the **apothem of the decagon** is the same as the height of the rectangle)*



The two end pieces meet together at a line on the apothem

(place end pieces of the rectangle such that the two longer legs of these two right-triangles meet at the apothem, a line perpendicular to and in the center of a side)

So we can say a rectangle and a decagon are equivalent if the base of the rectangle is equal to half the perimeter of the decagon and the height of the rectangle is equal to the apothem of the decagon.

Plate I

We also saw that this decagon and this rectangle (half) is equivalent.

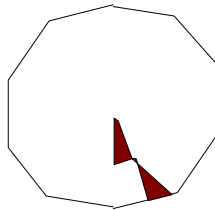
This time the **base of the rectangle is the perimeter of the decagon.**

What about the height of the rectangle? How does it compare to the decagon?

(rectangle height is half the apothem)



Use the two end pieces from the rectangle aligned along the apothem



Another means of viewing height on the decagon

So ... what's the relationship between these two rectangles?

Two rectangles are equivalent if the base of one is half the base of the other – and the height of one is double the height of the other.

December 15, 2004

Area of Plane Figures

Concept of Area Using Yellow Material

Rectangle

Material

Yellow material: 2 parallelograms (1 whole, 1 divided into 2 pieces), 4 rectangles, 3 acute-angled triangles (one whole, 2 divided into pieces), 2 right-angle triangles (1 whole, 1 divided into 2 pieces), 2 obtuse-angled triangles (1 whole, 1 divided into 3 pieces).

Prologue

First thing is to know how to put things away: Bottom = 2 obtuse triangles to form parallelogram; on top = 2 right triangles to form a square; in open space at end = first acute triangle with 2 smaller triangles filling up each end; on top = other 2 acute triangles to form a square; in open space at end = divided parallelogram and 2 pieces filling out each end; on top = whole parallelogram; across very top = layer of rectangles with 1 rectangle laid oddly at top.

By keeping the material in this order it fits into the box, but also the pieces are ready to be used in the presentation.

These are all small group presentations; 2-3 children. This is likely the first time we use the term area (otherwise known as surface). Technically, this space is called surface but when we measure it is referenced as area.

Prerequisites

Children would have done sensorial part of equivalent figures with the iron material. They would have done some work measuring length, work with angles and nomenclature of polygons.

Purposes

To give the concept of measuring area with a unit square. Understanding the pattern in a rectangle. Preparing for deriving the formula for area of a rectangle. Transforming other figures into rectangles and preparing for deriving those formulas.

Begin

Concept of Measuring Area Using the Rectangle

(review)

Remember that work we did with measurement? We talked about having a unit of measurement and we had to determine what that unit would be when we measure.

And then we see how many of those units fit into the space we're measuring. If measuring with meters, our answer is how many meters fit, for example.

We've also been working with the iron material. We saw a triangle can be made into an equivalent rectangle. We looked at equivalent rhombi and rectangles. We traced iron insets but also used graph paper, counting squares to compare the size of figures.

We've seen therefore how handy a square is as a unit of measurement. They fit together well so there is not waste of space when we measure. Squares work well for measurement.

Present Material

(Our ploy is to extend the lines: slide second rectangle to side from under the top – but slide back in since we don't get squares. Second move is to slide up the top 2 rectangles exposing the vertical lines behind. We need both sets of lines (horizontal and vertical) so we pull the top 3 rectangles off to expose the bottom rectangle which has all squares)

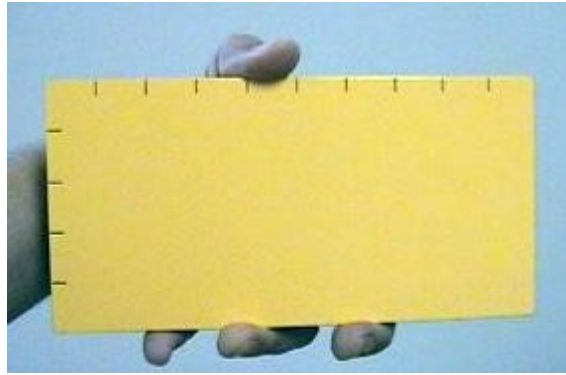


(The four stacked rectangles in order are:

- top = only hash marks on side and top edge.
- Second has horizontal lines.
- Third has vertical lines.
- Bottom or fourth = the squares)

(other materials are set well aside!)

Today, we'll look at measuring area. (hold up rectangles, **stacked, ordered - per above**)

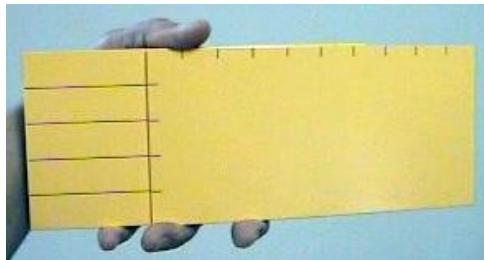


We want to measure the area – the whole surface of this rectangle.

We'll do that using a square as our unit of measurement. So I want to know how many squares will fit onto this rectangle.

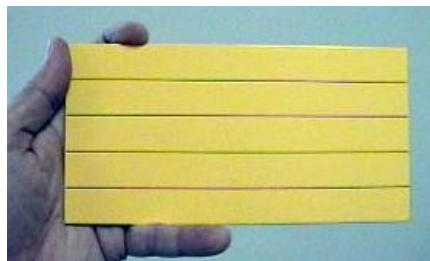
If you look closely, you should see some little lines across here (top length of rectangle) and across here (height of rectangle).

Maybe if I extend these lines running along the side, that will help me measure.

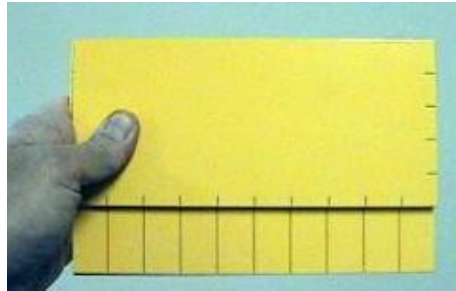


Now watch. (pull hidden rectangle sideways to right from behind the topmost rectangle)

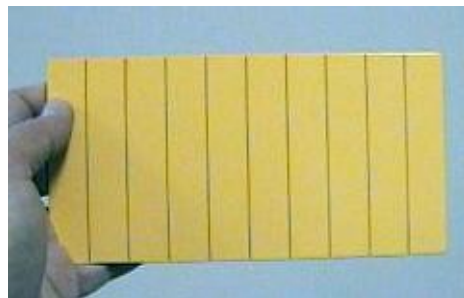
Do I have squares here I can count? No.



Maybe if I extend these lines from the top, downward, that will help. So watch.
(top 2 rectangles are slid upward to expose vertical lines on third rectangle)



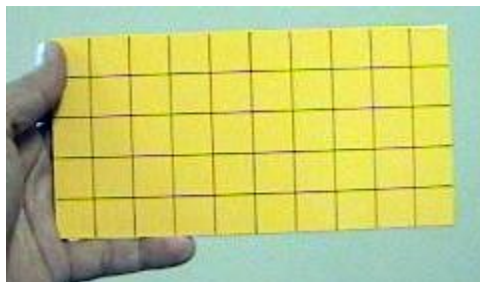
Well now I've extended these lines (top 2 rectangles are off the pile)



But I need to have the lines together don't I? I still don't have that!

This is what we have here.

(show bottom rectangle only – others in left hand)



So this rectangle is covered with squares! Now we only have to count the squares.

(count them all 1-by-1 - 1, 2, 3, 4 ... 49, 50)

There are 50 squares here.

So we can say the area of our rectangle measures 50 unit squares.

Maybe now we'd like to find the area of other rectangles. We've got all different sizes of graph paper. You can draw rectangles as big as you'd like and then measure them.

Remember the area is always expressed in unit squares; we can't just say 50 but we have to say 50 unit squares.

Epilogue

We hope they will set off counting squares but soon discover a faster way: multiplying number of squares of base and height. We let the child discover this.

December 15, 2004

Parallelogram

Prologue

Don't be afraid to review nomenclature; **always emphasize the height and base both before and after the transition to rectangle** is made.

Begin

We're going to look at this shape today. What is this? (parallelogram)

Remember how we saw we could find the area of a rectangle?

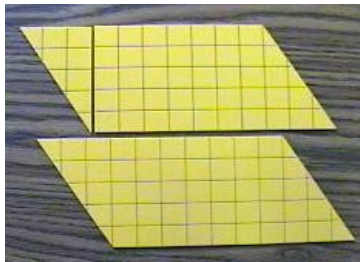
We discovered if we multiplied the base and height, we could find the area. Can I do this with a parallelogram too? My squares are not always whole – there are bits and pieces.

(show that parallelogram has grid marked with partial squares)

I have another parallelogram in my box. They look the same don't they? Let's make sure.

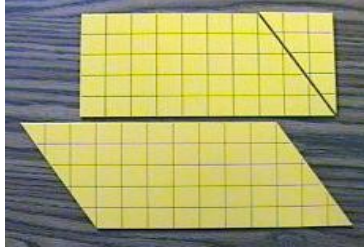
(superimpose one onto the other – see illustration below)

So we can say they are equivalent.



But we said we can't just count these squares because we have bits and pieces of squares.

Maybe if we do this, it will help. (move one corner triangle to the other end)



Now what shape do I have? (rectangle)

Now I have a rectangle. Do we know how to find its area now? (base times height)

So what's the area of my parallelogram? (fifty squared units)

That's how we find the area of a parallelogram.

Epilogue

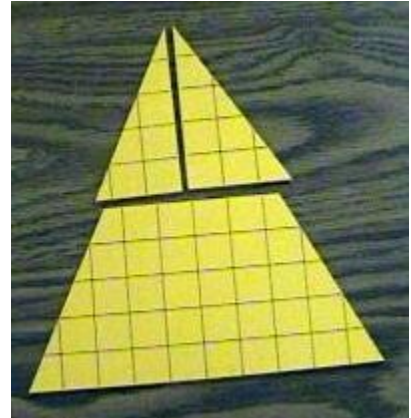
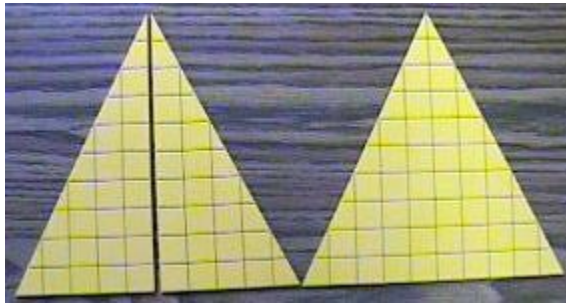
Non-whole parallelogram is divided such that one corner can be shifted to the other end to make a rectangle.

December 15, 2004

Triangles

Begin

Acute



Doubling Area

(second triangle is divided along the height line into 2 pieces)

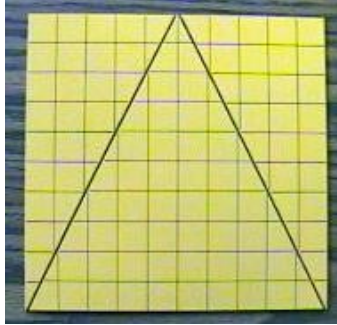
What do we have here? (acute-angled triangle)

It's decorated with squares. Can we just count them? No, we've got bits and pieces of squares to count.

I have another acute-angled triangle. They look the same. (superimpose onto one another)

Yes, they are equivalent. So now suppose this time we do this.

(separate the two triangles at the height line and turn them 180 degrees – placing each onto the 2 top sides of the first whole triangle)



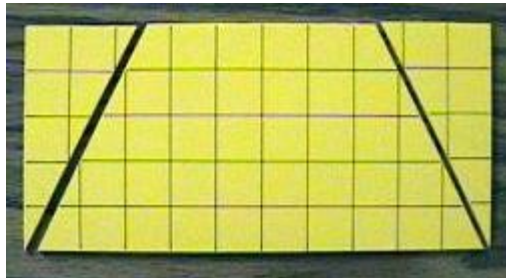
If we multiply base times height, we get 100 now. But we put 2 triangles together. So I need to divided my product by two. So the area of one triangle is 50 squared units.

Dividing Height

(begin with the third triangle divided into 3 pieces, bottom and two top pieces)

I have another acute triangle. Looks the same as the first triangles but let's compare to be sure.

(superimpose as did the first two triangles to show they are equivalent)



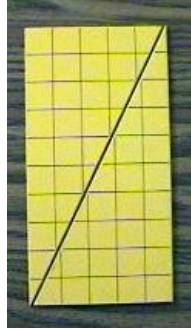
This time suppose we do this.

(Take 2 top pieces off the top and fold onto the sides.

Dividing Base

(using second acute triangle divided into 2 at the height line)

I wonder if we can use just this triangle to find the area.



(separate and flip the first over and upside down, laying it next to first, to form rectangle)

Now can I find the area? (yes)

What shape do I have here? (rectangle)

All we have to do is count base and height and multiply. We see it is again equal 50 units squared. This time, what did I do to my original triangle?

I divided the base!

Epilogue

Leave the other material neatly set aside when you are not using them.

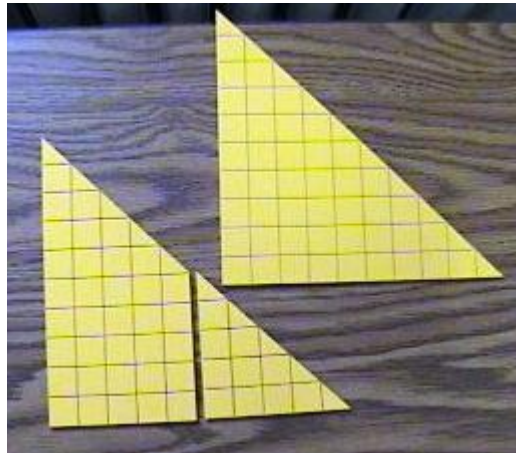
December 15, 2004

Right-Angle

Begin

Doubling Area

(laid on table – one whole and one in two pieces – handle them as two whole pieces)



It's decorated with squares, but we have bits and pieces of squares to count. Too tough.

In my box, I have another right-triangle. (place next to the first)

The two look the same but we'd better make sure. (superimpose one onto other)

We can say they are equivalent, right?

The first thing we can do is this. (turn 180 degrees and put the 2 together)

(join hypotenuses of the two right triangles to double triangle area in combined square)

Now can we find the area? What will we multiply? (base and height)

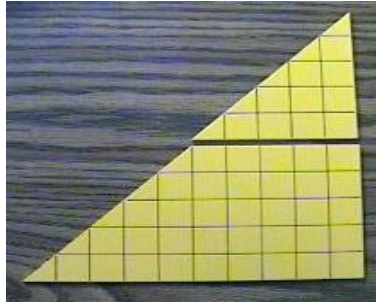
That will give us 100 units squared but is that the area of one triangle? (no)

So what do we do? (divide by 2 since there are 2 triangles)

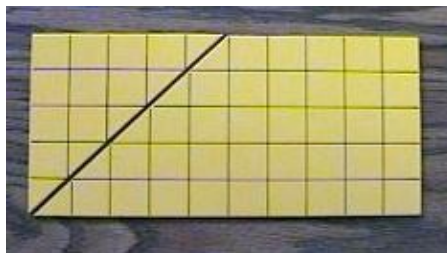
Dividing Height

Remember we saw that this is not the only way to find the area.

(keep only the divided triangle on the table)



Can we find the area with only this single triangle? (join hypotenuse sides of both pieces)



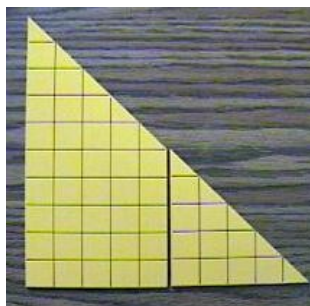
Now we can find the area. What's the area? (50 squared units)

What did I do to my original triangle here? (divided the height)

Dividing Base

(we only have two right triangles, so we use the second in pieces and turn it 90 degrees)

We only have this one triangle. Can we find the area?



Well watch what we can do

(rotate piece from bottom right to upper right corner of new rectangle)



What do you think the area is now? (50 squared units)

What are we doing to our original triangle? (dividing the base)

Epilogue

Show equivalence of beginning figures as always. Discuss whether squares can be counted (i.e., are there bits and pieces making it impractical)

December 15, 2004

Obtuse

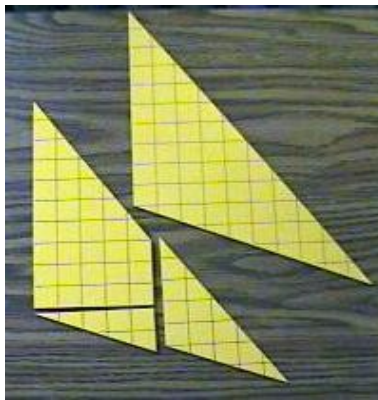
Begin

Doubling Area

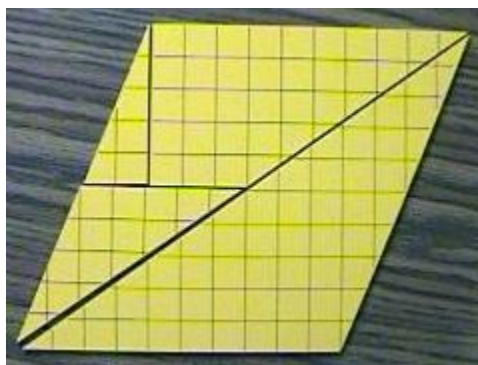
What do we have here? (obtuse-angled triangle)

I want to find the area. Can I do that? (no, bits and pieces)

In my box I have another obtuse triangle. They look the same but let's check.
(superimpose) Yes, they are equivalent.



Now suppose we do this. (make a parallelogram with the two triangles)



Remember when we worked with the rhombus – a special kind of parallelogram?

So what we saw with that rhombus and rectangle is going to help us here too.

A parallelogram and a rectangle – like a rhombus and rectangle – are equivalent if the

bases and heights are the same; we know they are equivalent if so.

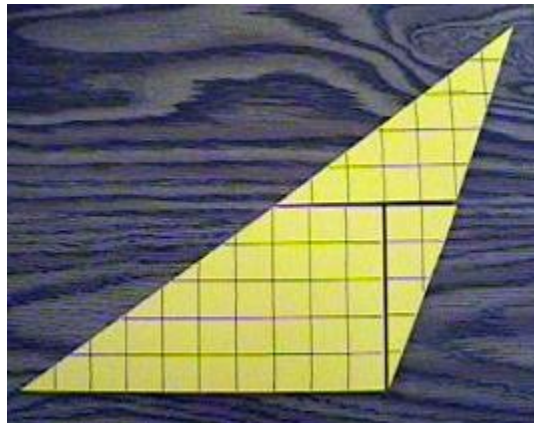
Can we find the base and height? (we can count them – 10)

How about the height? (count upward from base – 10)

So the area of our parallelogram equals 100 squared units. This means the area of our triangle is how many (50). Right, we have to divided our product by two since we have two triangles here.

Dividing the Height

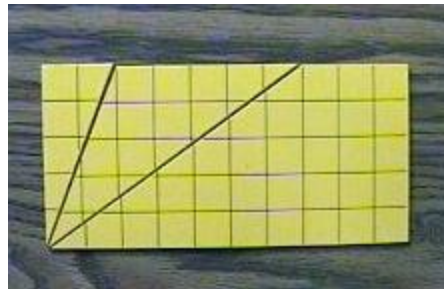
(keep divided triangle on the table)



I can't measure the squares because of the bits and pieces, can I?

But I can do this.

(rotate two pieces to form rectangle)



What did I do this time? (divided the height)

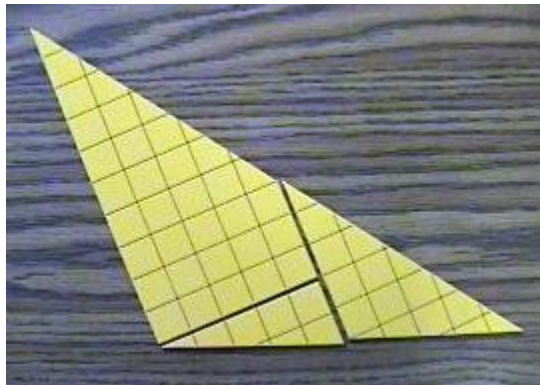
We divided our height externally – it was an external height.

(use the right triangle next to the original one if height is not perceived by the child)

(repeat movement of triangle pieces to the rectangle to reinforce height work)

Dividing Base

(keep only the divided triangle - turn it to use divided side as base)

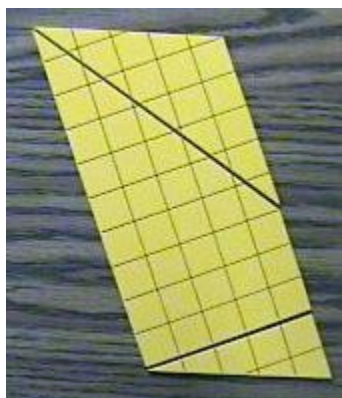


We want to find the area. But I can't move the rest of my base to divide it!

I can't even count my squares.

Since the vertical lines are at an angle, we can't count the height.

We can't count the squares since we have bits and pieces! What can we do?



We can move the acute vertex at the base upward to the top to form a parallelogram!
This allows us to use base and height again, doesn't it?

And we can conclude from our previous work that the area is again 50.

Epilogue

Show the children how to put materials away.

January 13, 2004

Deriving Formulas for Area - Yellow Material

Prologue

The children should have worked with the yellow material when we gave the concept of area. They've explored rectangles, parallelograms and triangles – with this material and with graph paper. With iron material, they've explore trapezoids and decagons.

Hopefully, they've realized there are two important lines here: base and height. With those, we can find area by multiplying. Up to this point, they've not used formulas. Now we will get to formulas. We'll derive them with the yellow material.

As we work, try to have only the tickets we are using, with the rest in a mixed up order and set aside.

Material

Yellow material.

An envelope labeled 'area' having tickets: one with capital 'A', a '+' and '=' sign, the number '2', a capital 'B' and 'D', lowercase 'a', 'b', 'd', 'h' and 'p', two division bars and two parentheses.

Begin

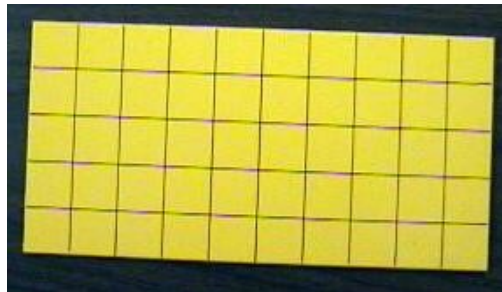
(place yellow materials box lower left)

You remember we worked with this material before finding area, right?

Rectangle

We want to look at it again today beginning with this

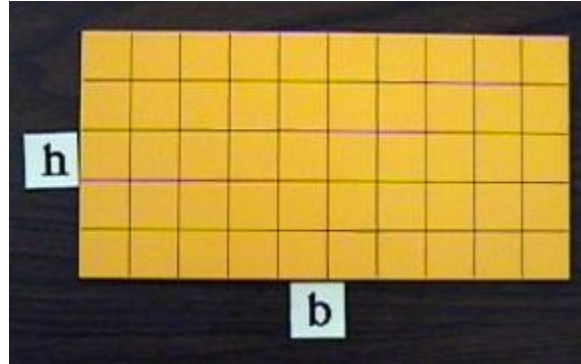
(rectangle – one of the small 4 – with squares)



What shape is this? (rectangle)

There were a couple important lines we looked for. What were they? (base and height)

Can you show me those lines?



Let's ticket those lines. (place lowercase b on the base and lowercase h at left side)

Now what did we do when we wanted to find area? (multiply base times height)

So we can say that area equals base times height

(build that expression just below the rectangle “ $A = bh$ ”)



Area equals base times height. (restate the completed formula)

This is only one shape. We have others.

(move tickets to the side, mixed up again)

Parallelogram

(place parallelogram on table, move rectangle next to box)

What shape is this? (parallelogram)

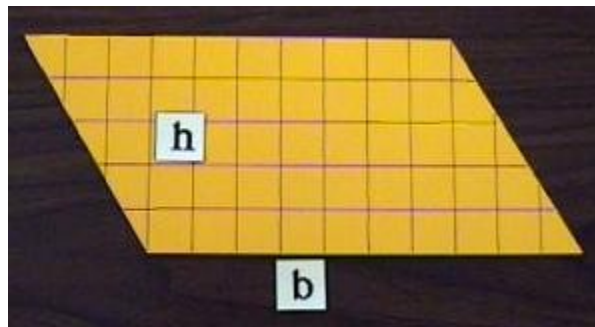
We found we could find the area of a parallelogram too.

What were the important lines for this one? (base and height)

Where do we find the base of our parallelogram? (point to base) Here.

Where do we find the height of our parallelogram? (point to a vertical line) Here.

(ticket b and h)



What did we do with base and height using a parallelogram? (multiplied)

So let's multiply the base and height of our parallelogram.

(build expression below again: " $A = bh$ ")

This seems to be the formula that works for these 4-sided figures.



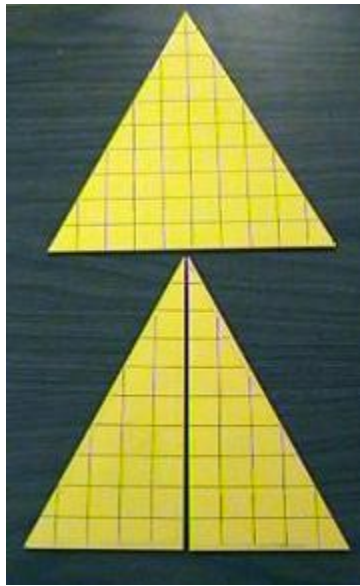
Acute Triangle (Doubling Area, Split Height and Base)

Let's discuss triangles!

(Place acute on the table)

What kind is this? (acute)

(place the acute split vertically – below the first whole one – they are aligned on table)



What is the relationship between these? (equivalent)

We verified that by placing one on the other, right? (do so)

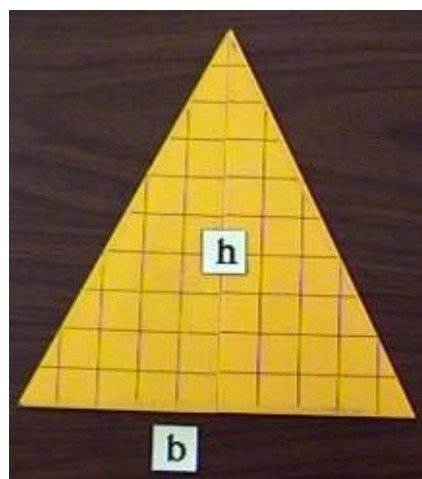
We also talked about important lines with triangles.

What were they? (base and height)

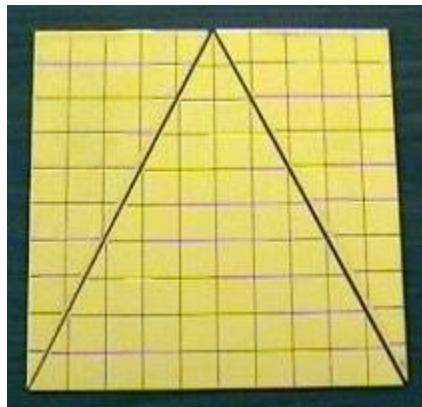
Where is the base? (point to bottom edge) Right here.

Where is the height? (point to middle vertical line) Right here.

Let's ticket those lines (place b and h tickets)

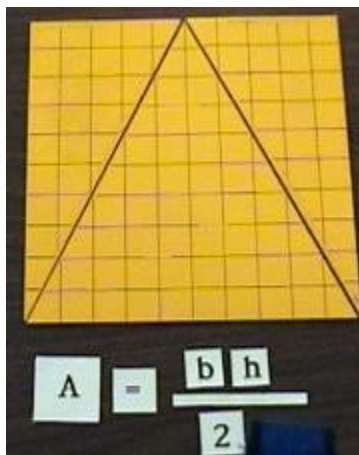


Remember the first thing we did to find area of this triangle was this (double the area forming a large square – put the two together)



Now can we find area? We can say area equals b times the h – but will that give me the area of the triangle? (no) No, I doubled the area. So what do I need to do? (divide by 2)
(build ticket expression while speaking: “ $A = bh / 2$ ” use division bar, fractional notation)

(height is not on the edge necessarily)



So this is one formula used to find the area of a triangle.

We saw we could also find the area using just one triangle too.

(take last acute triangle in 3 pieces)

Remember we also have this acute triangle.

We saw this triangle has what relationship to the other two triangles? (equivalent)

Again, what are the important lines? (base and height)

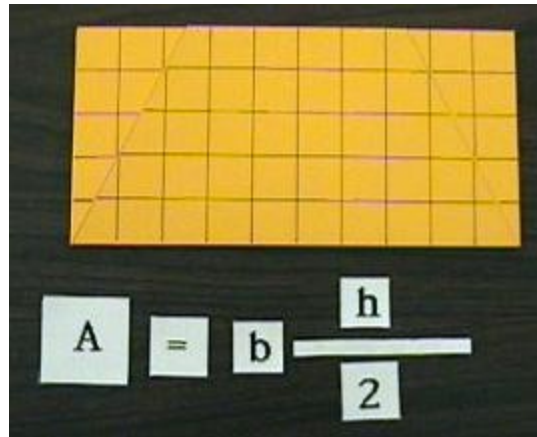
Who'd like to ticket them? (child does so at this point)

With this triangle, we can do this.

(move top of triangle to top two corners of newly formed rectangle)

Now we have a figure we can find the area of. What are we going to multiply?

What have we done here? (we split the height in half)



So, to express the formula for area we say area equals the base times half the height

(form $A = b \frac{h}{2}$ – using division bar under h with 2 directly beneath)

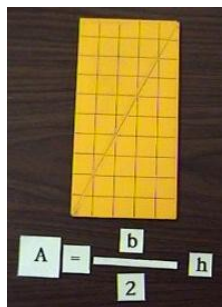
Split Acute Base

There's one more way to find area here too.

(place acute triangle in 2 pieces on table – others set aside)

Who can ticket base and height for me? (child does so)

This time to find the area we did this. (fold second piece to form a rectangle)



What part of my triangle did I divide this time? (base)

So, to express the formula for area we say area equals the base divided by 2 times the height (form $A = b/2 h$ – using division bar under b with 2 directly beneath)

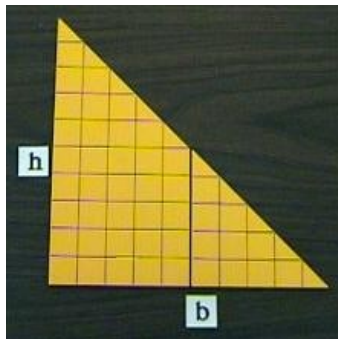
We have two more triangles, don't we. Let's do the same thing with each of them.

Right Triangle

What kind of triangle is this?

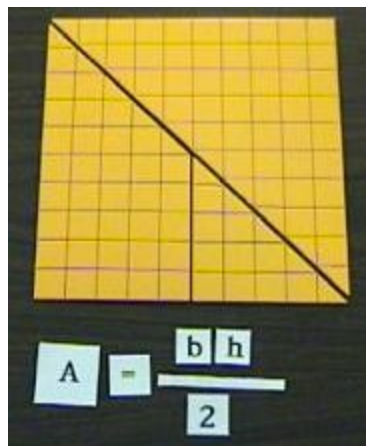
What are the important lines? (base and height)

Where is the base and height – can you ticket them?



We have this other right triangle too. What's the relation between them? (equivalent)

The first way we found area was to do this (double the area using both triangles for square)

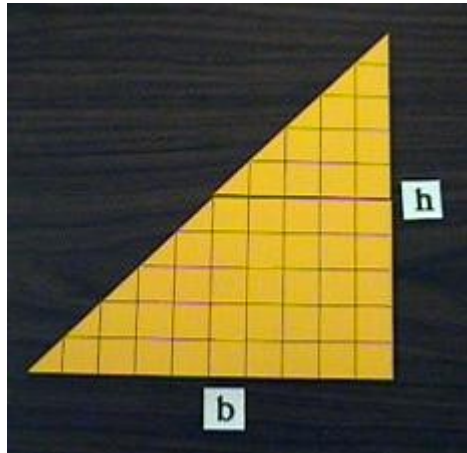


So, to express this formula, we'll say area equals what? (base times height, divided by 2)
(form $A = bh / 2$ – using division bar under bh with 2 directly beneath)

Split Height

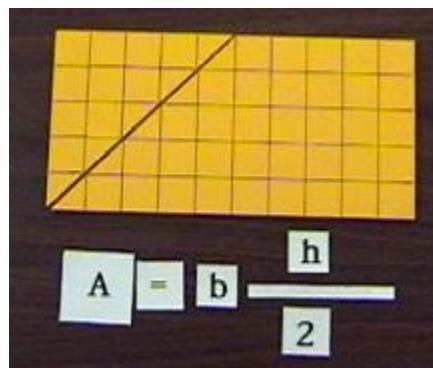
We can also use just one triangle. (use triangle in two pieces to split height)

Who can ticket base and height for me? (child does so)



With this one, we can do this. (rotate top vertex to corner to form rectangle)

Now we have a figure we can find an area of. What did we do to our triangle though? (divide height)

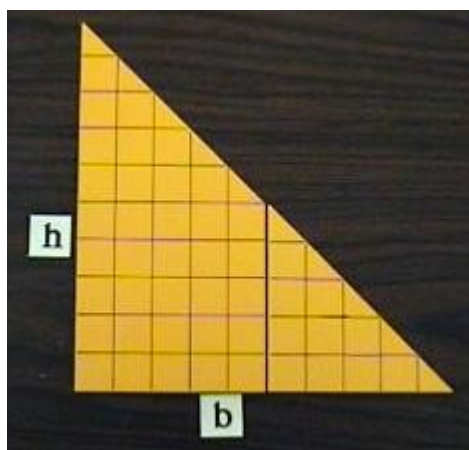


So to express this formula, we'll say area equals what? (base times, height divided by 2)
(form $A = b \cdot h/2$ – using division bar under h with 2 directly beneath)

Split Base

(reorient triangle to split base)

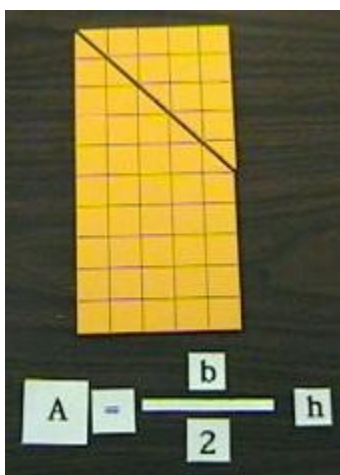
Let's label this triangle. (child does so)



With this one, we can do this. (rotate bottom vertex on hypotenuse to top corner to form rectangle)

Now we have a figure we can find an area of. What did we do to our triangle though? (divided base)

So to express this formula, we'll say area equals what? (base divided by 2, times height)



(form $A = \frac{b}{2} h$ – using division bar under b with 2 directly beneath)

Obtuse Triangle

What kind of triangle is this? (obtuse)

Remember we had another one too. What was their relationship? (verify as equivalent)

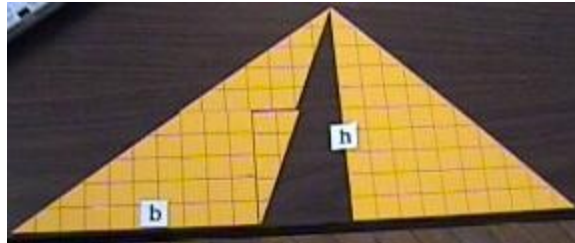
What are the two important lines here? (base and height)

Where's the base here? (point to bottom edge)

Let's ticket that.

Where's the height here? (point to external height – place a straight edge from top vertex)

Let's ticket that here (on table). That's the height – out there and external.

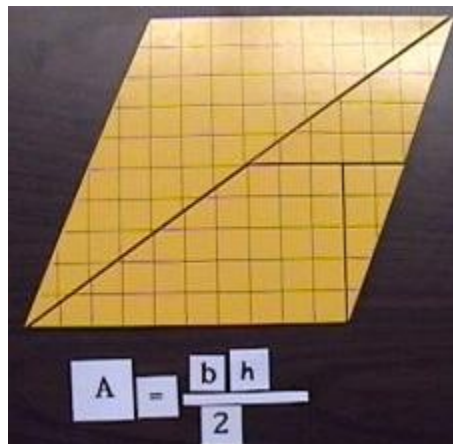


The first way we found area was to do this (double the area)

Now I can move the height ticket internal to the figure. (move ticket to center line of the two)

What do we do to find area.

(base times height – all divided by 2)



Let's ticket that. (area equals base times height, all divided by 2)

(form $A = bh / 2$ – using division bar under bh with 2 directly beneath)

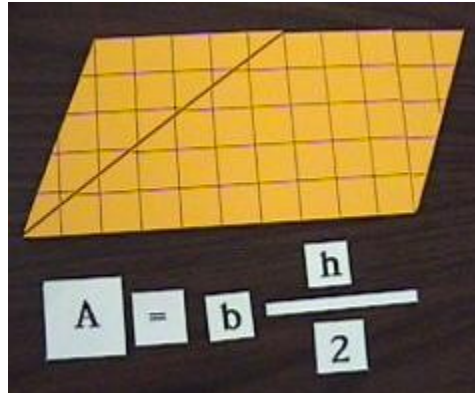
We can also find area with just one triangle too. (position obtuse to split height)

Who can label the base and height for me? (child does so)

Our height is external again – so we can use the straight edge of our right triangle to show the height and label it externally again.

But now we can do this. (form rectangle as in earlier lessons)

What did we do to height? (divided it)



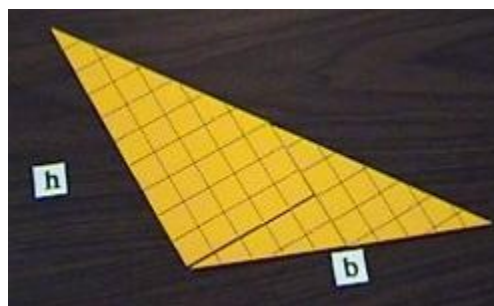
Let's form that expression – area equals base times, height divided by 2)

(form $A = b \cdot h/2$ – using division bar under h with 2 directly beneath)

We can also find area with this triangle another way too.

(position obtuse triangle to split base)

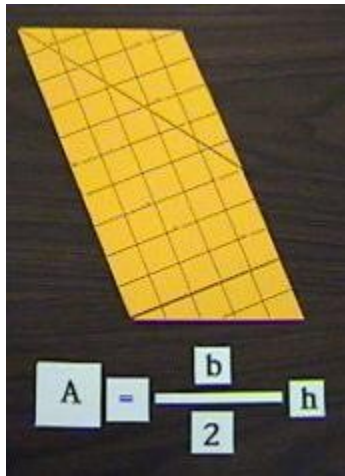
Who can label the base and height for me? (child does so)



Our height is external again so we label it externally again. (ticket base and height)

But now we can do this. (form rectangle splitting base - move to top - form parallelogram)

What did we do to our base? (divided it)



Let's form that expression – area equals base divided by 2, times height)
 (form $A = \frac{b}{2} h$ – using division bar under b with 2 directly beneath)

Summary

Three triangles – and each time 3 formulas!

Epilogue

Through this work, the children become familiar with the formulas. This is a good place for word problems, where they can plug in variables to solve. If they're exploring geometry, they've come across the formula for the area of a square (S^2) and they can begin to use that too.

For follow-up, the children can play games. One child can choose the pieces, set them up, do the manipulation and challenge the friend to make the formula. Then switch, taking turns – to build in repetition. Handwork on graph paper may continue too.

January 13, 2004

Deriving Formulas for Area - Iron Material

Materials

Iron material insets. Envelope of tickets labeled 'area' having tickets: one with capital 'A', a '+' and '=' sign, the number '2', a capital 'B' and 'D', lowercase 'a', 'b', 'd', 'h' and 'p'. We also have two division bars and two parentheses.

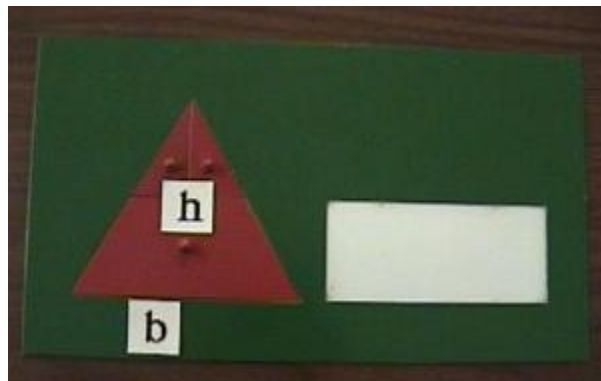
Begin

Let start with this plate. (left side triangle, right side empty rectangular space)

Triangles and Rectangles

What are some of the lines we know about a triangle? (base and height)

Where do we find the base? Where do we find the height?



What about our rectangle?

What are some of the lines we know about for a rectangle? (base and height)

Where do we find the base? Where do we find the height?

What do we know about the relationship between them (equivalent)

We show that by putting the pieces of the triangle into the rectangle. (do so and the put back)

Now let's ticket the base and height of the triangle. (place tickets b,h right on top of triangle)

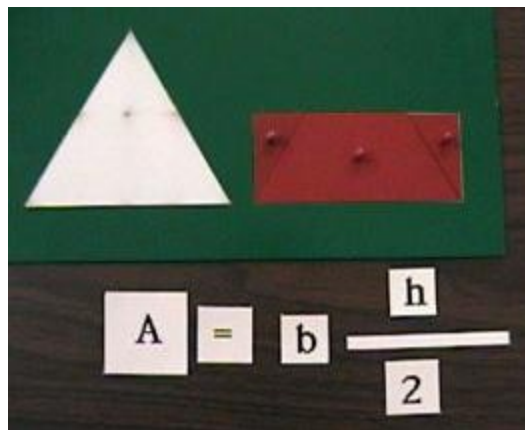
We know where the base and height of the rectangle are.

We know to find the area of the rectangle, we multiply base times height

When we put the pieces from the triangle into the rectangle, what happens to the base of the triangle? (becomes base of the rectangle)

What happens to the height of the triangle? (divided in half)

So to express the formula for finding the area of the triangle, we can say that area equals what?



Area equals base times height divided by 2.

(form expressions $A = b h/2$, using division bar and 2 directly under h)

Rhombi and Rectangles

What figures do we have here (rhombi and rectangles)

What is their relationship? (equivalent)

What lines do we know about these figures? (base and height)

Where's the base of one of the rectangles? (child points to a base)

Where's the height? (child points)

Where's the base of our rhombus? (child points)

Where's the height of our rhombus? (child points)

Let's ticket them. (do so)

What's the relationship between the bases and heights of our rhombi and rectangles (same)

How do we find the area? (multiply base times height)

Area equals base times height. That's the same as our formula for a parallelogram.
(form expression $A=bh$)

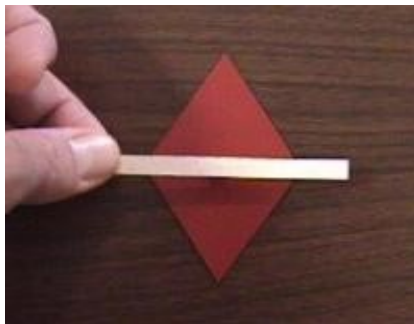
The rhombus is a special parallelogram so this formula works.

Different Rhombi Orientations

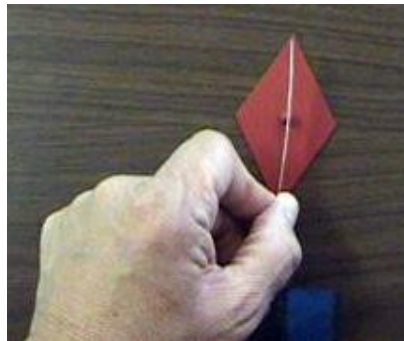
But a rhombus is not always in this position. Suppose it's like this (oriented like a diamond on table)

Like this it's not so easy to see base and height, is it? But remember we learned special rhombus parts. (use a division bar to hold across the rhombus)

If I hold my bar here what is this called? (minor diagonal)



If I hold my bar here what is this called? (major diagonal)

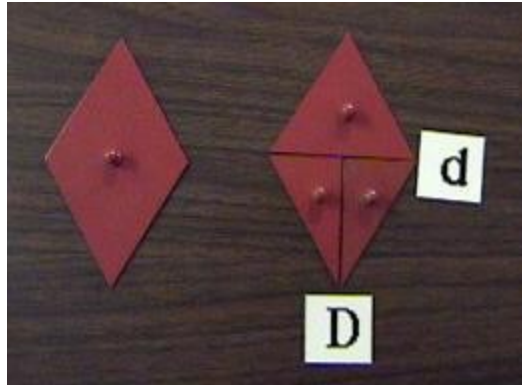


Split Rhombus Major

I'll build another rhombus like this one but from these pieces. (2 right and 1 equilateral)

What did we call this line (minor) Let's label this (small d ticket)

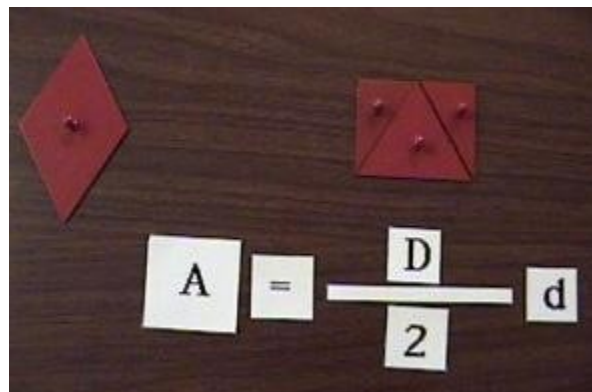
What did we call this line (major) Let's label this (cap D ticket)



Watch these lines as we do this.

(rotate corners upward to reform the rectangle)

Now I have a figure I can find the area of.

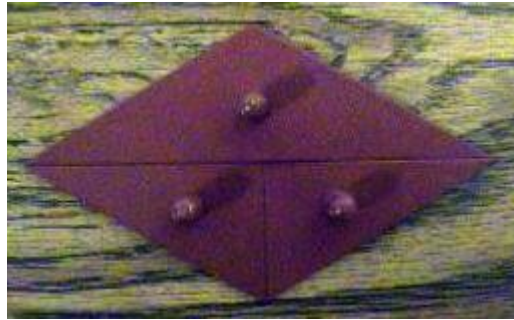


I can say area equals minor diagonal times half the major diagonal.

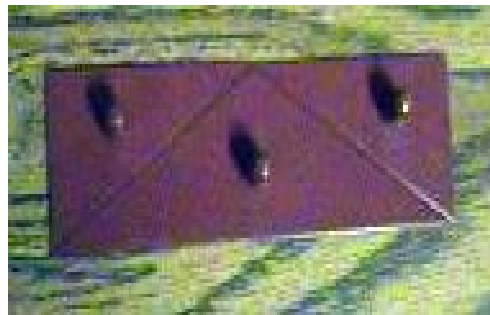
(form expression $A = d \text{ times } D/2$, using division bar and 2 directly under D)

Split Rhombus Minor Using Scalene

Let's build another rhombus like this.



Keep an eye on the minor diagonal as we transform it into a rectangle.



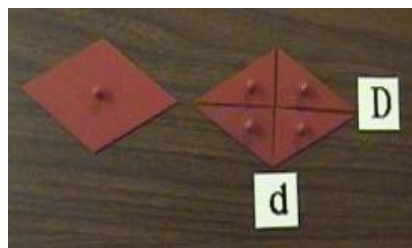
By moving the two pieces, we can make our rectangle and we can find area.

We can see that area equals major diagonal times, minor diagonal divided by 2

Let's look at it using another rhombus with four pieces and split the minor diagonal again.

Where is the major diagonal? (point to major diagonal) Let's ticket it.

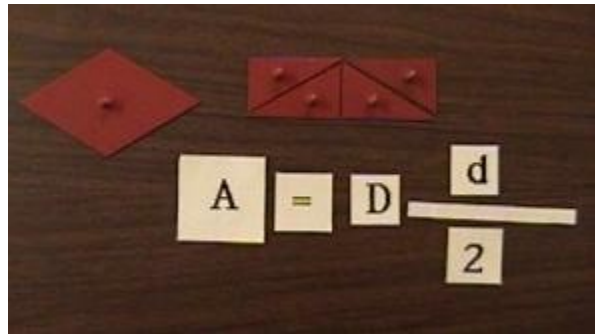
Where is the minor diagonal? (point to minor diagonal) Let's ticket that here too



Keep an eye on those two lines as we transform it into a rectangle.

This time the diagonals are used differently.

Area equals major diagonal times, minor diagonal divided by 2



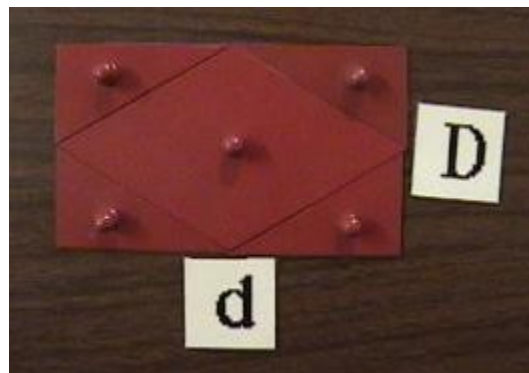
(form expression $A = D \text{ times } d/2$, using division bar and 2 directly under d)

Double Rhombus

What part of the inner rhombus gives us base (major diagonal)

What part of the inner rhombus gives us height (minor diagonal)

Let's ticket those.

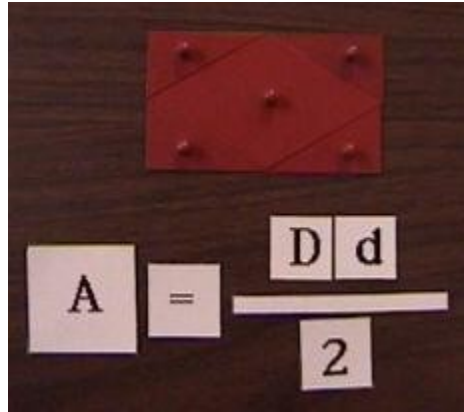


Normally we use base times height.

To express that here is major diagonal times minor diagonal.

But will that give us area. (no, we doubled it)

So we have to divide it by 2.



Area equals major diagonal times minor diagonal, divided by 2)
 (ticket $A = Dd$, all over division line and 2 underneath)

Trapezoid

What do we have here (trapezoid, divided trapezoid and empty rectangle)

What was their relationships (verify equivalence)

So all of these shapes are equivalent.

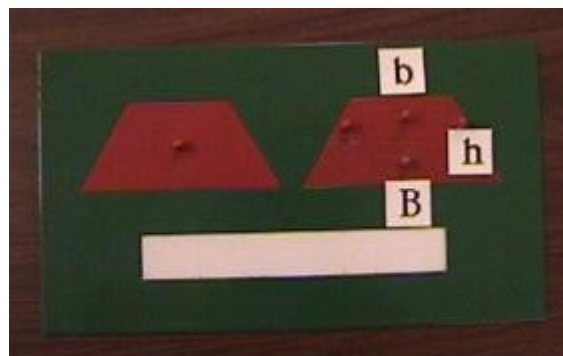
What are the parts of a trapezoid that we learned? (major and minor bases)

Where is the major base? (longer base)

And the minor base? (shorter base)

What other part can we find? (height)

Where is that? (perpendicular line from major base to minor base) Let's ticket those lines.



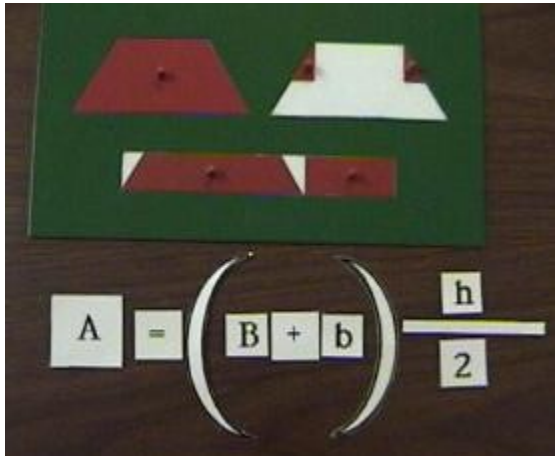
Now let's put the cut-up trapezoid into our rectangle.

What part of the trapezoid makes up the base of the rectangle

We can say major base plus minor base of trapezoid gives us the base of our rectangle.

What part of the trapezoid makes up the height of the rectangle (half the height)

We can say one-half the height of trapezoid gives us the height of our rectangle.



Move top corner pieces inline to emphasize half the height is used.

We can say the area equals major base plus minor base, times half the height.

$(A = (B+b) \cdot h/2)$ -- division bar and 2 directly under h)

Decagons And Tall Rectangle

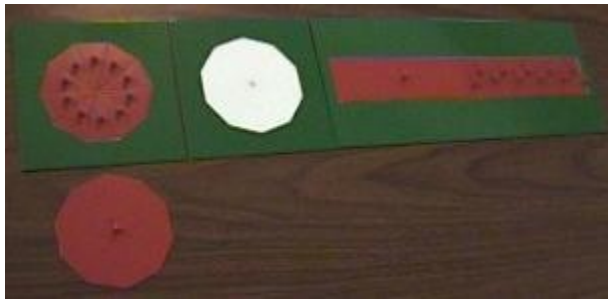
What kind of shape do we have here? (decagons)

Decagons allow us to look at formulas for regular polygons of 5 or more sides. They all use the same formula.

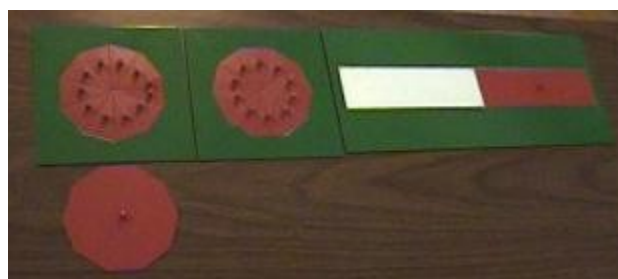
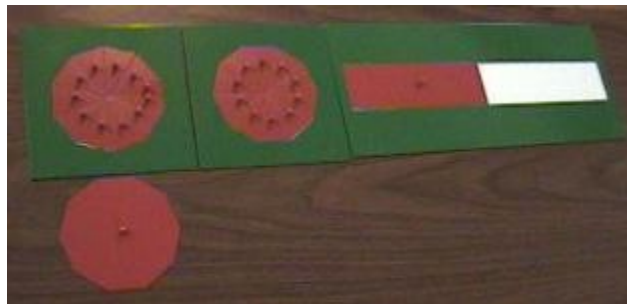


What's the relationship between these (equivalent)

Let's remind ourselves of that (verify by moving pieces between the decagons)



Remember we saw this rectangle is equivalent to the other half which was divided into triangles.

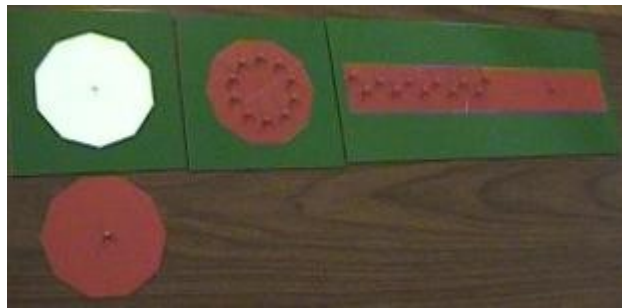


I can move this rectangle from the left to the right side to show that.

What was the relationship between the decagon and the rectangle?

We saw we can build all the triangular pieces from our rectangle into the decagon.

And if we put these decagon pieces back into our space (left side) in our rectangle space we again see the decagon and this rectangle on the left are equivalent too.



Remember, we use base and height – where do we find base of our rectangle (bottom)

Where do we find the height of our rectangle? (along side)

What parts did we learn for a decagon? (radius, apothem, center, perimeter)

Who can show me a radius? (child shows line between pieces)

How about the apothem? (child shows apothem - line from center of piece to center)

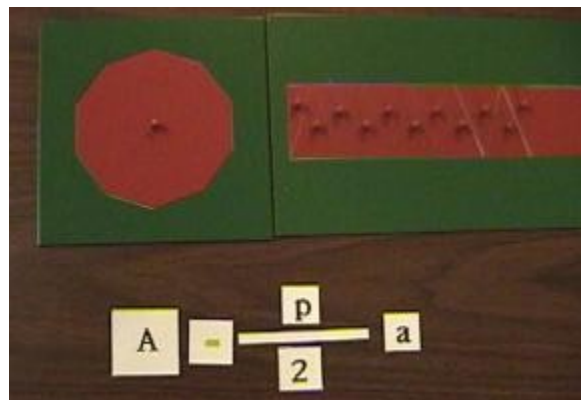
In our solid decagon, can you label our perimeter? (do so)

Let's look at the relationship between rectangle and decagon.

What part of the decagon give us the base of the rectangle (half perimeter)

What part of the decagon give us the height of the rectangle (apothem)

So we arrive at this formula



Area equals half the perimeter times the apothem.

($A = p/2 a$ – with division bar and 2 directly under p)

Decagon and Long Rectangle

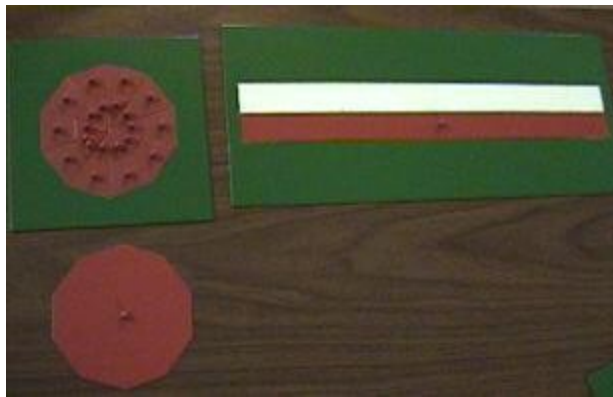
Now we also have this place (the thinner longer rectangular plate)

What's the relationship between this rectangle (bottom whole) and the decagon.

We saw they were equivalent.

And we saw that this rectangle is equivalent to the other one divided into pieces (hold to compare them)

Let's build divided rectangle pieces into our decagon.



So we see the divided rectangle is equivalent to the decagon.

Let's move the whole rectangle piece up and return the divided one onto the bottom space.

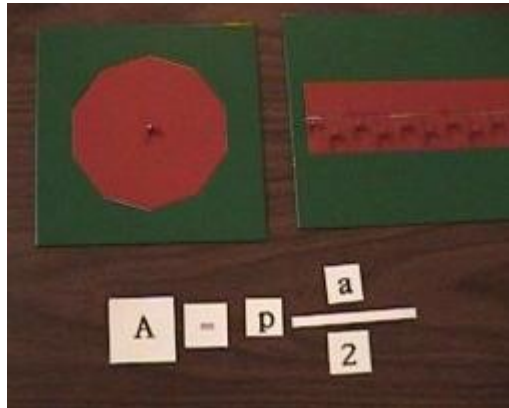


What are the important names we're using (perimeter and apothem)

What part of our decagon gives us the base of our rectangle (perimeter)

What part of our decagon gives us the height of our rectangle (half the apothem)

So we arrive at this formula



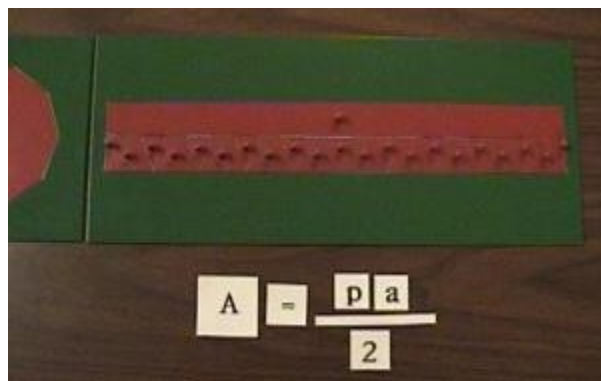
Area equals the perimeter times half the apothem.

($A = p \cdot a/2$ – with division bar and 2 directly under a)

But what if we wanted to consider this whole rectangular space – both parts.

Can we, using this whole rectangular space, express the formula for the **area of a regular polygon**?

(still uses a and p even though we've used rectangle to derive relations)



Yes, area equals ap, divided by 2.

Why we divide by 2? (because we used the entire rectangular space to find the area for one decagon – base of rectangle is perimeter of decagon, height is the apothem – base times height gives area of two decagons)

Epilogue

Word problems are great follow – give the children opportunities to use these formulas. We have a pentagon in the material, so they should be able to measure and find perimeter and apothem – and use them to create equivalent decagons.

January 14, 2004

Special Polygons, Circumference and Area

Nomenclature of the Circle

Age

Lower elementary.

Materials

Planar board, newsprint or piece of similar paper, box of geometry sticks, the yellow paper circumferences, fraction insets – $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{6}$. Plate from iron material (circle with triangle inscribed) and a pencil.

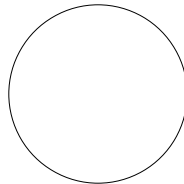
Prologue

This is the one plane geometry figure we've not seen yet. This follows the same pattern as other nomenclature work. Find out what the children know, give a lesson on what they don't know but need and then make sure there is some variation of a 3-period lesson. They may have encountered some of the terms in the primary classroom. Begin with a small group.

Begin

(fasten stick horizontally on the board at one end and use the other to pencil in – through the eyelet - a circle)

This whole shape, we call a **circle**.



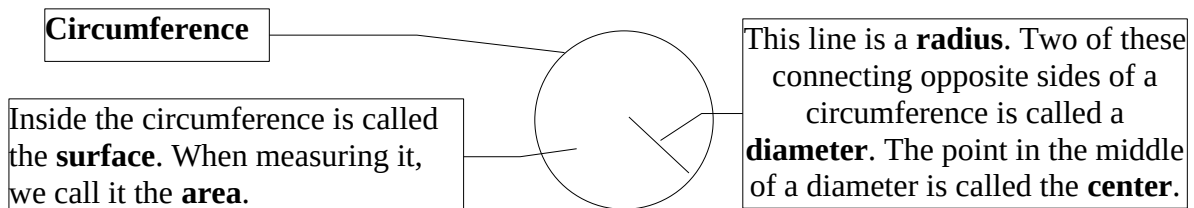
This line on the outside has a special name – called a **circumference**.

What do we call this? (**center**)

This line from the circumference to the center has a special name. It's called the **radius**.

Two radii that go in a straight line are called the **diameter**. (point each time to the circle)

The space inside is called the **surface** or the **area**. It's the area of the circle if measured.



(do second-period as long as we need it)

I have a circle like this (present circle inset $\frac{1}{2}$)

When I have half of it, it's called a **semi-circle**.

(point to the $\frac{1}{2}$ inset)



This part of the circumference is called a **semi-circumference**. (point to edge of $\frac{1}{2}$)

Any time we have a part of a circumference, like here (point to $\frac{1}{4}$ th piece)
we call it an arc. This is an **arc**.



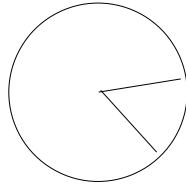
Refer to edge of inset piece

When you have a piece of a circle that goes from the center to the circumference, we call that a **circular sector**.



(point to the $\frac{1}{6}$ th inset)

I can make a **circular sector** on my circle here. What do I add? (add a second stick -radius)



Epilogue

The children can make these for themselves, with sticks, graph paper, on charts or in booklets. They can label individual parts of the circle.



Begin

(review terms given in previous lessons on circle nomenclature)

(create circle on paper, use stick as a compass-like tool, pencil in the eyelet of the stick)

Suppose I have some lines like this.

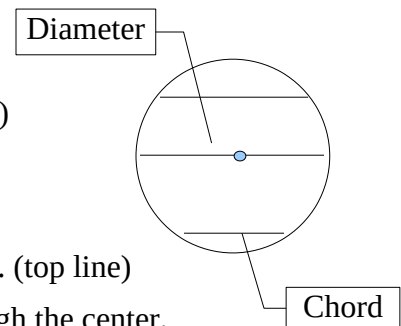
(3 sticks are laid across horizontally – one through the diameter)

This line goes through the center – it's called the **diameter**.

This line does not. But it has a special name. It's called a **chord**. (top line)

It begins and ends on the circumference but does not pass through the center.

And this line is called a chord. (bottom line)



So what do we see here? (point to the circle inset with triangle inset, triangle sides)

It's a chord too.

How many do we have? (3)



And I have this, part of a circle cut by a chord – is called a **segment**. It's the part near the circumference, cut by a chord.

Epilogue

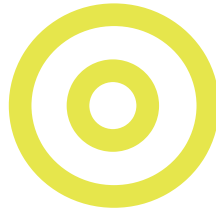
Segments are smaller than semi-circles. Let the children explore these materials. Same follow-up applies as the first nomenclature of circle presentation. Give them interesting paper to use when creating circular shapes with scissors – let them write their own definitions.



Begin

(lay circumference paper pieces on the board – 2 concentrically placed)

When I have 2 circles like this, this area (point to space between the 2 circumferences) is called a **concentric circular sector**.



Epilogue

Children should have learned concentric circles in primary but may need review.

January 14, 2004

Relationships Between Line Segments & Circumferences, Two Circumferences

Material

Two yellow paper circumferences (made from durable material but not too stiff; use a geometry stick to measure the circumference – which is actually the inside of the cutout – remember which sticks you used to make them)

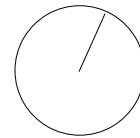
Begin

(lay out circumference)

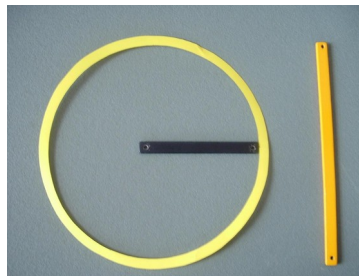
What is this? (circumference)

I want to know where my center is. I'll mark it with a radius.

(use stick end on circumference and other in center – eyeballed)

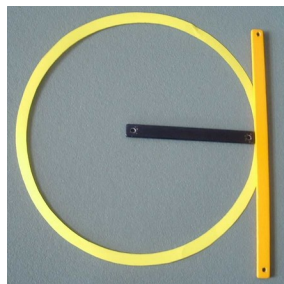


I have a line and I put it here (place new stick vertically to right of the circumference)

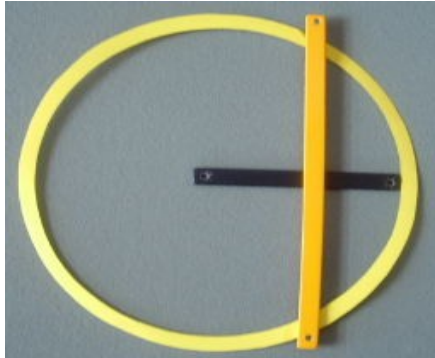


We say it is **external to the circle**.

If I have a line here like this (move that stick so it touches the circumference edge) we say it is **tangent**.



If I have a line here like this (move that stick so it intersects the circumference at two edges and continues outward at both ends) we say it is **secant**.



(do second-period)



Begin

I have a line like this. What's it called (external)

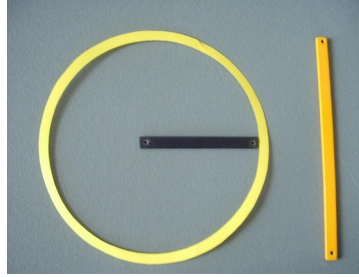
Where is the center of the circle?

What do we call this? (radius)

What's the relationship between the center of the circle and this line, relative to the radius?

(we have the 'external' image on the table)

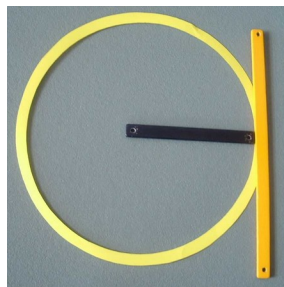
Is it the same as the radius? (no)



Is it longer or shorter?

We could say the distance from the center of the circle to the external line is greater than the radius of the circle.

(move to tangent position)

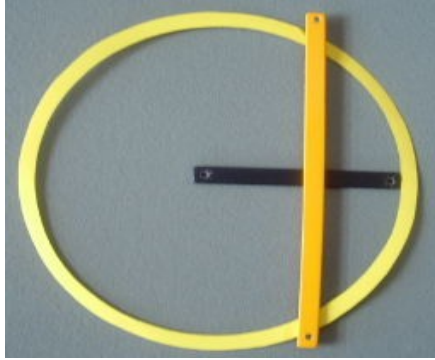


What can we say about the distance from the center to a line that is tangent, relative to the radius?

So when a line is a tangent the distance from the center of the circle to the tangent line is equal to the radius.

(move to secant position)

What can we say about the distance from the center to a line that is secant, relative to the radius?



It's the same as the radius? (shorter)

So **when a line is a secant the distance from the center of the circle to a secant line is less than the radius.**



Begin

(place 2 circumferences, lay first on the table)



Two circumferences can have a relationship too.

(set next to each other but separated – one left, one right)

We call these two circumferences **external**.

(make the 2 tangent)

What if I put them like this so their inner edges touch.

(inside edges touch so there is a slight overlap of the two shapes)

Is one inside the other? (no) But they touch, right?



So this circumference is externally tangent to this circumference.

(place them concentrically)

These two circles are still tangent.



But now they are **internally tangent**.

(place them intersecting)



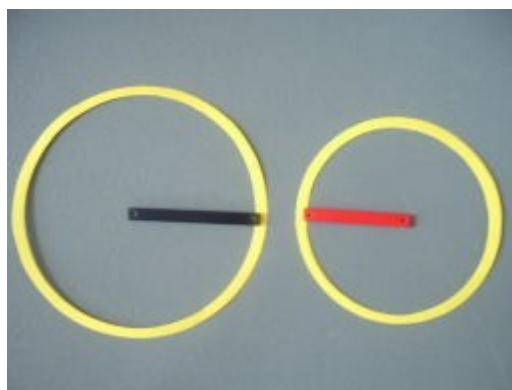
These two circles are **secant**.

(so a second-period)



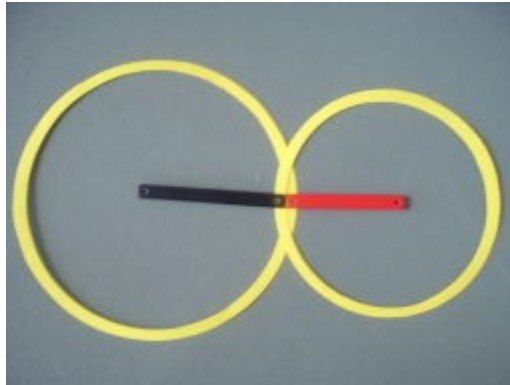
I have two circumferences. What is their relationship? I want to know where the center is so I'll place a radius on each one. (place radii - large radius is black, small radius is red)

What can we say about the distance from the center of one circle to the center of the other circle, with respect to the radii?



The distance is greater than the sum of the radii.

(keep radii in same orientation but place circumferences externally tangent)



What can we say about the distance from the center of one circle to the center of the other circle, with respect to the radii?

The distance is equal to the sum of the radii.

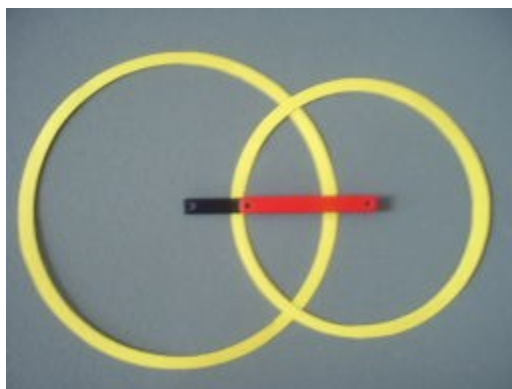
(keep radii in same orientation but place circumferences internally tangent)



What can we say about the distance from the center of one circle to the center of the other circle, with respect to the radii?

The distance is equal to the difference between the radii.

(keep radii in same orientation but place circumferences secant – intersecting)



What can we say about the distance from the center of one circle to the center of the other circle, with respect to the radii?

The distance is less than the sum of the two radii.

Epilogue

Nomenclature is useful in future work. This work also helps us proving geometric theorems later on. They can draw, label and create booklets as usual for repetition.

January 14, 2004

Area of a Circle

The Circle as a Special Polygon

Age

Upper elementary.

Material

Polygon drawer from geometric cabinet, frame from the unit of the circle fraction insets; next lesson uses a piece of newsprint, pencil, ruler, 3 small circles from iron inset materials and a piece of chalk.

Prologue

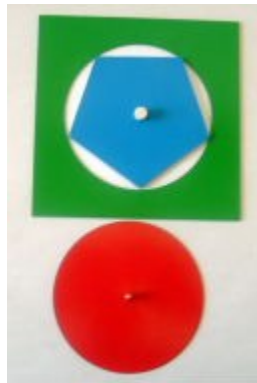
This is another small group lesson.

Begin

Here I have a circle cutout in this frame (hold up empty unit fraction inset)

I want to fit the polygons from this drawer into this.

Let's look at this pentagon. Look at the spaces here (pentagon is put into the circle inset)



The spaces are kind of big!

Let's look at this hexagon. Look at the spaces here (hexagon is put into the circle inset)

Let's look at this heptagon. Look at the spaces here (heptagon is put into the circle inset)

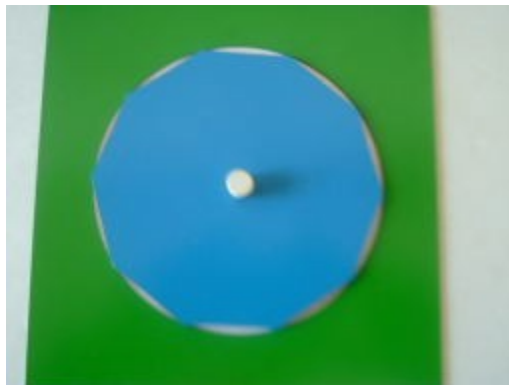
Let's look at this octagon. Look at the spaces here (octagon is put into the circle inset)



What do you notice about the spaces? (they're getting smaller)

Let's look at this nonagon. Look at the spaces here (nonagon is put into the circle inset)

Let's look at this decagon. Look at the spaces here (decagon is put into the circle inset)



Look how small the spaces are now.

Can you imagine how small those spaces would be if we had 20 sides? Or 50 sides! 1000 sides? Or even a million sides! What if we have an infinite number of sides?

Then it would be a circle. A circle is a special polygon having an infinite number of sides.

Epilogue

This lesson helps children see that a circle is a special polygon with an infinite number of sides.

Measuring the Circumference

Material

The unit of the circle fraction insets, a piece of newsprint, pencil, ruler, 3 small circles from iron inset materials and a piece of chalk.



Begin

What do we call the line around a circle? (circumference)

(circle (3) insets at right, paper on table in front of teacher)

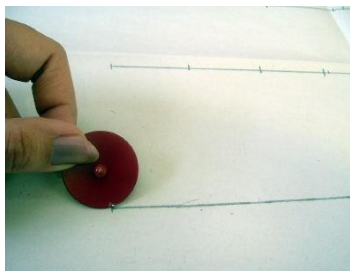
I wonder how we could measure the length of the circumference.

(draw 3 horizontal lines, using a ruler, onto the paper – same distance of 1 cm. from the edge – about 12 inches long)

(mark chalk at edge of circle to show beginning and ending point of circumference)

I'll take my circle and measure the circumference, marking with my pencil at the end.

I'll place the chalk mark on the line right at the very beginning. (roll circle along the line)

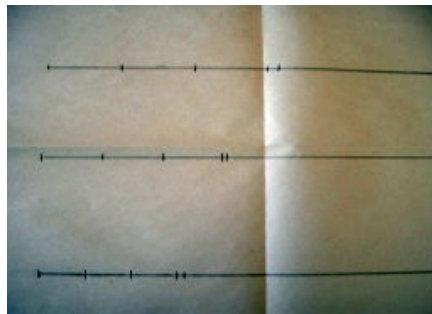


When my chalk mark is back on the line, that marks the end of my circumference.

I also want to measure the number of diameters that make up the length of my circumference. (set circle diameter on line and mark each diameter endpoint on the line)

(allow children to repeat procedure for other two circles – chalk mark on circle edge, roll inset on its edge – marking end of circumference and each diameter point.)

(hold up circle on the line, make each diameter along the line, moving circle each mark)



Mark each line from left: begin, 3 diameters, and a point at end of each circumference.

Each time we found out the diameter fits on the circumference 3 times with a little bit left over. This is true of every circle – it was discovered a long time ago.

We know that in 1560 BC, the Chinese used this number. But it was the Greeks who gave this number a name. They gave it the name of one of their letters – pi.

But their letter looked like this (write the symbol – squiggly horizontal line and two vertical lines dropping down from it)

Pi is a relationship between the circumference and diameter of a circle. They calculated it for many circles and decided the number to use was **3.14** – three and fourteen hundredths.

The number called **pi, 3.14 - has been rounded off, but it's not a repeating decimal. Mathematicians even now, have computers working day and night to see if they can get it to repeat.** But it hasn't yet.

I wonder if knowing about pi will help us find the circumference of a circle. Rolling the circle was kind of hard – and what if I had a really big circle?

What about a circle like this? (hold up unit) I happen to know the diameter of this circle is 10 centimeters. And we know the diameter will fit on the circumference 3.14 times. So how can we find the circumference? (multiply)

Yes, we multiply 10 centimeters times 3.14 – and my circumference is therefore 31.4 centimeters. This works for all circles!

So the formula is:

Circumference of a circle is equal to the diameter times pi

$$(C = d\pi)$$

Extend

We know that to find the circumference we use diameter times pi.

But what if I only know my radius? What can I do?

A diameter is the same as two radii – so we can say

the circumference is equal 2 times pi times the radius.

$$(C = 2\pi r)$$

Epilogue

Pi is a ratio of the diameter to the circumference of a circle. The children can now find the diameter and/or circumferences of circles in the room. They can make up word problems.

Formula for Area of a Circle

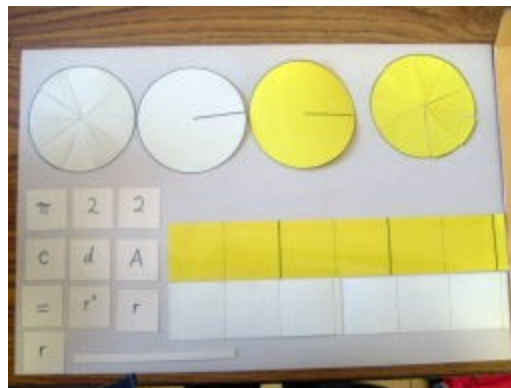
Material

Envelope (teacher-made) marked area of a circle containing:

- 2 different colored circles but same size – each with a radius marked and the circumference highlighted with a black edge,
- the same 2 circles – one cut into 10 equal pieces and the second cut into 11 pieces – 9 of which are equal with the tenth cut in half, circumference is also highlighted
- a long yellow rectangle – the height of which is equal to the radius of a circle and base

of which is equal to the circumference of the circle – diameters marked off in heavy black lines lengthwise across the rectangle plus radii also marked off in lighter lines

- 2 short white rectangles, height equal to the radius and base equal to $\frac{1}{2}$ circumference – so each white rectangle is one-half the yellow rectangle – radii are marked off.
- Tickets for the formula – uppercase A, and C, lowercase d, r (2), r^2 , one with ' π ' on one side and 3.14 on the other, the number 2 (2), equal sign and a division bar.

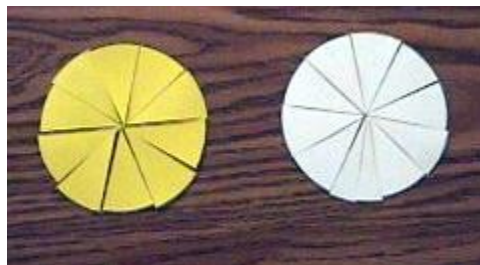


Begin

Who can form these pieces into a nice tight circle? (using yellow circle pieces)

While you do that, I'll make this white circle from these parts. (do so)

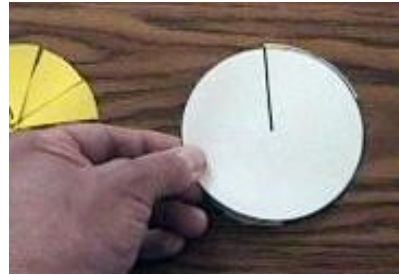
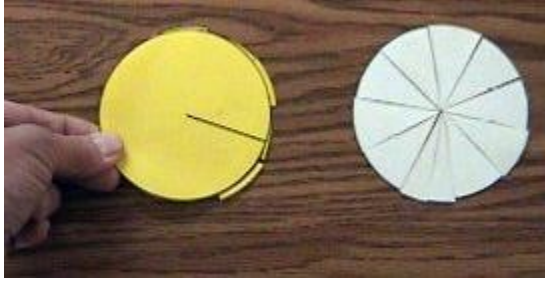
(two circles, one white and one yellow, are now constructed in pieces on the table)



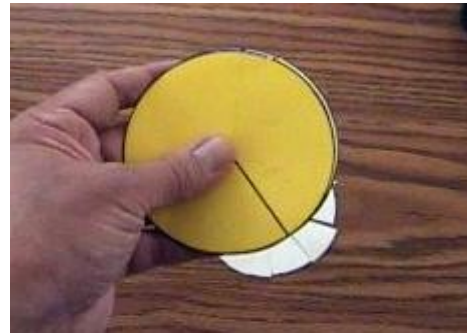
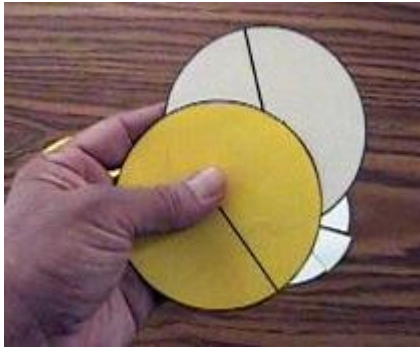
Let's look at these circles. They are congruent. (hold up whole circle versions)

Now what about these two? (superimpose the whole yellow onto the pieced yellow)

Now what about these two? (superimpose the whole white onto the pieced white)



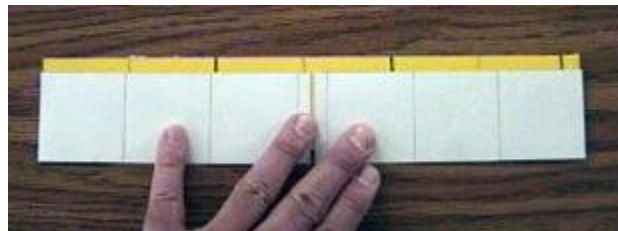
And the two whole circles are also equivalent. (yellow and white)



What does that tell me about the two pieced circles then? (also equivalent)

So all of these circles are equivalent.

I have these rectangle also. (superimpose the two white rectangles onto the yellow)



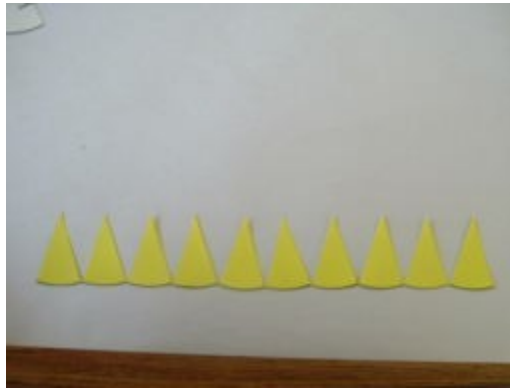
They are equivalent – the two white rectangles and the one yellow.

We want to use this material to see if we can learn how to calculate the area of a circle.

What other similar work have we done? What did we do with a triangle?

(turned them into rectangles and then multiplied base times height)

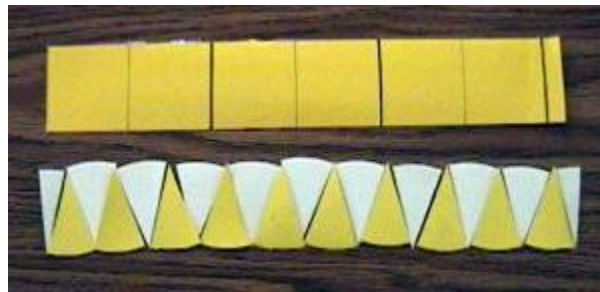
Watch.



(move pieces from yellow pieced circle onto a straight line – arraying left to right -with arc of circle on the straight line)

It's not a rectangle yet. Maybe this will help. Watch.

(place white pieces from white pieced triangle onto the arrayed yellow pieces with arc upward and each piece interleaved between colors)



If we took more time and cut the circle into more pieces, my edges would not be as wavy – but we can see this is beginning to look like a rectangle.

Can you imagine what this would look like if we kept cutting into more and more pieces?

Let's compare this yellow rectangle to the one we built. (superimpose)

They are very close – so we'll say they are equivalent.

How would I find the area of this whole yellow rectangle? (base times height)

What is this height with respect to the circle? (radius)

What about the base with respect to the circle (circumference)

So we could say the area of a circle equals the circumference times the radius.

Form $A = Cr$ using tickets

But how many circles did I use? (2) So let's divide this formula by 2.

form $A = \frac{(Cr)}{2}$

But what if I don't know the circumference. How can I find it using the diameter?

Well I have to multiply it times π .

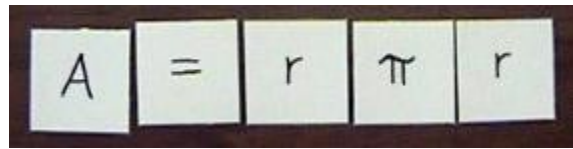
A visual representation of the formula $A = \frac{d \pi r}{2}$ using tickets. The formula is shown as a sequence of tickets: 'A', '=', 'd', 'π', 'r', followed by a horizontal line, and then '2' below the line.

What if I know the radius but not the diameter?

So I'll take out my d and make it $2r$

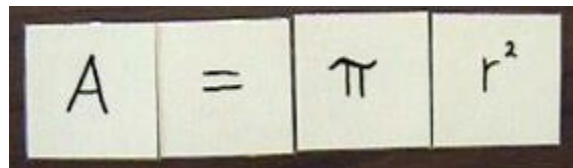
A visual representation of the formula $A = \frac{2r \pi r}{2}$ using tickets. The formula is shown as a sequence of tickets: 'A', '=', '2', 'r', 'π', 'r', followed by a horizontal line, and then '2' below the line.

But this is kind of awkward – so we'll get rid of this 2 over 2. That's the same as 1 anyway.



$$A = r \pi r$$

We also have r times r – so I can substitute r^2 for r times r.



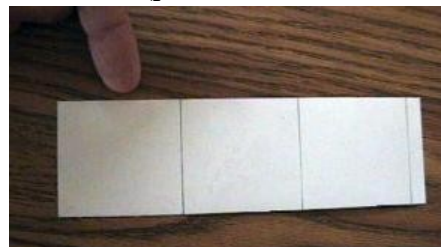
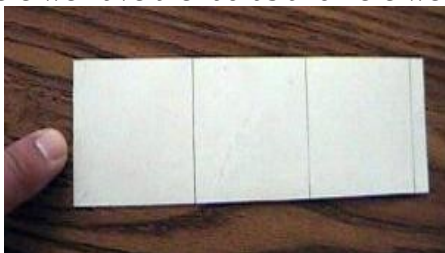
$$A = \pi r^2$$

We said this big rectangle has the area of two circles. What does that tell us about the area of the small rectangle? (it has area of 1 circle)



We looked at this rectangle – here on **this side is the radius** (hold up white small lengthwise - at side is 1 radius) And this **bottom was half the circumference**.

Here we have the radius and here we have the radius (point to side and to first mark)



This is radius squared. We have radius squared 1, 2 3 times and a little bit more.

So this is the formula – r^2 taken pi times. (refer to the small white rectangle)

Epilogue

Now children can calculate the area of circles, using word problems and other materials. The circle pieces are based on the tenths of the circle fraction pieces.

January 14, 2004

Relationship Between Apothem and Sides of a Polygon

Materials

- envelope labeled hexagons: 3 regular hexagons with 1 apothem marked in red on each. One has sides of 1 centimeter, another with sides of 1 decimeter, last with sides of $\frac{1}{2}$ decimeter (not used here)
- envelope labeled: fourth large hexagon in (6 equilateral triangles) parts. One side is 1 meter long. On one side the apothem is marked.
- Meter stick and metric ruler

Age

Usually upper elementary

Prologue

This is usually done after work with the circle and pi. Children become curious about relationship of circumference and diameter and they might wonder about other similar relationships. Another small group presentation.

Begin

You asked whether there is something like pi for other figures.

I am going to measure the side of this hexagon.

This side is 1 centimeter.

I'm going to measure its apothem too. (do so)

It seems to be about .8 centimeters

Well let's look at this hexagon.

I am going to measure the side of this hexagon.

This side is 1 decimeter.

I'm going to measure its apothem too. (do so)

It seems to be about .86 decimeters

Well let's look at this large hexagon. (put it together)

I am going to measure the side of this hexagon too.

This side is 1 meter.

I'm going to measure its apothem too. (do so)

It seems to be about .866 meters

So these apothem seem to be the same size for the apothem – allowing for the different circle sizes – but more precise each time.

People have studied this and they say the relationship between the apothem and the side of a regular hexagon can be expressed as

—
.86 where the 6 is a repeating decimal.

So if you have a regular hexagon and know the length of one side, you can multiply that number by .86 (or .866 or .8666, etc.) and that would give you the length of the apothem.

Area of a Regular Hexagon

Once you know the apothem, you can use that measure to find the area of a regular hexagon.

$$A = \frac{(ap)}{2}$$

So we have 0.86 apothem times 6 meters, all divided by 2 – which equals 2.58 m².

Epilogue

Children can use this to solve word problems – given all but one of a given variable in the formula. For example, give apothem and area and ask for side – or – give them the side and apothem and ask for the area.

Every regular polygon has this relationship though the number is different. The children can explore geometry books and find the number for each respective polygon.

January 24, 2005

Solid Geometry

The Concept of Volume

Abstract

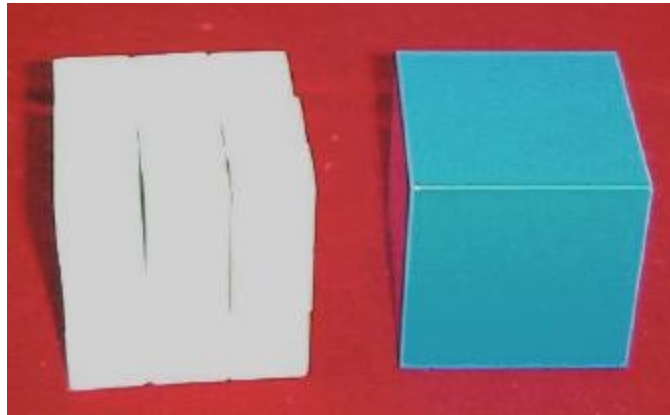
Model a blue cube by using 27 white 2 cm. cubes and discover how to express volume in cubic units.

Age

8 or 8 ½

Material

Geometric solids, 2 centimeter cubes, and a felt mat.



Prologue

We'll give the children concrete experience with solid geometry. At the primary level, they have had some experience with solid geometry: the pink tower, brown stair, red rods and various cylinders. They've taken in impressions of solids all their lives. Hopefully they've worked with these materials (geometric solids) and have matched up those solids with corresponding bases – leading to seeing the relationship between 2 and 3 dimensional – between plane and solid figures.

Some children are more experienced than others, including related language. Much of this work is concrete and sensorial, so we don't wait too long to introduce it. Because of the work needed before, the children are often 8 years old or older. They will

extend applying and using whatever formulas they have learned. Send them to the primary class to borrow the cubes and prisms from the pink tower, brown stair, or geometric solids.

The children will explore the fact that when dealing with a solids and their volume, one is referring to the space inside this solid figure. This is the space to be measured. If the borrow the pink tower cubes, use one of the even numbered cubes for the first presentation; along with whatever solids we use, we also use the box of 2 centimeter cubes (tiny white box). Don't use anything smaller in order to use the 2 cm. Cubes.

As you build the cube, simply build it to look like the original; don't intimate some structured method other than this; then compare at the end to assure it is the same.

These are all small group presentations.

Begin

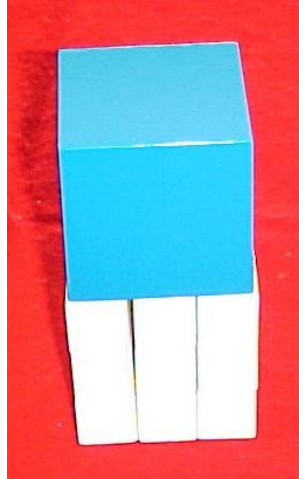
(hold up blue cube)

What shape is this? (cube)

Today, we'll learn how to measure the space inside this cube. (set down on the mat)

To do that, we'll use this (hold up small (2 cm.) white cube). This is going to be our unit of measurement.

Let's see if we can make a cube just like this one (blue one), out of these two centimeter cubes. (blue is in center of mat, collect 9 cubes to build bottom layer, same for middle and top layers- 27 total, and place blue cube on top of the white one we've built the white one)



I made a cube the same size. (place blue down again on mat)

If we want to express the volume of this cube, the space inside, we can do that in 27 unit cubes. We just have to count how many we have. I have 3 times 3 (count sides) times 3 – 27 unit cubes.

(hold up blue) So, we can say the volume of this cube is 27 unit cubes. We measured the space inside here.

Would you like to do another one?

Epilogue

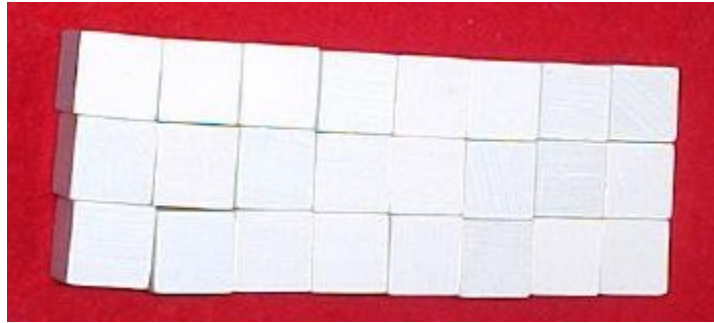
We can also try the blue prism as follow up. Remember to guide the children to **express the volume as some number of “unit cubes.”**

January 24, 2005

Equivalence as Related to Solid Figures

Material

Box of 2 cm. Cubes, felt mat, paper and pencil.



2 cm. Cubes in a 1 x 3 x 8 configuration

Prologue

All of these are small group presentations. **We are using factors** here.

Begin

(count out some number of cubes, choose 24)

Let's see what different solids I can make out of these 24 unit cubes.

I'll make the first one – then it's your turn. (form into a 4 x 2 x 3 (high))

My solid is 4 by 2 by 3. (point to the sides as you say each dimension)

Who has an idea of a different solid that you'd like to build from the same cubes?

(use same 24 cubes and children take turns)

How can we express your figure? (2 by 3 (high) by 4)

It's 2 by 3 by 4 – and it's equivalent to the one I built. Who can make another one?

Can we make another one equivalent to the one I made and the one we just made now?

How do you express this one? (1 (high) x 1 x 24)

Is yours equivalent too? Why? (because it has the same number of cubes)

Let's write that down (write $24 = 1 \times 1 \times 24$) How do we write the others we built?

Epilogue

This continues to use the concept of equivalence from prior lessons. If they pick irregular figures, we can still point out equivalence but cannot refer to dimensions.

January 24, 2005

Three Important Dimensions

Material

Box of 2 cm. cubes, felt mat, gray rectangular prism and 5 yellow rectangular prisms from the solid geometry material.

Prologue

Informally, we watch and assess if they get the idea of the 2 dimensions of the base (we need to get area of base) and the 3rd dimension of height. When they do, we can begin this lesson. It helps the children become consciously aware of the 3 dimensions used to measure volume. Take care in how to move gray prism so as not to double-count.

Begin

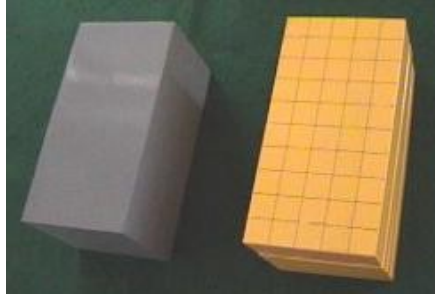
(gray prism in the center of mat)



What shape is this? (rectangular prism)

(bring yellows prisms to the mat)

Here I have some yellow slices that altogether are the same as this gray prism.

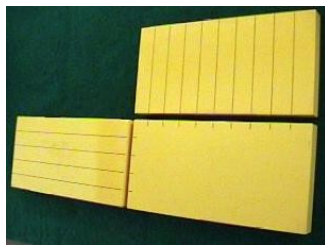
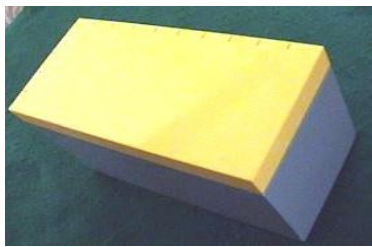


(set gray on top of stack of yellows to compare sensorially)



They're the same, right?

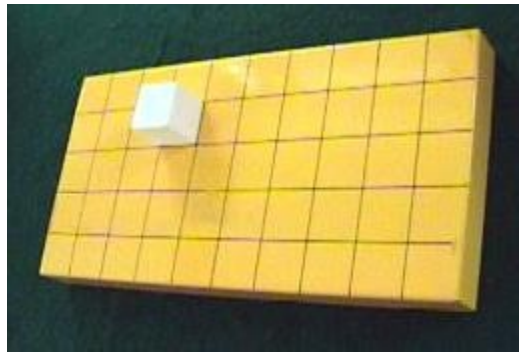
Let's see what we have here. (hold up one yellow rectangle, it has markings on the side)



(repeat another yellow rectangle, they now have lines across, up and down and finally one with a grid – there are 5 total – hold up grid)



This one is covered with squares.



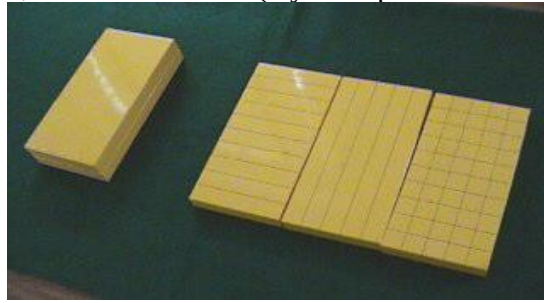
One of these 2 centimeter cubes fits exactly on one of these squares drawn here, and it's exactly the same height as my yellow prism.

So we can say this yellow prism (single) is made up of unit cubes. How many do we have? We have 1,2,3,4 ... 9, 10 times 1,2,3,4,5 – so we have 50 unit cubes.



50 unit cubes makes up the base of my prism. (four yellow lay vertically, at right is the grid prism horizontally)

We have 50 taken 1 time, 50 taken 2 times (lay next prism from left atop the grid)



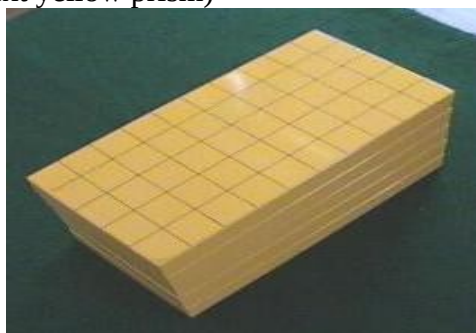
50 taken 3 times (place next yellow prism)



50 taken 4 times (place next yellow prism)



50 taken 5 times (place next yellow prism)



So altogether – 50 taken 5 times give us 250 unit cubes. That is the same as this gray one because they're equivalent.

Now that we know how to count and multiply to find the volume, let's see if we can do this with just the gray prism – not using the yellow prisms.

(yellow at bottom left, stacked up and put aside, gray in the center mat)



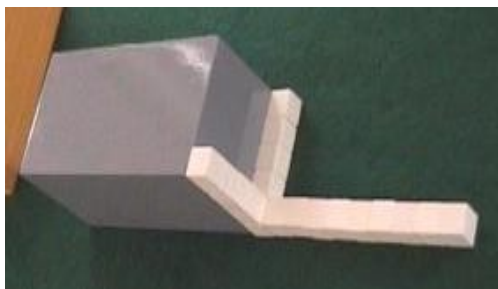
First we count how many we have here. (lay white unit cubes along the bottom long edge)

That's one side of the base.

(move prism so we can see how many at short edge bottom – lay cubes along that edge))

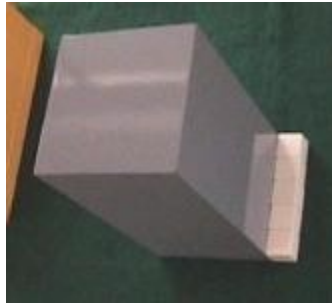


But then we also need how many we have here (lay cubes up the side for the height and remove the gray cube leaving the 3 dimensions marks in a structure of white cubes)



Now to know the volume, we count here 10, and here ... 5 5 times 10 gives us the area of the base – and 1,2,3,4,5 height is 250 – that's gives us 250 unit cubes.

Suppose my prism is in this orientation (set on end) Let's see if we can find those 3 dimensions again.



(find first side at bottom, move prism to allow construction of the 3 stacks of cubes that form the structure of the gray prism, move gray again to construct the white cubes along the edge upward to find the height – looks like a tower with 2 feet at end)



To express our volume, we first find the area of the base which is 5 times 5, giving us 25 – times 10 for the height – which gives us 250 unit cubes. (look at the structure as you go)

So now when you want to find the volume of a solid like this, you don't have to build the whole thing out of unit cubes – just find this three important dimensions and multiply them.

January 24, 2005

Equivalence Between Prisms with Various Bases

Material

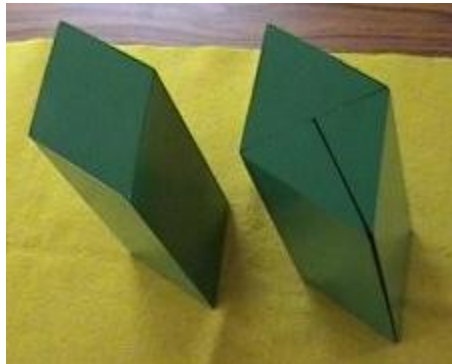
Two Green rhomboidal prisms and two blue hexagonal prisms – all on a floor mat on the floor or – on a felt mat on a low table. Green set has a solid rhomboidal prism and one that's unequally divided into 3 pieces height-wise. Hexagonal set has a solid whole and one that is unequally divided into 4 pieces height-wise. There is variation in the way these pieces are cut so review their shapes beforehand – use a rubber band to protect them from falling.

Prologue

This needs to be done on the floor on a floor mat or on a low table – so the children can clearly see the tops of the figures. These are all small group presentations.

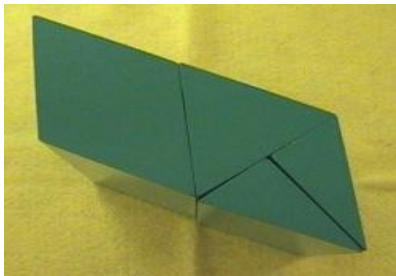
Begin

(begin with both green)



Here we have a rhomboidal prism. And here's another. (point to each)

This one is cut up into pieces. It looks to me they are the same but let's compare to make sure.

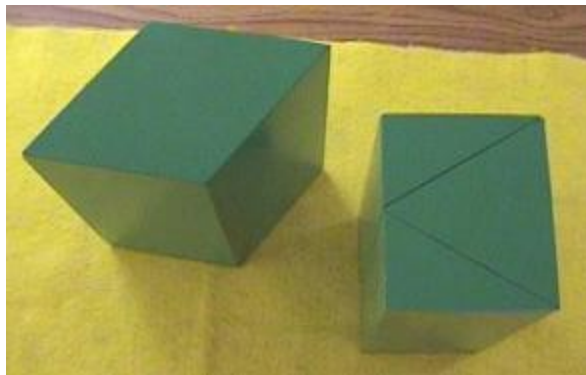


(set both on their ends height-wise, place each side against it's pair side, then set base on base)

Yes, they are the same – each face is the same.

So anything we build out of these cut-up pieces will be equivalent to the whole one.

I can move these pieces here (make a rectangle from two small pieces plus the whole – standing on the end)



What's the relationship between these two prisms? (equivalent)

How about this one? (two small pieces together set on end, stacked atop the 3rd larger

piece on the bottom, like a tower 2-high – not including the whole solid)



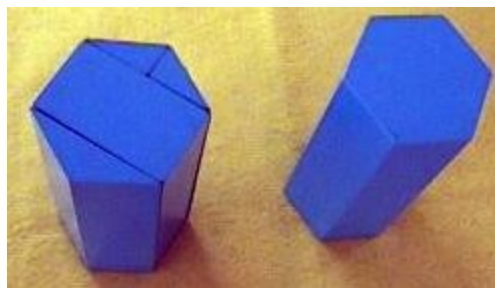
(repeat other variations)

If we wanted the volume of these what do we need? (find area of the base and multiply that times the height).

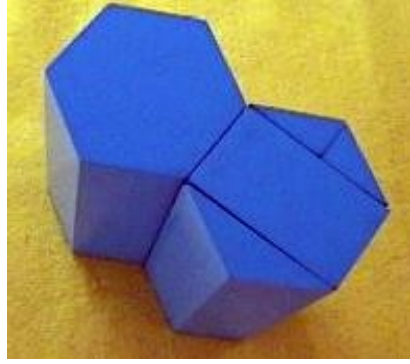
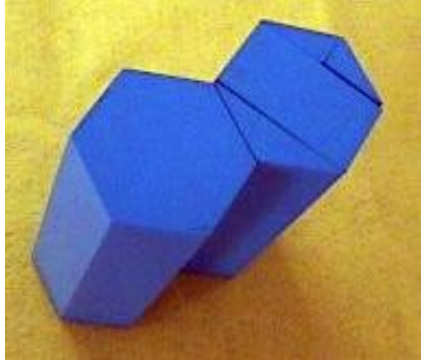
===

We also have this blue set of hexagonal prisms.

Here we have a hexagonal prism and here we have one that looks like it's the same. Let's compare them to be sure.

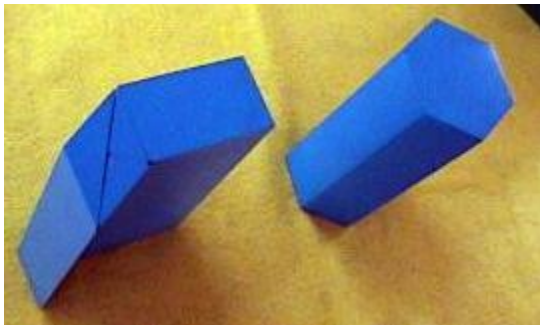
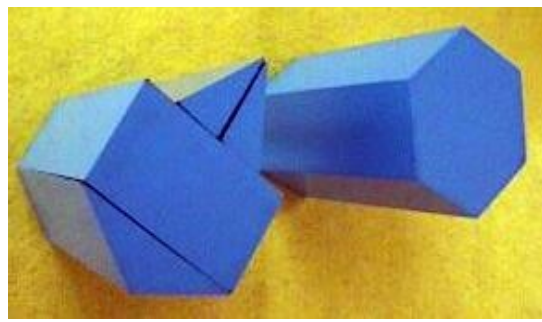
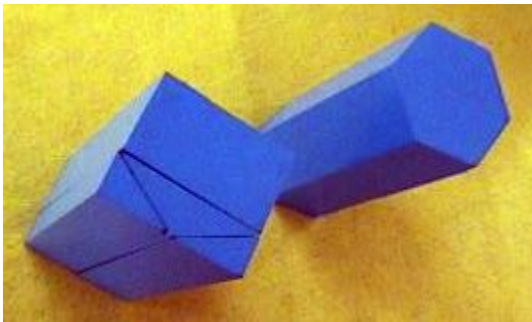


(set all faces against the other to assure the figures are equivalent)



So anything we build of the cut-up pieces will be equivalent to the solid piece.

(create shapes before, again setting the prisms on end, combining the pieces in different ways as with the green material)



Are these figures equivalent? What if we want to find the volume? (find the area of the base and multiply by the height)

Epilogue

We should be aware the divided prism – the two smaller pieces are equivalent to the third piece.

January 24, 2005

Derivation of the Formula

Material

Two Green rhomboidal prisms and two blue hexagonal prisms – all on a floor mat on the floor or – on a felt mat on a low table.

Tickets for volume (in a clear box labeled volume): 1 'V', 1 "Ab", 1 '3', 1 'h' 1 '=' and a division line.

Prerequisites

Children must have had all solid geometry lessons up to this point. Other lessons required are formulas for area, factors, multiples, equivalence and measurement.

Begin

(mix up tickets to be used to the right side of mat – both green rhomboidal prisms on mat, on end)

What kind of prisms are these? (rhomboidal prisms) What's their relationship? (equivalent)

This is the base (pick up and touch bottom, then set down)

This (touch top) is in a way also a base – and we can see this one.

Any kind of solid we make from these cut-out pieces has what relationship to the whole? (equivalent)

If we want to find out the volume of this prisms (whole) we first need to find what? (area of base)

Then we'd multiply that times what? (height)

So we can say volume equals area of the base times the height. This is our formula for finding volume.

What formula gives us the area of the base?

(base times height of the rhomboidal prism's base – verbalize formula) **$Ab = bh$**

So base times height of the rhomboidal prism's base gives us the area of its base. We then multiply that times its height.

So you could get our metric ruler and measure the base and height of the rhomboidal prism's base and plug those numbers into the formula for area of a rhombus. Then we'd measure the height of the rhomboidal prism and plug that number in.

Form formula with tickets:

$$V = Ab \times h$$

The final answer is expressed in units cubed – with a metric ruler it will be centimeters cube.

===

Let's look at our hexagonal solids (note equivalence of blue prisms as done above)

And anytime we want the volume of regular solids, first we need area of base and then multiply that times height.

So volume equals area of the base times the height. Now what formula would we need to plug in for the area of the base this time?

(apothem times perimeter – all over 2 – (formula for regular polygon) – for area of base)

How will we find the apothem here? (refer to the lesson relating side and apothem)

(0.86 times length of side gives apothem, gives length of the apothem)

Then multiply area of the base times height. Express the final answer in cubic centimeters.

Form formula with tickets:

$$V = Ab \times h$$

And for this hexagonal solid, we said this was equal to $(\frac{ap}{2}) \times h$

Epilogue

We want the children to apply these formulas. After measuring these particular prisms, they can explore others.

January 24, 2005

Solids of Rotation

Material

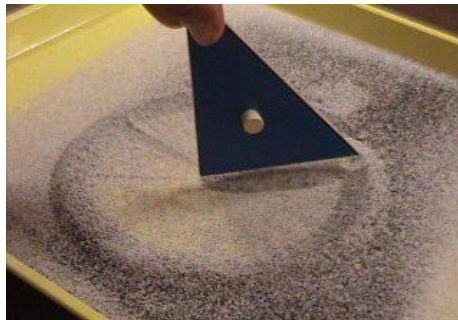
A tray with thin layer of fine sand in it. Either right-angled triangle from the geometry cabinet or green right-angle triangle in the constructive box of triangles. A rectangle also taken from the geometry cabinet.

Begin

There's an interesting relationship between plane figures and solid figures we work with.

I'm going to show you something about the relationship between this triangle (hold up blue right triangle) and a solid figure you know.

(hold up triangle on it's bottom edge, at top with index finger, and rotate it around in the sand)



What shape is in my tray now? (circle) But if we filled in all of the space my triangle just passed through, what figure would we have? (cone)

A cone, yes. A **cone is one of the solids of rotation.**

(shake tray to make sand level and evenly distributed again)

(hold up rectangle on its edge in same manner and rotate around)

What shape do we have now? (circle)

If we filled in the area covered by the rectangle as I rotated it, what shape do we have? (cylinder)

A **cylinder is another one of the solids of rotation.**

What about a sphere? What figure would you need to rotate to get a sphere? (circle)

Ovoid – made from rotating an oval

Ellipsoid is made from rotating an ellipse.

Epilogue

This is required before speaking about volume of a cylinder or a cone.

January 24, 2005

Volume of a Pyramid

Material

Metal solid blue hollow-volume material, tall rectangular prism, sand (or water, but carefully drying objects after lesson if using water) – 4-sided hollow pyramid, and shorter rectangular prism. Tickets for volume.

Begin

We'll use this material to help us find the formula for the volume of a pyramid. We already know the formula for the volume of the prism, right?

What formula do we need?

(area of base times height)

How about for this figure (pyramid)? Can't do it, right?

(put tall rectangular prism and pyramid together, side by side)

Let's put them right next to each other. What can we say we know? (heights are same)

Let's compare the bases (put pyramid on top of tall rectangular prism – both on end to make them tall)

So they have the same base and same height.

I wonder how many pyramids worth of sand it will take to fill up my tall rectangular prism (begin fill the pyramid with sand, level off with lid – dump the sand into the tall rectangular prism)

(hold up to show inside of rectangular prism)

It's not full yet. Let's try again. (fill pyramid again to top and pour into rectangular prism)

(show rectangular prism inside again)

It's not full yet. Let's try again. (fill pyramid again to top and pour into rectangular prism)

How many will it take do you think? (continue process of filling pyramid, pouring into rectangular prism)

(at three times, the rectangular prism is full – note the materials are inexact)

So it takes 3 pyramids of sand to fill up this tall rectangular prism.

If we want to express the formula for finding the volume of our pyramid in relation to this rectangular prism, what would it be?

(Area of the base times the height, divided by 3)

$$V = \frac{Abh}{3}$$

===

We also have this shorter rectangular prism. I wonder what it's relationship is to the taller rectangular prism. Let's compare the heights. (mark off height with finder against tall prism)

There 1, 2 and 3 times the shorter height – is needed to be as tall as our taller prism.

Let's compare out bases – they're exactly the same.

So how many pyramids will it take to fill this shorter prism? (1)

Yes, it should take one. (fill pyramid with sand and then transfer sand to short prism)

It fits exactly. Now let's see if we can use this to express the formula for the volume of our pyramid – this time in relationship to this shorter prism.

Volume equals what? (volume is area of base times the 1/3 height) because the height of the shorter prism is 1/3 the height of the pyramid.

$$V = Ab\left(\frac{h}{3}\right)$$

January 24, 2005

Volume of Cylinders and Cones

Material

Blue solid cylinder and cone and box of tickets for volume labels.

Begin

Remember our solids of rotation we talked about? Here are two of them, a cone and cylinder.

Let's compare the two of them (place cone on cylinder)

What can we say about the bases? (same)

(place next to each other on table)

What about their heights? (also same)

If we want to express the volume of the cylinder, how are we going to do that?

Volume equals area of the base times the height. $V = Abh$

This time what formula would we plug in for the area of the base? (π times r^2)

We also have this cone; we can assume the cone has the same relationship to the cylinder as the pyramid did to the tall rectangular prism. (same height and same base in both cases)

So what does that do to our formula? (have to divide it all (volume of cylinder) by 3)

$$V = \frac{(Abh)}{3}$$

I divide by 3 to express the formula for the volume of the cone. There are also formulas for finding volume of the sphere, ellipsoid and ovoid.

Now, you can look them up and apply them.

January 25, 2005

Polyhedrons

Nomenclature

Material

Blue solid geometry material: square pyramid, triangular prism, tetrahedron
straw triangle outline plus thin cardboard face with height marked in red, (regular
tetrahedron – from 4 equilateral triangles) triangular pyramid made from constructive
triangle set, plumb line.

Prologue

Polyhedrons are solids bounded by planes.

Purpose

Introduce the term slant height.

Begin

(begin with tray and polyhedrons on the table – no mat)

This is a polyhedron (hold up triangular prism)

The word polyhedron comes from '**poly**' which means **many**, and the Greek word '**hedra**' which means **base or seat**. A triangular prism could have many bases depending on how you set it.

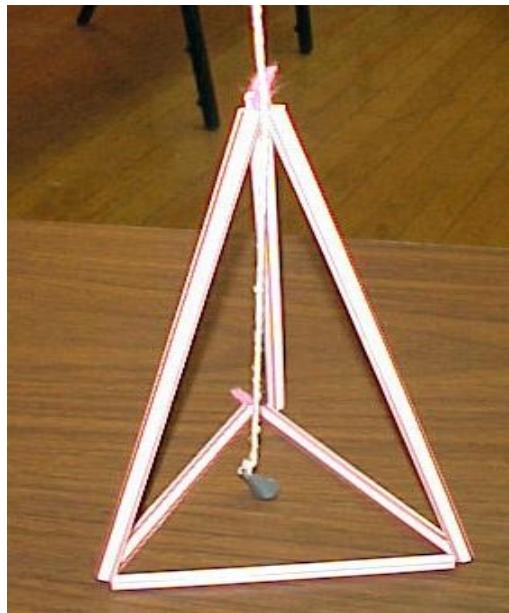
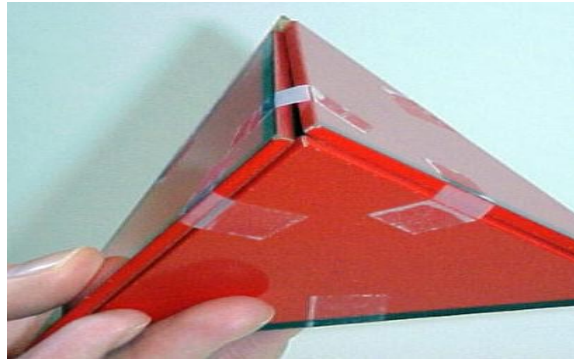
This is a regular polyhedron because each surface is the same. It's a regular tetrahedron – tetra meaning 4. It has 4 equal bases.

We'll look at nomenclature for polyhedrons. We have surfaces and a base. There is volume.

What would we call this point? (vertex) (show blue square pyramid)

We also have perimeter of the base here (bottom of pyramid)

Here's a polyhedron made from straws. I made this one out of constructive triangles.



Using my plumb line and my straw polyhedron, I can show the height.
(hold plumb line from the vertex until the weight touches the table)

Polyhedrons that come to a point have 2 heights.

Looking at this white triangle, we **can see a height marked down the center**. This triangle is made to fit exactly here (against the face of straw polyhedron)



This red line shows us what is called the **slant height**.
(repeat for 2nd period nomenclature practice)

Which height is greater? (slant height)

Epilogue

Children may want to make polyhedrons of their own using straws and so forth.

January 25, 2005

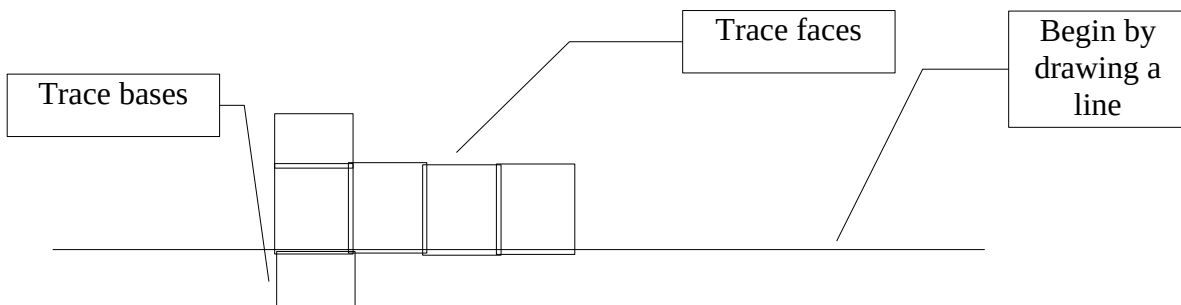
Surface Area

Prologue

We'll derive formulas for lateral area and total area. This is a very sensorial presentation that enables us to derive formulas for lateral and total area.

Material

Geometric solids, pieces of plain paper or newsprint, ruler, lead pencil, piece of chalk and crayons.



Begin

(hold up rectangular prism)

If we have a polyhedron like this – we learned how to measure the volume, right?

Volume refers to the space inside, right?

Today, let's look at surface area.

Lateral area refers to the area of these sides. It does not include the base or bases.

Total area does include the base or bases.

We'll derive the formulas for each type of area using a solid and piece of paper.

Let's start with a rectangular prism. We want these surfaces on the outside.

We know how to get volume – the space on the inside; but we want the outside.

Lateral Area

Lateral area is the area around the four sides. We want to find that area.

We could measure one side and multiply it by 4. But instead of doing that, we arrive at a formula instead.

Start by drawing a line on the paper and line up the prism with that line, with one face down on the paper. I can trace around the face now.

Now I turn it to the next face and trace it (continue tracing each side)
(now we have the four faces traced along the horizontal line)

We know how to find the area of a rectangle using base times height. What part of my prism (hold up blue rectangular prism) gives us the base?
(perimeter)

So we can say **lateral area equals the perimeter times – what part gives us the height – times the height: that's the lateral area or...**

$$LA = ph$$

Total Area

Now suppose I want to include the area of the bases. That will give us the total area.

First thing to do is to trace the bases onto the paper. I'll add one here at the top of my left hand rectangle. (see paper notes) And then, I'll add the other base below it.

The **total area is equal to the lateral area plus – how many bases (2) – plus the area of 2 bases.**

We know that to find the lateral area, we multiplied the perimeter times the height. What do we need to do to find the area of one of the bases?

(base times height of the base)

So we need 2 times base times height altogether. We write lateral area this way:

Formulas

$$LA = ph \quad \text{(lateral area formula)}$$

$$TA = LA + 2B = ph + 2bh \quad (\text{total area formula})$$

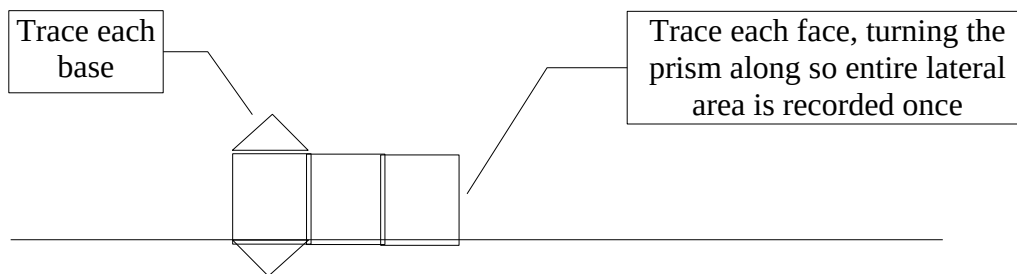
(write both formulas on paper, including $TA = ph + 2s^2$ for where bases are square)

On another day --

Suppose we have a triangular prism. Can we find the lateral and total areas for this?

(draw line on paper as before – lay prism down, lining up bottom with the line and trace first face)

We're not done yet! (roll figure to next side and trace – continue for 3 sides)



On our paper, we have a rectangle and we know how to find an area of that.

The lateral area equals – what part of our prism gives us the base of the rectangle (perimeter of the base).

What part of the prism gives us the height of the rectangle? (height)

So we have perimeter times height for our lateral area.

So now we want the total area so we'll add on our bases – adding one here above our first left rectangle and another just below it.

Our total area equals the lateral area plus 2 bases.

This time what formula do we need to find the area of the base? ($2bh$ over 2)

The 2s cancel out.

$$LA = ph$$

$$TA = LA + 2B = ph + 2 \frac{(bh)}{2} = ph + bh$$

(p = perimeter of the base)

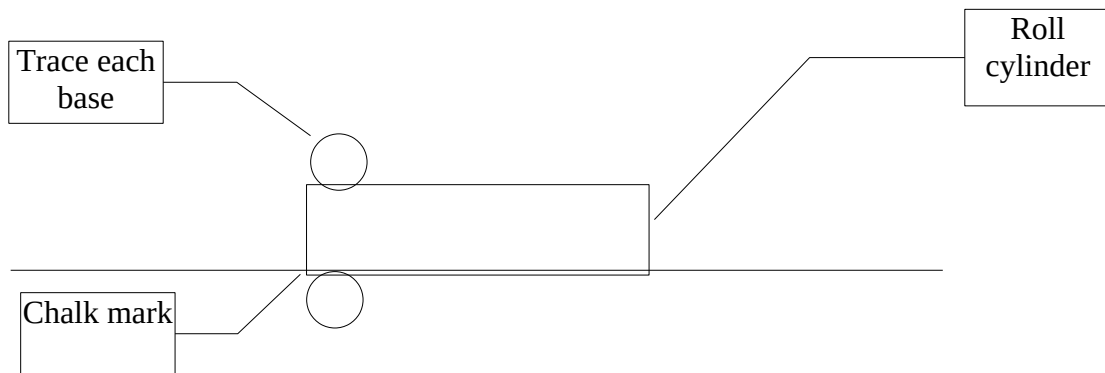
SO the 2s cancel out. So we have $ph + bh$

(ph is perimeter of base times height and bh is base times height)

=== on another day ..

What about this one? (round cylinder)

Can we find the lateral and total area for this too?



Let's start the same way – by drawing a line on paper.

Here I have to use my chalk to mark a little spot on the edge or circumference of the circle-base.

I'll align my cylinder with my line and mark the top of my cylinder – where it's touching the paper.

I mark the top and bottom of the cylinder and roll it across the page – then I can connect

my four points to make a rectangle from the lateral area.

So lateral area equals – what part of the cylinder give base of rectangle (circumference of base) – circumference uses 'C' – what part of the cylinder gives us the height of the rectangle (height). And that's how we find lateral area.

Now I need to find the total area. So we have to add on our bases. We have to mark them on our paper. One here and one down here.

Total area equals lateral area plus 2 bases.

Lateral area = Ch

Area of base = $2(\pi \times r^2)$

(2 comes from the 2 bases)

So lateral area = Ch

(circumference times height)

Total area = $LA + 2B = Ch + 2(\pi \times r^2)$

(lateral area plus area of the 2 bases)

===

What about this? (square pyramid)

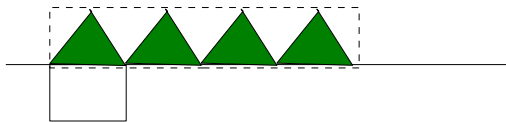
Can we find the lateral and total areas of this?

(start off again drawing a line, tracing each side, etc.)

I need these faces for the lateral area. Let's line the base of the pyramid edge on the line and trace around my first face.

(continue for all 4 faces)

I want to keep track of what I've traced so I'm going to use a crayon to color in the surface area I've outlined. We can't just multiply this time, can we?



We've been finding rectangles so far so let's outline a rectangle here by enclosing what we've traced.

Now we have a shape we can find the area of – base times height.

What part of our pyramid gives us the base of our rectangle? (perimeter)

What part of our pyramid gives us the height of the rectangle? (slant height)

So perimeter of the base times the slant height – if I just multiply perimeter of the base times the slant height – will that give me the lateral area of my pyramid? (no, we have 2 of them – by enclosing the rectangle). So I have to divide by 2 to get the **lateral area**.

So what about total area. We need to add the base on. We know that:

$$TA = LA + 1B$$

So we plug in lateral area formula = perimeter times slant height divided by 2 plus area of the base (base times height of the base) – that will give us the total area.

$$LA = \text{perimeter of base times slant height} = p \times sH - \text{all over } 2)$$

$$TA = LA + B = \frac{(p \times sH)}{2} + bh$$

===

How about this cone? Can we find lateral and total area for this?

(start off drawing line on the paper)

We have a circular base again, so I'll use my chalk to make a mark on the circumference.

This time I'll line up the cone sideways along the line.

And I'll mark where my vertex ended – right on the line.

The spot I made with my chalk is right on the line.

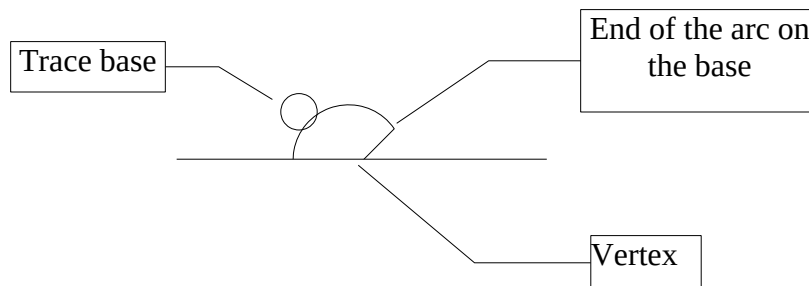
I'm going to hold my vertex steady – while I roll the figure and trace at the same time – until that little dot arrives back down to the paper.

(turn paper counter-clockwise 90 degrees to give impression of figure as triangle)

What does this most look like? (triangle)



How would we find area for a triangle? (base times height divided by 2)



What part (circumference of base)

What part of the cone gives us the height of the triangle? (slant height)

So circumference times slant height – divided by two since it's a triangle – give us lateral area.

$$LA = (C \times sH) / 2$$

Now total area means we need to add on our base – somewhere along the curved line.

Total area equals lateral area plus – in this case, 1 base

$$TA = LA + B = (\text{circumference of base times slant height of cone} / 2) \text{ plus } (\pi \text{ times } r^2)$$

Epilogue

Be aware there are different standards for notation (e.g., H_{prism}). By recording the bases where we are, the child can cut out the shape and construct the solid.

A great book for geometry is called “Master Math Geometry”, by Debra Anne Ross, published 2005, ISBN 1564146677

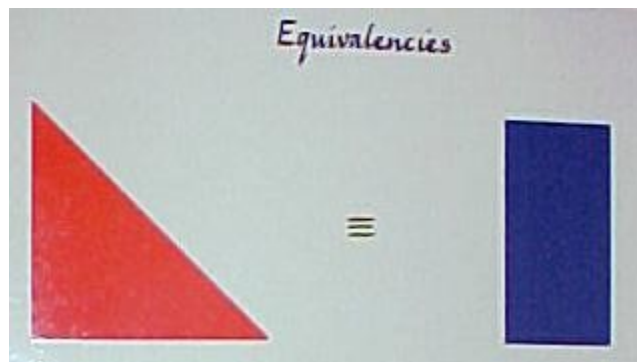
Constructing the solid geometric figures from stiff construction paper is an excellent follow-up.

January 25, 2005

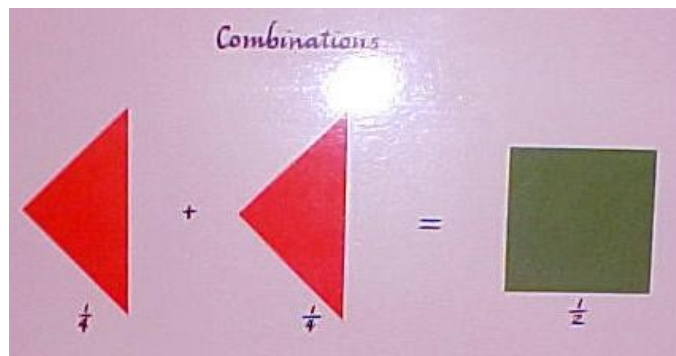
Geometry Charts

Consist of two sets: squares and triangles. We've seen circle already with fractions. The charts are made on half sheets of poster board using fade-less art paper. They are meant to be brought out after a presentation to remind the children of the lesson and inspire follow-up work – though the charts are never used in the lessons.

Suppose you presented equivalence; here is a chart that reinforces that concept.

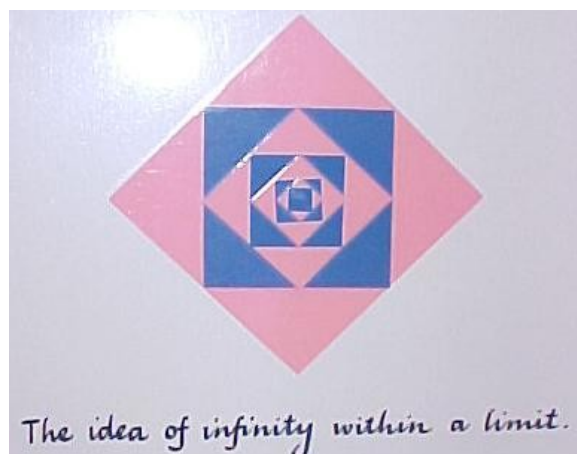


This one shows combinations: two triangles make up a square.



There are some interesting charts that go into the concept of “infinity within a limit”.

These (hold up three charts) are representations trying to show something that goes on and on forever. The children should never be required to copy or remake the charts.



There are many of these geometry charts. As new teachers, you need to prioritize material-making. These are low-priority since they are not required for the lessons and these concepts are not part of the public school curriculum. When you get to making these, realize that a few can be made at a time.

Include illustrations of these charts to be able to make them in the future. The colors don't matter as long as we are consistent; the exception is that the unit is typically gold to denote its importance.

See appendix A for remaining charts.

January 25, 2005

Geometry Nomenclature Material

The geometry nomenclature is made following the same pattern as other nomenclature material. We have pictures without the name, loose labels with names only, definitions with the names missing. Loose labels can be used with either the pictures or definitions. The control of error is the booklet containing all of the above.

The material has been organized here into different sets or racks by letter.



The first set A is “from a point a solid.” The second rack B has to do with lines. Third set C has to do with angles. Set D is for general plane figures. Set E is for triangles, F is for quadrilaterals, G is for polygons and set H is for circles.

Just as the other nomenclature material we've seen, this is introduced to the children after the appropriate lesson. We don't expect the children to learn the terms from this material; it is learned from the lessons. They do provide a way of reviewing a lesson. They provide ideas for follow-up work and independent work.

The definitions provide a way to review what we've introduced in the lessons. We aren't looking for the children to memorize definitions – preferring the children create their own. Once they've written their own definition, they can compare it with the definition in the nomenclature material.

This material is meant to be early reading material. So definitions that are here are simplified. Once the children can read and understand the scientific definitions they don't need this material any longer since they'll use resource books instead.

The material is made with each set having a different color cover. But there is no required coding of colors.

January 25, 2005

Geometry Commands

Geometry commands are written follow-up work that give the children some ideas of what to do for follow-up work. These command should use the same nomenclature that we've introduced in lessons – any other wording in the command card must be clear and straight-forward for understanding. Try to incorporate commands that involve handwork and an exploration of the environment.

The ideal circumstance is that we give a clear exciting presentation and the children can't wait to get to work, and full of ideas. In spite of this, matters are sometime different and the children aren't used to choosing their own follow-up work. We might take over a class that has been run by a teacher who assigned follow-up work after every presentation – and this has killed their volition and enthusiasm. They may have no clue how to begin. We have to help the children work toward the ability to freely choose their own follow-up work.

On the other hand, we might begin a brand new lower elementary class in which there are no older children inspiring and modeling for the younger ones. Again, the teacher must help the children realize what they can do on their own of their own volition.

The geometry command cards can be a useful tool to get the children working and engaged – they can't be swinging on the chandeliers. They are used as follow-up to presentations – though we may not assign a command card as such. But if we have given a lesson and see a child doing nothing – invite the child to choose an angle or some element from the lessons and suggest to them they try out one of the command cards.

If we find ourselves using these tools 5 years later with the same children, we should ask why. Why weren't we able to get the child working away from them on his own. We might suggest they write their own command card and place it in their work in progress file for later use. The cards should not be used as a crutch.

The commands are organized by topic: command related to angles, etc.

January 25, 2005

Geometry Summary

In the elementary class, we want to transmit to the children a love for geometry. We want to do that by putting them in contact with all of these wonderful materials and ideas, related to geometry. Then – leave them free to have their own experiences with the material. The materials will serve in this way, as a foundation in building up their knowledge that will serve them later on. That formal study is at a more abstract level. We give the concrete sensorial foundation that make the abstractions understandable.

Our primary purpose is not so much to teach the children geometrical information by rather to utilize the subject of geometry to enable the child to use his mind for logical reasoning, deductive reasoning, for finding relationships. The area of geometry is especially useful as a catalyst for this sort of thing.

Whenever possible, bring in historical aspects – the people involved in the development of geometry. But as well as the historical aspects, we want to relate geometry to biology and zoology, nature and architecture, art and music. This is a wonderful book to make some of these connections to nature (“By Nature's Design”, ISBN 0811803295). One example discusses the angles made by cracks that are relieving pressure on a uniform material, from shrinkage – 120 degrees. That is compared with honeycomb angles.

Any research we can inspire in the children will be of benefit to them. They may become interested in studying machines and the geometry involved there. Or perhaps traffic signs that use geometry can be used in some manner. These are things that help to keep geometry alive in the classroom – rather than letting it become a subject of a curriculum. We need to make sure they learn the public curriculum but beyond this we can suggest beyond that.

We have built up from plane figures to solids. At some point in the children's experience, we want them to be able to reflect on the spectrum of geometric constructions that begin with the point, go on to the line (one dimension), to a surface or plane (two dimensions), and then on to solids (three dimensions). The child has already had quite a bit of sensorial preparation for this. Points were seen in the form of golden bead material: the unit (point), ten (line), hundred (square plane) and thousand (cube solid). We also have our first nomenclature booklet (from a point to a solid); often the children discover the nomenclature booklets but if not we call it to their attention.

Appendix A

